

High-Speed Silicon Electro-Optic Modulator for Electronic Photonic Integrated Circuits

by

Fuwan Gan

B.S (EE), Shanghai Jiao Tong University (1997)

M.S (EE), Shanghai Jiao Tong University (2000)

Submitted to the Department of Electrical Engineering and Computer
Science

in partial fulfillment of the requirements for the degree of

Doctor of Philosophy

at the

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

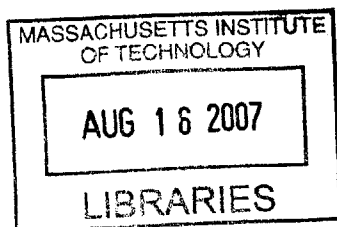
June 2007

© Massachusetts Institute of Technology 2007. All rights reserved.

Author
Department of Electrical Engineering and Computer Science
June 8, 2007

Certified by
Franz X. Kärtner
Professor of Electrical Engineering
Thesis Supervisor

Accepted by
Arthur C. Smith
Chairman, Department Committee on Graduate Students



BARKER

High-Speed Silicon Electro-Optic Modulator for Electronic Photonic Integrated Circuits

by

Fuwan Gan

Submitted to the Department of Electrical Engineering and Computer Science
on June 8, 2007, in partial fulfillment of the
requirements for the degree of
Doctor of Philosophy

Abstract

The development of future electronic-photonic integrated circuits (EPIC) based on silicon technology critically depends on the availability of CMOS-compatible high-speed modulators that enable the interaction of electronic and optical signals. This thesis investigates electrically driven Mach-Zehnder modulators based on high-index contrast silicon waveguide technology and electronic carrier injection.

Modulators based on four different structures are investigated: the forward-biased PIN diode with and without lifetime reduction, the reverse-biased PIN/PN diode and a metal-oxide-semiconductor (MOS) structure. These devices are compared with each other in terms of achievable performance. A modulator based on the forward-biased PIN diode with lifetime reduction is designed to reach 34GHz bandwidth and a low figure of merit $V_\pi \cdot L = 0.6V \cdot cm$ using a carrier lifetime reduction and a graded doping profile. A bandwidth of 1-2GHz has been demonstrated so far which is considerably smaller than the design bandwidth due to high series resistance. Modulators based on the forward-biased PIN structure without lifetime reduction have a low figure of merit, very low voltage and extremely low power consumption in the low frequency regime. The measurements demonstrate a RF power consumption of 100mW for 25% modulation depth and a figure of merit of $V_\pi \cdot L = 0.28V \cdot cm$ at frequencies up to 10GHz. A pre-compensation technique, using a high pass filter which consists of a parallel resistor and capacitor, extends the modulator bandwidth from 100MHz to 5GHz experimentally. Further it is shown that, modulators based on the reverse-biased structure can in principle reach very high speed, up to 40-80GHz in design but it's difficult to reduce $V_\pi \cdot L$ values close to or even below $1V \cdot cm$ and the necessary drive voltage is higher than the voltage provided by the CMOS technology. For the measured bandwidth of the fabricated devices so far only 1-2GHz has been demonstrated. This discrepancy is caused by the RC delay due to the experimental setup and high contact resistance. Finally, the performance of the modulator based on the metal-on-semiconductor (MOS) structure is analyzed.

Furthermore, an electrically driven Mach-Zehnder waveguide modulator based on a high-index contrast silicon split-ridge waveguide (SRW) technology and electronic

carrier injection is proposed. The excellent optical and carrier confinement possible in high-index contrast waveguide devices, together with the forward biased operation and the good thermal heat sinking due to the silicon slab close to the waveguide, enables high speed modulation with small signal modulation bandwidths beyond 20GHz, a V_π times length figure of merit of $V_\pi \cdot L = 0.5\text{Vcm}$ and an insertion loss of about 5.3 dB.

Finally, all-optical switches based on optical carrier-injection in high index contrast Si/SiO_2 split-ridge-waveguide (SRW) couplers are proposed. The waveguide devices are suitable for the construction of low-loss optical switch matrices as well as fast optical switching. These devices exhibit robustness against fabrication tolerances, improved heat sinking, good carrier confinement and high uniformity in transmission over the entire C-band of optical communications in contrast to comparable devices based on buried or ridge waveguides. A reasonably low electrical switching power of 1-10mW is predicted for switching frequencies in the 1MHz-1GHz range. Faster switching speed can be achieved by carrier lifetime reduction.

Thesis Supervisor: Franz X. Kärtner
Title: Professor of Electrical Engineering

Acknowledgments

I have been fortunate enough to work with many wonderful people during my research at MIT. My thesis supervisor, Prof. Franz X. Kärtner, has inspired me to this exciting field with his enthusiasm and guided me through my thesis work with his kindness, openness, brightness and georgeous thinking. He has been tremendously patient and encouraging when I encountered difficulties. I would like to thank Prof. Erich P. Ippen and Prof. Rajeev J. Ram for serving as my thesis committee and I had many valuable discussion with them.

I sincerely thank Dr. Steven J. Spector at Lincoln Laboratory for device fabrication and Dr. Michael W. Geis and Robert T. Schulein for device characterization. It was due to their excellent work that my designs were turned into practical devices. I thank Dr. Jung U. Yoon, Dr. Matthew E. Grein, Dr. Theodore M. Lyszczarz and all other collaboratos at Lincoln Laboratory for their fabulous work. I'm thankful to Dr. Felix Jan Grawert for guiding me with his expertise in laboratory and helping me to set up the carrier lifetime measurement. Ali R. Motamedi and Hyunil Byun have spent tremendous effort on the pump-probe measurement of carrier lifetimes. Dr. Jürgen Michel and Prof. Kazumi Wada were tremendously helpful in the design of all-optical switches and the RF-PCD lifetime measurements. I also want to thank Dr. Shoji Akiyama for the passivation of wafers used in carrier lifetime measurements carried out by myself. I benefited a lot from Jan-Mattle Schley's thesis on all-optical switching in silicon. Furthermore, I would like to thank Dr. Michael Watts, Miloš A. Popović, Anatoly M. Khilo, M. S. Dahlem and many others for many valuable discussion.

I like to thank my group mates for being good friends and making my MIT life colorful and enjoyable. My friends, Xudong Chen, Dandong Feng, Jianping Fu, Ji Li, Jifeng Liu, Ling Qin, Jianrong Pei, Wei Zhang, Jinshan Xu, Lirong Zeng and many others, have been so supportive and we enjoyed a wonderful time together.

Most importantly, I like to thank my girl friend Jing Zhou for her love and support. She has always been interested in my work and helped tremendously on my thesis

writing. We have shared every moment together, pleasant or difficult, and so will we in the future.

Finally, I thank my father, Jingheng Gan, and my mother, Xinfeng Huang, my brothers, sisters and all other family members for their love and encouragement over the years. I could not have reached this point without their support.

Contents

1	Introduction	21
1.1	Background and Motivation	21
1.2	Outline of Thesis	27
2	Overview and Theory of Optical Modulators	29
2.1	Physical Effects for Optical Modulation	30
2.1.1	Electrooptic Effects (Pockels and Kerr effects)	30
2.1.2	Franz-Keldysh Effect	31
2.1.3	Quantum Confined Stark Effect (QCSE)	32
2.1.4	Plasma Dispersion Effect	34
2.1.5	Other Effects	39
2.2	Materials for Optical Modulators	39
2.2.1	Dielectric Crystal and IIIV-Semiconductor Materials	40
2.2.2	Group-IV Materials	41
2.3	Prior Art of Silicon-Based Optical Modulators	41
2.4	Theory of Optical Modulators	47
2.4.1	PN Diode Theory	48
2.4.2	PIN Diode Theory	52
2.4.3	Transmission Characteristics of Mach-Zehnder Interferometer	55
3	High-Speed Silicon Modulators for High-Index-Contrast (HIC) Waveguides	59
3.1	Investigation of Modulation	60

3.1.1	Modulator Based on Forward-Biased PIN Diodes with Lifetime Reduction	61
3.1.2	Modulator Based on Forward-Biased PIN Diodes without Lifetime Reduction	69
3.1.3	Modulator Based on Reverse-Biased Diodes	82
3.1.4	Modulator Based on Metal-Oxide-Semiconductor (MOS) . . .	86
3.1.5	Comparison of Forward-Biased PIN, Reverse-Biased Diodes and MOS Structures	88
3.2	Large-Signal Modeling of PIN Diodes Using SPICE	90
3.2.1	SPICE Model Implementation	90
3.2.2	SPICE Model Results	93
3.3	Fabrication and Measurement Setup	97
3.3.1	Device Fabrication	97
3.3.2	Experimental Setup	98
4	Electro-Optic Modulators and All-Optical Switches Based on Carrier-Injection in Si/SiO_2 Split-Ridge Waveguides (SRW's)	103
4.1	Waveguide and Coupler Design	104
4.1.1	Split-Ridge Waveguide (SRW)	104
4.1.2	Broadband Coupler with Split-Ridge Waveguide (SRW)	107
4.2	Electro-Optic Modulators Based on SRW	111
4.2.1	Device Structure	111
4.2.2	Device Performance	113
4.3	All-Optical Switches Based on SRW	117
4.3.1	Switching Principle and Device Structures	118
4.3.2	Two-Waveguide All-Optical Switch	119
4.3.3	Three-Waveguide All-Optical Switch	121
4.3.4	Switching Speed	123
4.4	Summary	123

5	Carrier Lifetime in High-Index-Contrast Waveguides	127
5.1	Recombination Mechanisms	128
5.1.1	Radiative Recombination	128
5.1.2	Auger Recombination	130
5.1.3	Shockley-Read-Hall (SRH) Recombination	132
5.1.4	Surface Recombination	134
5.2	Carrier Lifetime Measurement Model	135
5.2.1	Lifetime Model	135
5.2.2	Effective Lifetime	138
5.3	Carrier Lifetime Measurement	140
5.4	Carrier Lifetime Extraction in a Waveguide with PIN Diode	144
5.4.1	Kuno's Method (Reverse Recovery)	144
5.4.2	Caverly's Method (RF Technique)	146
6	Conclusions and Future Work	149
6.1	Conclusions	149
6.2	Future Work	151
A	DC/AC Characteristics of PIN Diode	153
B	PIN Diode SPICE Model	159
C	PiN Diode SPICE Model Code	169

List of Figures

1-1	Power dissipation of microprocessors for different technology nodes [3].	22
1-2	Timing-jitter requirement for ADCs as a function of sampling rate and number of effective bits [8].	23
1-3	Optical intra-chip architecture for multi-core microprocessors [9]. . . .	24
1-4	Optical analog-to-digital converter [10].	25
2-1	(a) Equilibrium state, (b) Franz-Keldysh effect and the tilting of energy bands in a uniform electric field [20].	32
2-2	Quantum-well energy subbands and wave functions (a) in the absence of an applied electric field, and (b) in the presence of an applied electric field [37].	33
2-3	(a) The subband energy of Ge/SiGe quantum wells, (b) effective absorption coefficient spectra [78].	34
2-4	(a) Experimental optical absorption spectra of c-silicon as a function of electron concentration, (b) calculated refractive index change as a function of electron concentration based on Kramers-Kronig theorem at wavelength $\lambda = 1.55\mu m$. The circle lines are for holes and the solid lines are for electrons [35].	37
2-5	(a) Refractive index change based on Soref model (solid, triangle) and Drude model(dashed, circle), (b) absorption coefficient change based on Soref model (solid, triangle) and Drude model(dashed, circle) at wavelength $\lambda = 1.55\mu m$	38

2-6	(a) PN junction with the depletion region from $-x_p$ to x_n , (b) the band structure of a forward-biased diode [33].	48
2-7	Normalized PN diode diffusion capacitance and conductance as a function of $\omega\tau_p$ (p^+n diode) or $\omega\tau_n$ (n^+p diode) [33].	52
2-8	Diagram of PIN diode structure, carrier distribution at intrinsic region and the current density at the edges $x = \pm d$ when assuming that the injection efficiency at both ends is unity. The small subscripts n/p indicate electrons and holes, and the capital subscripts N/P indicate the current flows from the P/N-region.	53
2-9	Transmission of Mach-Zehnder interferometer with push-pull method and $\phi_B = 89.1^\circ$: $\frac{1}{2}e^{-2\alpha L_{DC}} \cos(2\Delta\phi - \phi_B)$ (red solid-dot), $\frac{1}{2}e^{-2\alpha L_{DC}} \cosh(\Delta\phi/4)$	57
3-1	(a) Mach-Zehnder modulator with forward biased PIN-phase modulators, (b) different doping profiles for the Si-buried waveguide pin-phase modulators.	62
3-2	(a) Optical loss due to doping (solid) and with carrier injection (dashed), (b) f_{3dB} (solid) and $V_\pi L$ (dashed) of structure A with different width w_r , $\tau = 10ps$, $V_{DC} = 3V$	63
3-3	Optical loss due to doping (solid) and with carrier injection (dashed) of structure C (a) with $w_d = 0nm$ and (b) with $w_d = 50nm$, f_{3dB} (solid) and $V_\pi L$ (dashed) of structure C (c) with $w_d = 0nm$ and (d) with $w_d = 50nm$. ($\tau = 10ps$, $V_{DC} = 2.5V$)	64
3-4	(a) Modulator I-V characteristics; (b) effective index change due to plasma and thermal effects for $\tau = 10ps$ (solid), 100ps (dashed), 1ns (dotted) in structure C. ($w = 450nm$, $h = 250nm$, $h_s = 50nm$, $w_r = 200nm$, $w_d = 50nm$)	65
3-5	Modeled performance of the modulators based on different structures.	66
3-6	(a) Measured I-V curves of PIN diodes with three different processes, (b) comparison between the modulators with silicon ion implantation (blue line) and light n-type doping (red line).	67

3-7	Diode current in terms of input optical power. The waveguide is silicon ion implanted.	68
3-8	(a) Measured I-V curves of PIN diodes at different temperatures, (b) extracted total resistance and series resistance R_s from measured I-V curves. (6mm long, 20 seconds annealing in vacuum $450^\circ C$, the waveguide is not silicon ion implanted and has no sidewall doping.) .	70
3-9	Comparison of I-V curves between measurement and modeling. (6mm long, 20 seconds annealing in vacuum $450^\circ C$, the waveguide is not silicon ion implanted and has no sidewall doping.)	71
3-10	Power consumption of a 500- μm -long Mach-Zehnder interferometer in forward bias with 25% modulation depth: measurement (solid), result from analytic modeling (dash) and MEDICI numerical modeling (dash-dotted).	74
3-11	Measured RF power input in terms of driving frequency for phase shifters with length 0.5mm (square line) and 2mm (cross line). The measurement was done with the modulation depth to be 25% for both modulators.	75
3-12	Measured modulation depth of different structures by applying the same RF power.	79
3-13	(a) Frequency response of Mach-Zehnder modulators: measurement under forward biasing (solid); MEDICI simulation without pre-compensation circuit (dash-dotted); MEDICI modeling with pre-compensation circuit (dashed); measurement of reverse biased device (dotted); analytical modeling under forward biasing (short dashed), (b) PIN diode circuit model with a simple RC pre-compensation circuit. All forward-biased diodes are 250- μm -long and reverse-biased diode is 6-mm long.	81
3-14	(a) Measured modulation at 10GHz without pre-compensation. The modulation depth is 25%, (b) measured modulation depth with different RF power and frequencies. The length is 500 μm	82

3-15	Measured frequency response of the Mach-Zehnder modulator. The plots are normalized to be 0dB at the lowest frequency [46].	83
3-16	(a) Refractive index change due to plasma effect, (b) refractive index change due to plasma and thermal effects. The series resistance $R_s = 4\Omega$, the length is $L = 6mm$	84
3-17	Carrier distribution in the base region during forward conduction (solid line) and reverse-recovery (dashed) [49].	91
3-18	Carrier distribution in base-region for (a) DC/small-signal analysis and (b) large-signal analysis [49].	92
3-19	DC characteristics of MEDICI simulation (solid) and SPICE modeling (dashed).	94
3-20	(a) Total resistance and (b) total reactive impedance versus frequency at forward bias $V_{DC} = 0.5V$, (c) diffusion capacitance versus forward bias at frequency $f = 100Hz$	95
3-21	(a) 4th order PIN subcircuit,(b) another equivalent 4th order PIN subcircuit, (c) carrier distribution in PIN intrinsic region [49].	96
3-22	(a) Transient MEDICI current (solid) and HSPICE current (dashed), (b) transient MEDICI total carriers (solid) and HSPICE total carriers current (dashed). The total carriers include holes and electrons. The random signal is [0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 1, 0] with one bit period to be 100ps.	97
3-23	(a) high n-type doping, (b) high p-type doping, (c) low n-doping or silicon ion implantation, (d) final structure after process.	98
3-24	(a) Mach-Zehnder interferometer, (b) SEM image of waveguide with metal contact, (c) SEM image of waveguide cross-section.	99
3-25	Schematic diagram of the experimental setup for modulator RF measurement.	101
3-26	(a) Experimental setup, (b) Mach-Zehnder modulator with probes.	101

4-1	Structure layout and single-mode profile for: (a) buried waveguide, (b) ridge waveguide, (c) split-ridge waveguide (SRW)	105
4-2	Relative change of cladding decay coefficient $\Delta\alpha/\alpha_0$ as a function of slab waveguide width with a waveguide width variation of 2.5 nm at the wavelength of $\lambda = 1.55\mu m$	108
4-3	64 x 64 all optical router.	110
4-4	Bandwidth for two-waveguide couplers based on buried (dashed) and SRW (solid) waveguides	111
4-5	(a) Externally-biased Mach-Zehnder modulator with coplanar RF-feeder, (b) split-ridge waveguide pin-phase modulator with single mode intensity profile.	112
4-6	(a) Modulator I-V-characteristics and maximum device temperature T_m at different operating points and carrier lifetimes $\tau=10ps$ (solid), $100ps$ (dashed), $1ns$ (dotted); (b) effective index change due to plasma and thermal effects for $\tau=10ps$ (solid), $100ps$ (dashed), $1ns$ (dotted). .	114
4-7	Sketch of the proposed switches (a) two-waveguide Switch, (b) three-waveguide Switch.	118
4-8	Two-waveguide switch performance (a) transmission to port D (%), (b) transmission to port C (%), (c) ON-OFF transmission ratio (dB)	125
4-9	Three-waveguide switch performance (a) transmission of port D (%), (b) transmission of port C (%), (c) ON-OFF transmission ratio (dB)	126
5-1	The Band-to-band radiative recombination processes in indirect band-gap semiconductors [20].	128
5-2	Band-to-band Auger recombination processes (a) eeh process and (b) ehg process [37].	130
5-3	(a) High injection Auger lifetime measurements for high resistivity n-type silicon ($\sim 90Vcm$) and p-type silicon ($\sim 150Vcm$), (b) values for the ambipolar Auger coefficient as a function of excess carrier density [59].	131

5-4	The energy band diagrams for the Shockley-Read-Hall generation-recombination processes: (a) electron capture $R_n = c_n n N_t (1 - f_t)$, (b) electron emission $G_n = e_n N_t f_t$, (c) hole capture $R_p = c_p p N_t f_t$, and (d) hole capture $G_p = e_p N_t (1 - f_t)$ [37].	133
5-5	Sample configuration	135
5-6	Effective lifetime with the wafer thickness (a) $d = 625\mu m$ and (b) $d = 1.5\mu m$	140
5-7	Implementation of the dual-color carrier lifetime measurement.	141
5-8	Modulated reflectivity versus frequency for $1.5\mu m$ thick SOI wafers passivated by thermal and LTO oxides, respectively.	143
5-9	(a) Diode switching circuit, (b) transient reverse recovery of a PIN diode with thick base region[69] and (c) with a thin base region.	145
5-10	Reactive impedance as a function of frequency.	147
A-1	Diagram of PIN diode structure, carrier distribution at intrinsic region and the current density at the edges $x = \pm d$ when assuming that the injection efficiency at both ends is not unity.	156
B-1	Carrier distribution in the base region during forward conduction (solid) and reverse-recovery (dashed) [49].	160
B-2	The three-stage equivalent circuit and model equations interpretation (a linear distributed dynamic system and two nonlinear static blocks) for PIN Spice Model [49].	163
B-3	Large-signal analysis with sinusoidal source $V(t) = V_0 + V_1 \sin(2\pi ft - \pi/2)$ ($V_0 = 0.4V$, $V_1 = 0.4V$).	164
B-4	(a) Carriers obtained from HSPICE simulations, (b) carriers obtained from MEDICI simulations, (c) comparison between carries from HSPICE and MEDICI . The square pulse train starts with 1V and varies between +1V and 0V, the period is $T = 1/5GHz$ and PIN diode length is $L = 500\mu m$	165

- B-5 (a) Modeled current versus time from MEDICI (solid) and HSPICE (dashed). (b) total carriers (half electrons and half holes) variation in the waveguide versus time of MEDICI (solid) and HSPICE (dashed). The square pulse train starts with 1V and varies between +1V and 0V, period is $T = 1/5GHz$ and PIN diode length is $L = 500\mu m$ 167
- B-6 (a) Modeled current versus time of MEDICI (solid) and HSPICE (dashed). (b) total carriers (half electrons and half holes) variation in the waveguide versus time of MEDICI (solid) and HSPICE (dashed). The square pulse train starts with 1V and varies between +1V and -1V , period is $T = 1/5GHz$ and PIN diode length is $L = 500\mu m$ 167
- B-7 (a) Modeled current versus time of MEDICI (solid) and HSPICE (dashed). (b) total carriers (half electrons and half holes) Variation in the Waveguide versus time of MEDICI (solid) and HSPICE (dashed). The square pulse train starts with 1V and varies between +1V and 0V, the period is $T = 1/10GHz$ and PIN diode length is $L = 500\mu m$ 168
- B-8 (a) Modeled current versus time of MEDICI (solid) and HSPICE (dashed). (b) total carriers (half electrons and half holes) Variation in the waveguide versus time of MEDICI (solid) and HSPICE (dashed). The square pulse train starts with 1V and varies between +1V and -1V, period is $T = 1/10GHz$ and PIN diode length is $L = 500\mu m$ 168

List of Tables

2.1	Materials for modulation	40
2.2	Electro-optic modulators (1980–1989)	42
2.3	Electro-optic modulators (1990–2004)	43
2.4	Electro-optic modulators (2004–2007)	44
3.1	Modulation efficiency at 10GHz	78
3.2	Comparison of forward-biased, reverse-biased PIN/PN diodes and MOS structures	89
3.3	SPICE model parameters of PIN diode	93
5.1	B-values for silicon at $300^{\circ}K$	129
5.2	Carrier lifetime measurement results	142

Chapter 1

Introduction

1.1 Background and Motivation

Modern microelectronics is driving the computing power by integrating more and more complementary metal oxide semiconductor (CMOS) transistors onto a single chip. For example, the Intel Itanium chip contains 410 million transistors per cm^2 [1]. It is expected that an integrated circuit will have 18,497 million transistors per cm^2 by 2020 [2]. The performance of integrated circuits has been exponentially improved by shrinking the transistor size and increasing the transistor density. However, when the feature size is below 100nm a simple scaling of transistor size does not lead to as much gain in computing as they had in the past. There are at least two challenges in increasing transistor density: power dissipation and interconnect latency [3]. The power dissipation, consisting of active power and leakage power, grows over the technology nodes because of the increase of the transistor density, as shown in Fig.1-1. The leakage power increases much faster than the active power for each technology node. The power dissipation in microprocessors is expected to be around 450W before 2010 when the industry has planned nodes at 45nm [1]. This is unacceptable for a chip with an area of around 1 inch². Secondly, the interconnect latency due to the RC and transmission line delay becomes severe in integrated circuits with a small feature size and a high operating frequency. The latency is mainly caused by the long global interconnections on the IC. The total length of metal interconnects can approximately

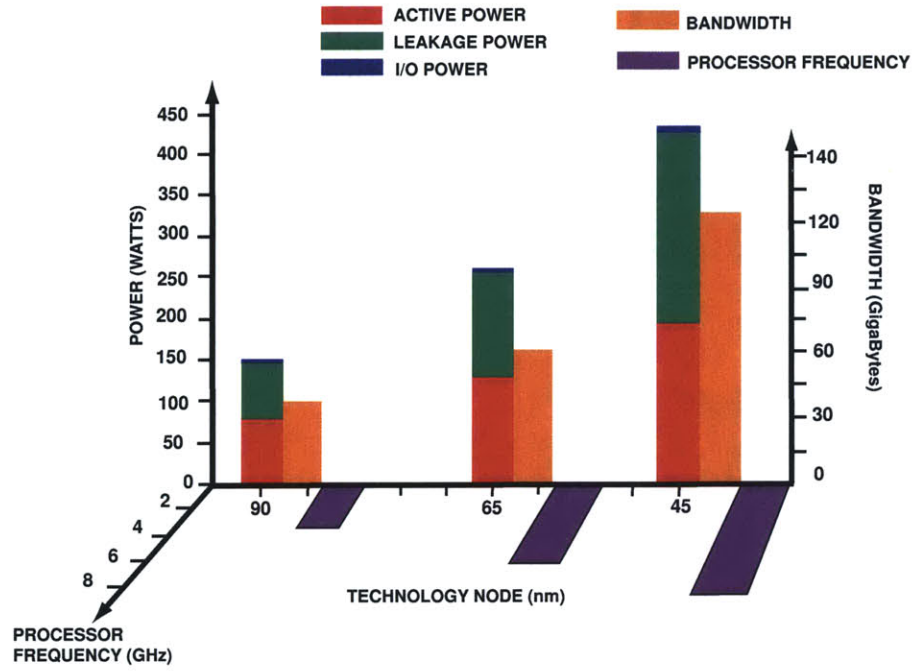


Figure 1-1: Power dissipation of microprocessors for different technology nodes [3].

be equal to the edge dimension of the chip which is around 15 mm, 20 mm and 30 mm for the 250-, 130- and 50-nm technology nodes, respectively [4]. The delay time of these wires is around 150-250ps which increases as the wires are longer and finer. This range of time delay is unacceptable for high-performance integrated circuits. Therefore, more intelligent schemes are necessary to improve integrated circuits and reduce power dissipation and interconnection latency.

Recently silicon photonics has been emerging as a promising solution in solving some of the dilemmas that electronic integrated circuits are facing. Silicon photonics processes and transmits signals by integrating electronics and photonics into a single Si-chip based on CMOS-compatible processes, also called silicon electronic photonic integrated circuits (EPIC). Silicon photonics can not only enhance the performance at high operating frequencies, but also provide larger bandwidth than electronic integrated circuits. Firstly, photonics is immune to electro-magnetic interference (EMI) which is an intrinsic advantage for large scale integration, contrary to pure electronics where the EMI becomes a serious problem especially in high-bandwidth applications.

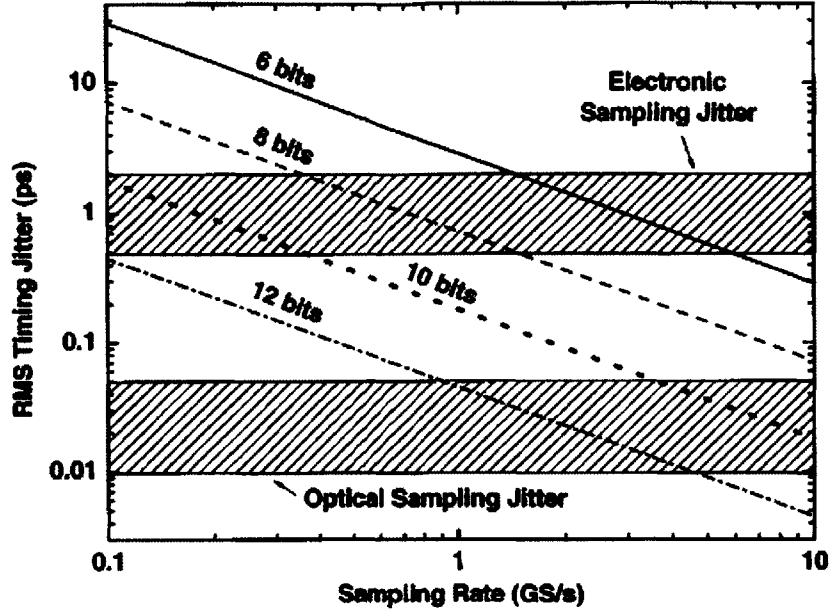


Figure 1-2: Timing-jitter requirement for ADCs as a function of sampling rate and number of effective bits [8].

EMI causes the crosstalk from neighboring traces which distorts electrical signals [5]. This can be understood that the electric field of an electron decays proportionally to $\frac{1}{r^2}$ at the distance r [6]. A study showed that about 30% of voltage variation is coupled from the neighbouring parallel wire for two parallel wires with a space of $0.4\mu m$, a width of $0.4\mu m$, a thickness of $0.6\mu m$ and a length of $2mm$ [7]. The crosstalk becomes severe when thousands of cores are integrated within a small area on the order of $\sim cm^2$. On the contrary, the electric field of an optical mode is proportional to $e^{-\alpha r}$ and decays very fast over short distance, where α is the electric field decay coefficient at the cladding. Waveguides can be spaced from each other by 2-4 micrometers to allow <30dB crosstalk between adjacent waveguides over a 4 cm chip propagation length.

Secondly, photonics offers low timing-jitter, making photonic circuits very attractive over the purely electronic circuits. Photonics can provide 1fs or even sub-femtosecond timing-jitter, about 2-3 orders higher than electronics, as shown in Fig.1-2. The silicon EPIC technology may provide over 8 bits resolution at a bandwidth of

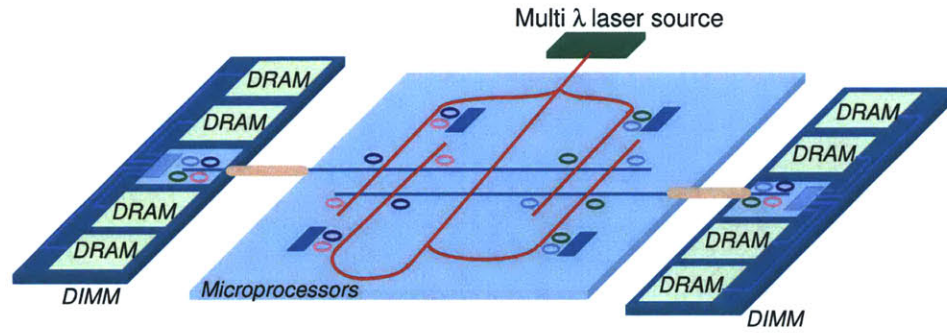


Figure 1-3: Optical intra-chip architecture for multi-core microprocessors [9].

40Gbps for the application of analog-to-digital convertors.

Thirdly, the silicon EPIC may provide very high bandwidth in signal processing and transmission by using dense wavelength division method (DWDM). DWDM is a well-known and well-established method in optical communications to increase total bandwidth. This concept is envisioned to be used in intra-chip, as shown in Fig.1-3, and in optical analog-to-digital converters (ADC), as shown in Fig.1-4.

In a intra-chip architecture for multi-core microprocessors, a photonic network connects all electronic microprocessors and memory modules. An off-chip or on-chip source provides multi-wavelength laser to the photonic network. Every electronic module, such as the microprocessor and memory, can receive and transmit optical signals. Microring filters can filter the desired wavelength and microring modulators can modulate imprint signals to narrow optical bandwidth light. If there is 100 wavelengths and every wavelength carries 10Gbps signal, the total bandwidth of the intra-chip can reach 1000Gbps.

As shown in Fig.1-4, the optical ADC channelizes the optical clock in time using precisely-tuned WDM filters to create time-interleaved optical sampling signals, each operating at the rate of the optical clock. The total sampling rate is then the optical clock rate times the number of WDM channels. A low timing-jitter mode-locked laser is designed to generate an optical pulse with a repetition rate of 1GHz. The pulse is chirped with a fiber spool or other dispersion devices. Then the chirped light is modulated by a high-speed silicon modulator and fed to a demux module based

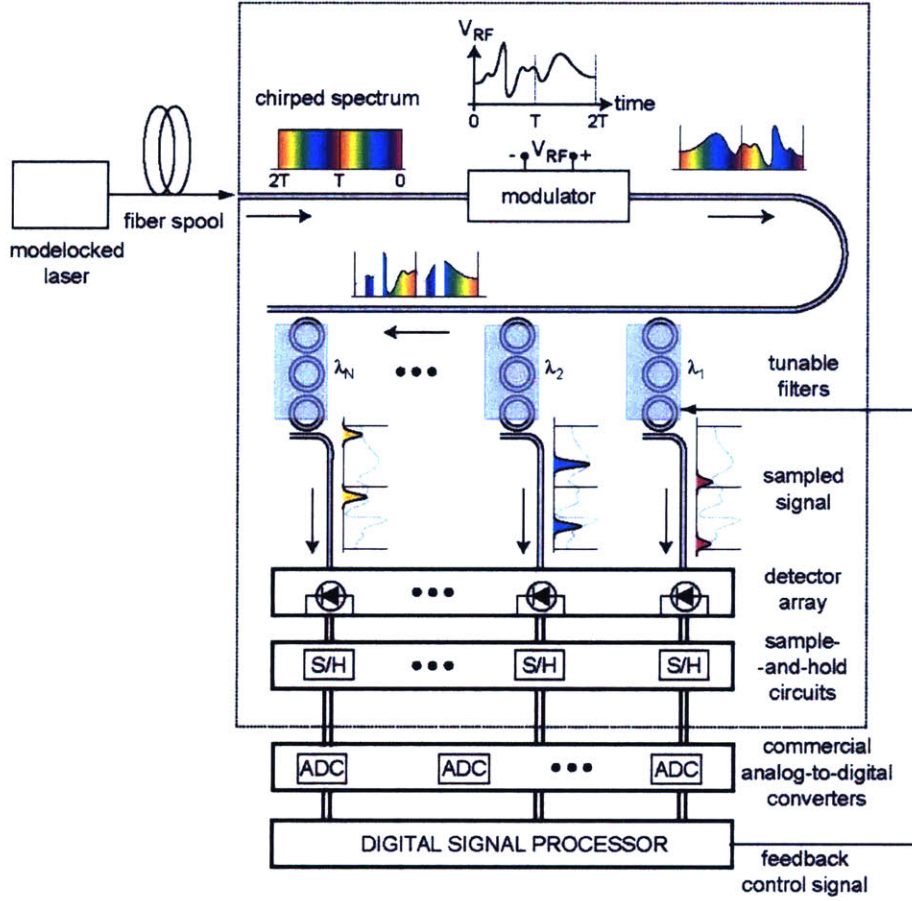


Figure 1-4: Optical analog-to-digital converter [10].

on ring resonators. Every channel only drops one wavelength which only carries a 1GHz signal. The photo-detectors and electronic ADC only need to process 1GHz signal and thus reduce the difficulties in high-speed circuits. This optical ADC can be designed to operate at 100Gbps and at least 7 effective number of bits (ENOB). This is a highly compact electronic photonic integrated circuit.

For the time being, a sub-micrometer technology node is available, such as 65nm node in industry. The e-beam technology pushes the smallest feature size less than 10nm [11]. The innovation in fabrication techniques open an opportunity to make ultra-compact photonics and make photonic circuits with large scale integration. With the progress in fabrication, a lot of high-performance photonic devices were

demonstrated in recent years. The first GHz silicon Mach-Zehnder optical modulator was reported by Intel [12]. 20GHz was also demonstrated recently [13][14]. Years' of research on Germanium photo-detectors has provided a deep understanding on the Ge-detector and a lot of high-performance devices have been fabricated [15]. Most importantly, they can be fabricated with CMOS-compatible techniques. There is also a lot of progress in laser sources. Intel and UC Santa Babara demonstrated a laser by bonding IIIV material to silicon waveguide at low temperature at which the performance of electronic circuits is not deteriorated [16]. Other devices, such as silicon ring filters, ring modulators and fiber-chip couplers, are also demonstrated with high performance and large scale integration. All these high-performance CMOS-compatible photonic devices have made it promising to integrate photonics and electronics into a single chip at low cost.

Silicon is the dominant material of the microelectronics industry. The availability of the well-established, cost-effective fabrication processes renders silicon an attractive choice for optoelectronics, for all applications where infrared light beyond the silicon absorption edge is to be guided, filtered, modulated and switched [17][18]. In addition, optical components that can be manufactured in a CMOS compatible fabrication process, promise full integrability with CMOS microelectronic circuits. Thus, silicon offers a potential platform to integrate microelectronic and optical circuits.

Rapidly increasing network traffic resulting from the worldwide increase in data communications demands faster, cheaper and more reliable switches and modulation devices. Optical switches and modulators are the fundamental components of optoelectronic integrated circuits. It translates the electronic signal into an optical signal. It's difficult to modulate light sources directly at high speed because the cavity in laser source, such as distributed feedback(DFB) structure, is usually sensitive to fabrication error and external environment. What's more, there is no on-chip silicon laser available for EPIC applications so far because silicon can not effectively generate light [16]. Off-chip laser and external modulation are necessary.

In order to achieve high-performance EPIC circuits, high quality electro-optic modulators (EOM) are needed. Firstly, the electro-optic modulators (EOM) should

be CMOS-compatible such that all processes can be done with standard cost-effective silicon processes. The silicon process has been studied for several decades and is very well-known. More importantly, it would be very advantageous to fabricate the electronics and photonics in the same well-established process at the same time. Secondly, a large bandwidth is desired for many applications. The speed of modulators is the bottleneck in EPIC circuits. The thermo-optic has been widely used because it can induce large refractive index change without optical loss, but the switching speed is at the order of μs . A switching speed of 970ns has been demonstrated based on thermo-optic effect [19]. The plasma dispersion effect induces the change of refractive index and optical absorption coefficient. Modulation speed based on the plasma effect is mainly determined by carrier lifetimes and carrier drift velocity. Other effects, such as the Stark effect, can be employed for fast modulation but are too weak to provide high modulation efficiency in silicon. Thirdly, a low power dissipation is required to reduce the total power consumption in an EPIC chip. Furthermore, low insertion loss, high modulation efficiency, high extinction ratio and very low nonlinearity are also needed.

1.2 Outline of Thesis

This thesis investigates electrically driven Mach-Zehnder modulators based on high-index contrast silicon waveguide technology and electronic carrier injection. It is organized as follows.

In Chapter 2, we review the theoretical work of optical modulation and the prior art of silicon-based optical modulators. In particular, various modulation effects are discussed and special attention is paid to four effects: the electro-optic effect, the Franz-Keldysh effect, the quantum confined Stark effect (QCSE) and the plasma dispersion effect. The properties of different materials for modulation are studied. The PN junction, PIN diode theories and Mach-Zehnder interferometer transmission characteristics are discussed, since PIN/PN diodes are widely used to inject carriers and the Mach-Zehnder interferometer is employed to achieve broad optical bandwidth.

In Chapter 3, we investigate high-speed electro-optic modulators based on high-index-contrast silicon buried waveguides. The modulators based on four different structures, the forward-biased PIN diode with lifetime reduction, the forward-biased diode without lifetime reduction, the reverse-biased PIN/PN diode and the metal-oxide-semiconductor (MOS) structure, are presented. The electrical properties of such modulators are simulated with the 2-dimensional semiconductor simulator MEDICI and the experimental results are analytically and numerically analyzed. The fabrication and measurement setup are illustrated and discussed.

An alternative waveguide structure, Si/SiO_2 split-ridge waveguides (SRW), is studied in Chapter 4. The SRW is explored and its properties of good heat sinking, large fabrication tolerance and broadband coupling are discussed. An electro-optic modulator is realized based on the SRW waveguide. Two architectures for all-optical carrier injection switches using two or three waveguides are proposed.

Carrier lifetimes in waveguides play an important role in the performance of electro-optic modulators. In Chapter 5, carrier lifetimes in bulk and SOI silicon wafers and high-index-contrast waveguides is measured. Carrier recombination mechanisms in semiconductors and high-index-contrast waveguides are discussed. Carrier lifetimes and surface recombination velocities of bulk and SOI silicon wafers with different surface passivation is measured with radio-frequency photo-conductance delay spectroscopy (RF-PCD) and dual-color pump-probe methods.

Finally, in Chapter 6, the conclusions are presented and future work is proposed.

Chapter 2

Overview and Theory of Optical Modulators

As discussed in chapter 1, optical modulators have been an active research topic for several decades because of their important applications. Researchers have been investigating different modulation effects, exploring possible modulation materials and various structures. Since this thesis investigates high-performance silicon electro-optic modulators, we review the theoretical work on optical modulation and the prior arts of silicon-based optical modulators before getting into the detailed discussed of the main work. It provides a guidance to the innovative designs of high-speed silicon-based optical modulators discussed later in this thesis.

This chapter is organized as follows. In Sec. 2.1, various modulation effects are discussed. Special attention is paid to four effects: the electro-optic effect, the Franz-Keldysh effect, the quantum confined Stark effect (QCSE) and the plasma dispersion effect. In Sec. 2.2, the properties of different materials for modulation are studied. In Sec. 2.3, the prior arts of silicon-based optical modulators are reviewed. In Sec. 2.4, the PN junction, PIN diode theories and Mach-Zehnder interferometer transmission characteristics are discussed, since PIN/PN diodes are later used to inject carriers and the Mach-Zehnder interferometer is employed to achieve broadband optical modulation.

2.1 Physical Effects for Optical Modulation

The modulation of optical signals, such as modulation of phase and amplitude, can be achieved through changes in refractive index and optical absorption of materials. Such changes can be induced by applying external electric fields, injecting free carriers, propagating acoustic waves or controlling the temperature in materials. Here, we focus our discussion on four different effects applicable for electro-optic modulators: the electrooptic effect, the Franz-Keldysh effect, the quantum confined Stark effect (QCSE) and the plasma dispersion effect. In particular, since silicon will be used to implement electro-optic modulators in this thesis, the properties of these four effects in silicon are discussed.

2.1.1 Electrooptic Effects (Pockels and Kerr effects)

The electrooptic effect describes the change in refractive index due of a material to applied electric fields [20][37][76]. The field-dependent change of refractive index is described by the sum of linear field-dependent terms and higher order terms,

$$\Delta\left(\frac{1}{n_r^2}\right) = \Sigma_k r_{ijk} E_k + \Sigma_{kl} s_{ijkl} E_k E_l + \dots \quad (2.1)$$

where n_r is the refractive index, $E_{k/l}$ is the electric field component in different polarization directions, such as x , y and z in Cartesian coordinate system, r_{ijk} are the linear electrooptic coefficients, i.e., the Pockels coefficients, and s_{ijkl} are the quadratic electrooptic coefficients, i.e., the Kerr coefficients. For example, the Pockels coefficient is $1.6 \times 10^{-10} \text{cm/V}$ for *GaAs* and $3.08 \times 10^{-9} \text{cm/V}$ for *LiNbO₃* [20]. The Pockels effect occurs only in crystals that lack inversion symmetry, such as lithium niobate and gallium arsenide. The Kerr effect is dominant in centrosymmetric materials. Because there is no carrier transport involved in the medium, modulators based on these two effects usually consume very little electric power and are more preferable in high speed modulation.

For crystalline silicon which is inversion symmetric, the linear electro-optic effect

(Pockels effect) vanishes and only the quadratic electro-optic effect (Kerr effect) is leftover. A theoretical prediction gives that the Kerr effect can cause refractive index changes up to $\Delta n = -10^{-6}$ and -10^{-4} in the electric field $E = 10^5 V/cm$ and $10^6 V/cm$, respectively, in silicon at wavelength $\lambda = 1.3 \mu m$ [35]. The later calculated refractive index change is not achievable because the required electric field exceeds the dielectric breakdown electric field of silicon $4 \times 10^{15} V/cm$ at free carrier concentration $N = 10^5/cm^3$ [77]. There is no linear electrooptic effect in bulk Si. Recent research has shown that a linear electrooptic effect can be generated by breaking the cubic symmetry of Si using strain [30].

2.1.2 Franz-Keldysh Effect

The Franz-Keldysh effect is caused by tilting of the energy bands of a semiconductor due to a uniform applied electric field. As shown in Fig.2-1(a), an electron can transit from the valence band to the conduction band by absorbing a photon with energy $h\nu = E_g$ in the absence of an external field. In an external field, the maximum of the valence band, the minimum of the conduction band and the wavefunction of holes and electrons may extend to the band gap between valence and conduction bands. An electronic transition with lower energy than the bandgap energy E_g can happen by photo-assisted tunnelling, shown in Fig.2-1(b). If the energy of the incident light is near the bandgap E_g , the absorption spectrum shifts to longer wavelength as a uniform electric field is applied and this effect can be used for electroabsorption modulators. If the incident light energy is smaller than the bandgap energy E_g , the refractive index change is more dominant than the absorption coefficient change and thus a phase modulator can be designed based on the Franz-Keldysh effect.

The Franz-Keldysh effect in silicon has been measured experimentally. The absorption spectra induced by electric-field effects in silicon is measured and then the refractive index change is calculated based on the measured absorption spectra using Kramers-Kronig relation. The refractive index change is computed to be $\Delta n = +1.3 \times 10^{-5}$ when the applied electric field is $E = 10^5 V/cm$ at wavelength $\lambda = 1.07 \mu m$ [35]. In this measurement, the refractive index change is mainly induced

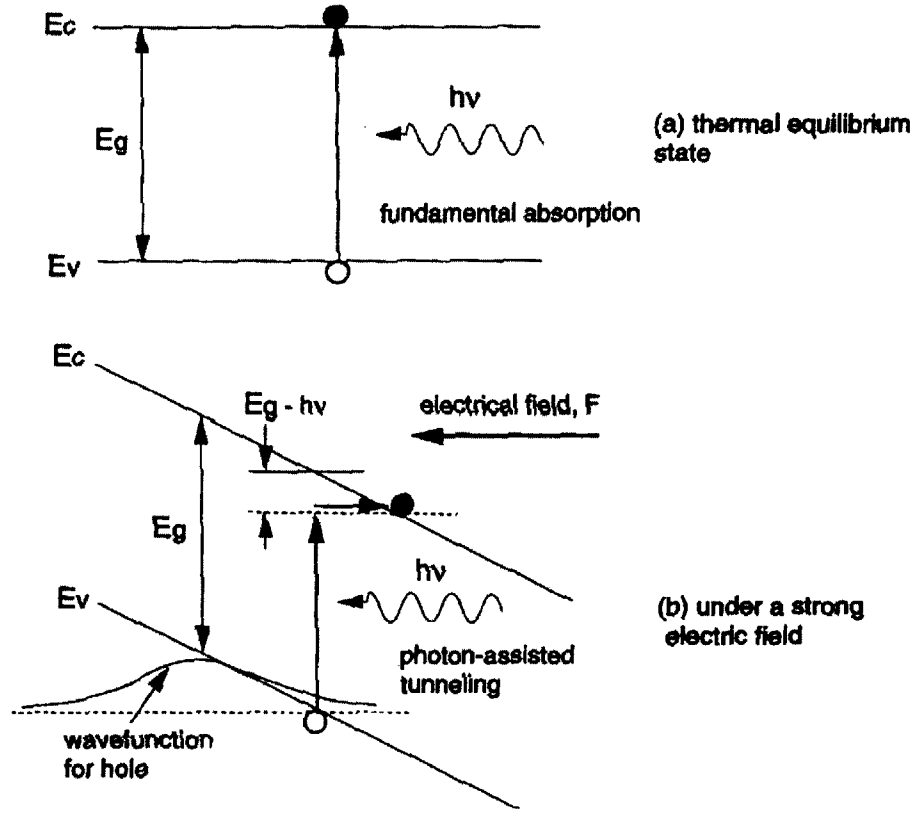


Figure 2-1: (a) Equilibrium state, (b) Franz-Keldysh effect and the tilting of energy bands in a uniform electric field [20].

by the Franz-Keldysh and electro-optic effects because the refractive index change due to the electro-optic effects is $\Delta n = -10^{-6}$ as discussed in previous section which is negligible. It is worth noting that the Franz-Keldysh effect is relatively weak compared with the plasma dispersion effect in silicon which is discussed in Sec.2.1.4.

2.1.3 Quantum Confined Stark Effect (QCSE)

The quantum confined Stark effect (QCSE) happens when an electric field is applied perpendicular to the quantum well, as shown in Fig. 2-2. The transition of electrons in quantum well structures is governed by the equation[20]

$$\hbar\omega = E_g + E_{el} - E_{hl} - E_{ex} \quad (2.2)$$

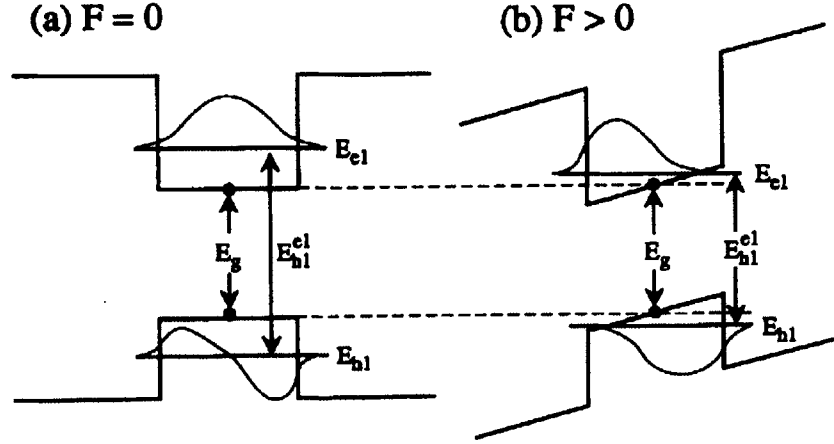


Figure 2-2: Quantum-well energy subbands and wave functions (a) in the absence of an applied electric field, and (b) in the presence of an applied electric field [37].

where $\hbar\omega$, E_g , E_{el} , E_{hl} and E_{ex} are the photon energy, the band-gap energy of the well material, the electron subband energy, the hole subband energy, and the binding energy of the exciton, respectively. The wavefunctions of an electron and a hole in the quantum-well are pushed to the well wall in opposite direction by the applied electric field F . The electron subband energy E_{el} decreases while the hole subband energy E_{hl} increases, thus the difference between these two energies $E_{hl}^{el} = E_{el} - E_{hl}$ reduces. The exciton binding energy E_{ex} also decreases. Therefore, the effective bandgap reduces. The electric field shifts the absorption spectra to longer wavelength.

The quantum-confined Stark effect is widely used in InGaAs/InAlAs quantum well structures, because it usually induces larger refractive index change in InGaAs/InAlAs quantum well structures than in the silicon-based materials [20]. A strong Quantum-confined Stark Effect has been recently found in germanium and silicon germanium quantum-wells [78]. The subband energy structure is shown in Fig.2-3(a) and the measured effective absorption coefficient spectra is plotted in Fig. 2-3(b). The maximum effective absorption coefficient change is 2,800 /cm at 1,438nm under 3 V bias. This achievement may lead to a short length and very high-speed optical modulator. However, it's challenging in fabricating multi-stack silicon germanium structures. Furthermore, it's difficult to couple the optical signal from the thin silicon waveguide

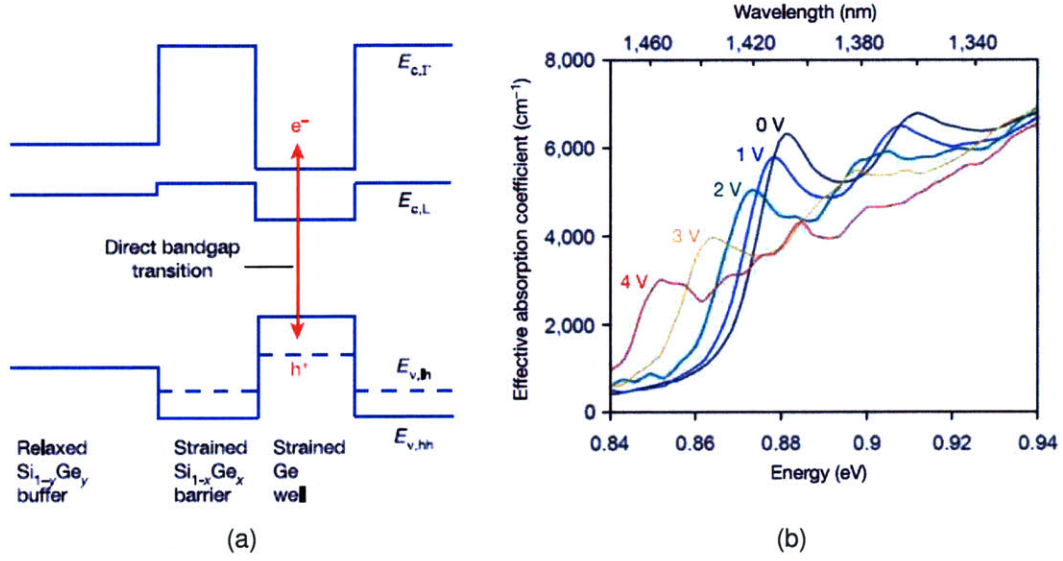


Figure 2-3: (a) The subband energy of Ge/SiGe quantum wells, (b) effective absorption coefficient spectra [78].

(250 nm) to the thick silicon germanium waveguide ($> 1\mu m$) which is located around $1\mu m$ above the silicon waveguide in the multi-stack structure.

2.1.4 Plasma Dispersion Effect

The plasma effect describes the change in refractive index and optical absorption coefficient induced by free carriers in a semiconductor. The motion of an electron in silicon structures is governed by Newton's Law,

$$\mathbf{F}_{tot} = m \frac{d\mathbf{v}}{dt} = \mathbf{F}_{ext} - \mathbf{F}_{damp} \quad (2.3)$$

where $\mathbf{r}$ is the position of the electron. $\mathbf{F}_{tot}$ is the total force applied on the electron. $\mathbf{F}_{ext} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \simeq q\mathbf{E}$ is the external force on the electron. The second term is negligible for $v \ll c_l$, as $\mathbf{E} = \mathbf{E}/c_l$ for a plane wave travelling at a speed of c_l , and $\mathbf{v} \times \mathbf{B} \simeq (v/c_l)\mathbf{E} \ll \mathbf{E}$ [6]. The damping force $\mathbf{F}_{damping}$ is induced from the momentum change in a band [79],

$$\mathbf{F}_{damping} = \frac{m\mathbf{v}}{\tau} \quad (2.4)$$

where τ is the relaxation time constant. Then the equation of motion of an electron can be rewritten as[80],

$$q\mathbf{E} = m\frac{d\mathbf{v}}{dt} + \frac{m\mathbf{v}}{\tau} \quad (2.5)$$

We assume that the electric field is $\mathbf{E} = \mathbf{E}_0 e^{-\omega t}$, then the electron velocity can be explicitly expressed as

$$\mathbf{v} = \frac{q\mathbf{E}}{-i\omega m + m/\tau} \quad (2.6)$$

Putting this electron velocity into Maxwell's equation, we obtain

$$\begin{aligned} \nabla \times \mathbf{H} &= \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J}_p \\ &= -i\omega\epsilon\mathbf{E} + Nq\mathbf{v} \\ &= -i\omega\epsilon_0[\epsilon_r + i\epsilon_i - \frac{(\omega_p\tau)^2}{(\omega\tau)^2} \frac{(\omega\tau)^2 - i\omega\tau}{(\omega\tau)^2 + 1}]\mathbf{E} \\ &= -i\omega\epsilon_0\epsilon\mathbf{E} \end{aligned} \quad (2.7)$$

and

$$\epsilon \equiv (n + ik)^2 = \epsilon_r + i\epsilon_i - \frac{(\omega_p\tau)^2}{(\omega\tau)^2} \frac{(\omega\tau)^2 - i\omega\tau}{(\omega\tau)^2 + 1} \quad (2.8)$$

where $\omega_p = \sqrt{\frac{q^2 N}{m\epsilon_0}}$ is the electron plasma frequency, $\mathbf{J}_p$ is the polarization current density, N is the free carrier density, m is the carrier effective mass, ϵ_r and ϵ_i are the real and imaginary parts of relative permittivity without polarization current, ϵ_0 and $\epsilon = (n + ik)^2$ are the permittivity of free space and the effective permittivity in the medium, respectively. Separating the real part and the imaginary part of ϵ , we obtain the relation between the real part n and imaginary part k of the refractive index

$$n^2 - k^2 = \epsilon_r - \frac{(\omega_p\tau)^2}{(\omega\tau)^2 + 1} \quad (2.9)$$

$$2nk = \epsilon_i + \frac{(\omega_p \tau)^2}{\omega \tau} \frac{1}{(\omega \tau)^2 + 1} \quad (2.10)$$

Setting the total force to zero, the electron mobility $\mu = q\tau/m$ can be obtained from equations (2.3) and (2.5). Substituting $\tau = m\mu/q$, $n = n_0 + \Delta n$ and $k = k_0 + \Delta k$ into equations (2.9) and (2.10), the real part change of the refractive index and the change of the electric power absorption coefficient are derived,

$$\Delta n = -\frac{e^2 \lambda^2}{8\pi^2 c^2 \epsilon_0 n_0} \frac{N}{m} \quad (2.11)$$

$$\Delta \alpha = \frac{4\pi \Delta k}{\lambda} = \frac{e^3 \lambda^2}{4\pi^2 c^3 \epsilon_0 n_0} \frac{N}{m^2 \mu} \quad (2.12)$$

where $\omega \tau \gg 1$, $n \gg k$ and $n_0 \gg \Delta n$ are used for infrared frequencies [81]. n_0 and k_0 are the real and imaginary parts of the refractive index without free carriers. Taking into account the contribution from electrons and holes, the total refractive index change is

$$\Delta n = -\frac{e^2 \lambda^2}{8\pi^2 c^2 \epsilon_0 n} \left(\frac{\Delta n_e}{m_e} + \frac{\Delta n_h}{m_h} \right) \quad (2.13)$$

and the absorption coefficient change is

$$\Delta \alpha = \frac{4\pi \Delta k}{\lambda} = \frac{e^3 \lambda^2}{4\pi^2 c^3 \epsilon_0 n} \left(\frac{\Delta n_e}{m_e^2 \mu_e} + \frac{\Delta n_h}{m_h^2 \mu_h} \right) \quad (2.14)$$

where Δn_e and Δn_h are the free electron and hole densities, and the subscripts "e" and "h" refer to electrons and holes, respectively.

An alternative way to predict the refractive index change is to convert from experimental electroabsorption data using the Kramers-Kronig relation [35],

$$\Delta n(\omega) = (c/\pi)P \int_0^\infty \frac{\Delta \alpha(\omega') d\omega'}{\omega'^2 - \omega^2} = 6.3 \times 10^{-6} P \int_0^\infty \frac{\Delta \alpha(V') dV'}{V'^2 - V^2} \quad (2.15)$$

where ω , $V = \hbar\omega/q$ and c are the angular frequency, the normalized photon en-

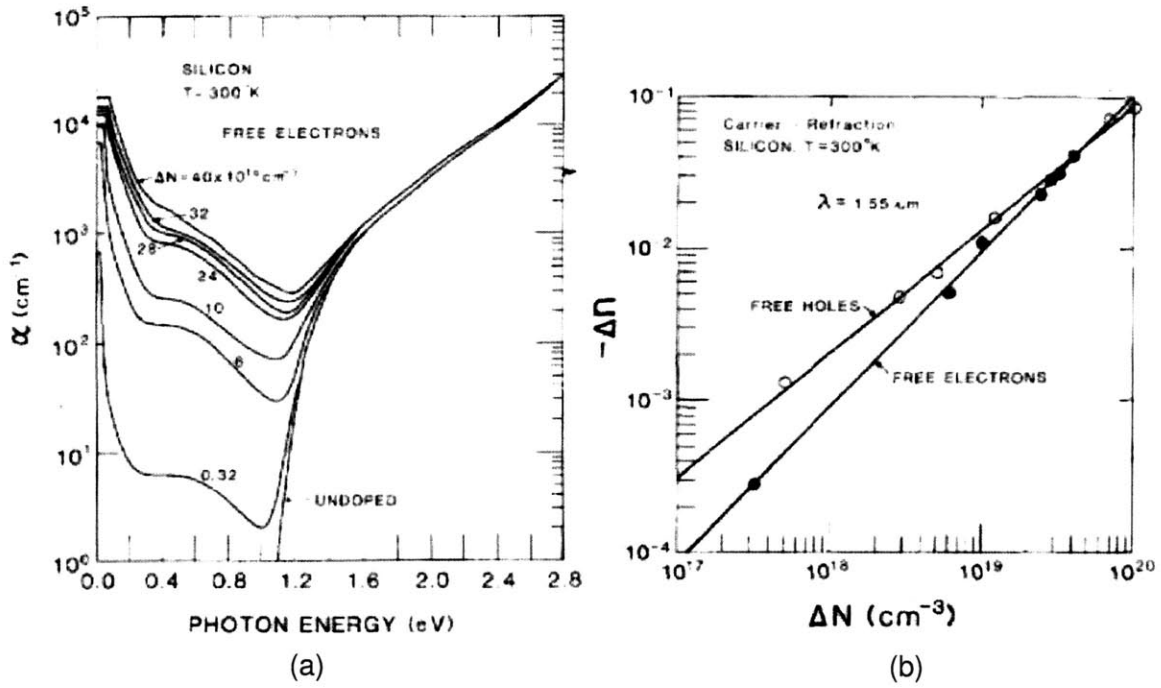


Figure 2-4: (a) Experimental optical absorption spectra of c-silicon as a function of electron concentration, (b) calculated refractive index change as a function of electron concentration based on Kramers-Kronig theorem at wavelength $\lambda = 1.55\mu m$. The circle lines are for holes and the solid lines are for electrons [35].

ergy and the speed of light, respectively. Fig.2-4(a) gives the experimental optical absorption spectra for different n-type doping. Because the absorption coefficient approaches that of intrinsic crystal silicon at higher energy, it is adequate to integrate $\Delta\alpha$ from energy 0 to 2.8eV in equation (2.15). Fig.2-4(b) shows the calculated refractive index change as a function of electron and hole concentration at wavelength $\lambda = 1.55\mu m$. A fitting of the refractive index change Δn in Fig.2-4(b) indicates that Δn depends linearly on the electron density and depends on the hole density proportional to $\Delta N_h^{0.8}$,

$$\Delta n = -8.8 \times 10^{-22} cm^3 \Delta n_e - 8.5 \times 10^{-18} cm^{2.4} \Delta n_h^{0.8} \quad (2.16)$$

$$\Delta\alpha = 8.5 \times 10^{-18} cm^2 \Delta n_e + 6.0 \times 10^{-18} cm^2 \Delta n_h \quad (2.17)$$

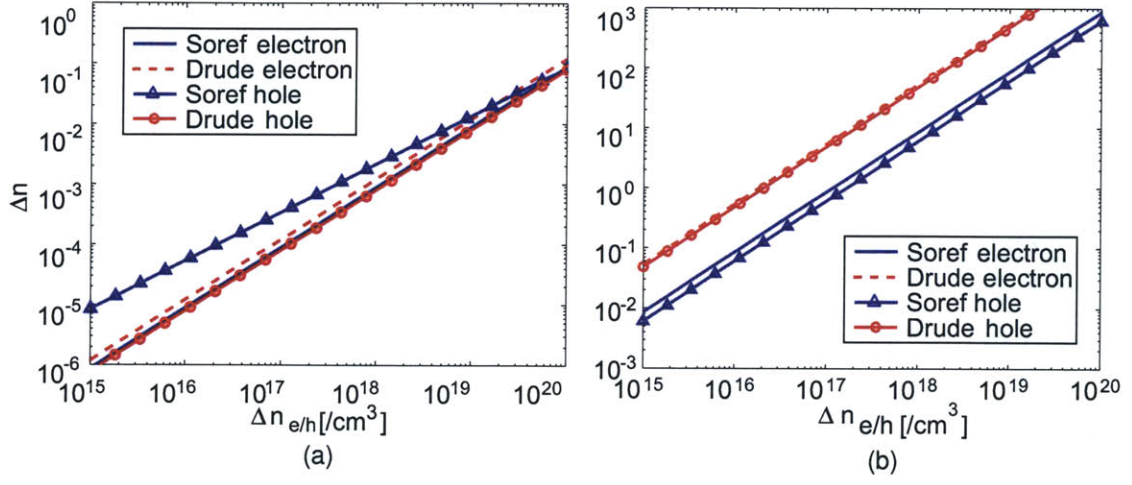


Figure 2-5: (a) Refractive index change based on Soref model (solid, triangle) and Drude model(dashed, circle), (b) absorption coefficient change based on Soref model (solid, triangle) and Drude model(dashed, circle) at wavelength $\lambda = 1.55\mu m$.

Compared with the Soref model in equations (2.16) and (2.17), the classical Drude model in equations (2.13) and (2.14) predicts different scaling of the changes of refractive index and optical absorption coefficients. In Fig.2-5, the refractive index and absorption changes from electrons and holes are plotted for the two models. We choose the parameters for the Drude model: $m_e = 0.26m_0$, $m_h = 0.39m_0$, $\mu_n = 1000cm^2 \cdot V/s$ and $\mu_h = 500cm^2 \cdot V/s$. It can be seen from Fig.2-5(a) that the Soref and Drude models match very well in the refractive index change induced by the electron density but differ in that induced by the hole density. There is almost an order difference at low free hole concentration around $\Delta_h = 10^{15} \sim 10^{17}/cm^3$. In addition, the Soref model predicts more refractive index change due to free holes than that due to free electrons while the Drude model predicts the same level changes due to electrons and holes. Moreover, the optical absorption coefficient change is an order of magnitude higher in the Drude model than in the Soref model at free carrier concentration around $10^{15} - 10^{18}/cm^3$ in Fig.2-5(b).

2.1.5 Other Effects

Besides the four modulation effects discussed in the previous subsections, there are other modulation effects widely used in optical modulation. The acoustooptic effect induces refractive index changes by an acoustic wave. The acoustic wave can propagate in a medium, such as a crystal, and induce periodic refractive index changes in the medium such that the incident light can scatter off from the index grating [37]. Secondly, the thermooptic effect is a large effect in silicon. The thermooptic coefficient is $2 \times 10^{-4}/K$ such that a 10 degree temperature change can induce 0.01 refractive index change which is at least 2-3 orders larger than the other effects, such as the electrooptic effects and the Franz-Keldysh effect in an external electric field of $\sim 10^5 V/cm$ and the plasma dispersion effect with a carrier injection of $\sim 10^{17}/cm^3$. However, the thermooptic modulation is a slow mechanism on the order of μs to ms [27]. Thirdly, Micro-Electro-Mechanical Systems (MEMS) is another slow mechanism inducing large refractive index change [82]. A moving component of the MEMS system can change the mode profile or introduce optical loss for light modulation. Finally, other effects, such as the field effect in liquid crystals [83], have been investigated to achieve high-performance modulation.

2.2 Materials for Optical Modulators

The performance of optical modulation not only depends on the modulation effects discussed in the previous section, but also depends on the material properties. There are many materials which can be used to modulate light. We mention a few, as shown in Table 2.1, for the purpose of the present thesis. In particular, we are interested in three kinds of materials, dielectric crystals, IIIV materials and silicon-based materials, which can be employed in electro-optic modulators.

Table 2.1: Materials for modulation

Materials	Dominant effects	Band energy	Optical applications	Advantages	Disadvantages
Crystal (like $LiNbO_3$)	electrooptic[84]		modulator	high modulation quality	expensive, large size, no electronics
GaAs	electrooptic[85] Franz-Keldysh[86] QCSE*[87]	direct bangap $\lambda_g = 872nm$	modulator, detector and laser	integration	expensive
InP	electrooptic[37] Franz-Keldysh[86] QCSE[87]	direct bangap $\lambda_g = 923nm$	modulator, detector and laser	integration	expensive
Ge/ $Si_{1-x}Ge_x$	QCSE[78] Franz-kelsysh[15]	direct bandgap $\lambda_g = 1.87\mu m$	broadband detector and modulator	CMOS-integration	
Silicon	thermooptic[27] plasma[21]	indirect bandgap $\lambda_g = 1.1\mu m$	modulation and passive devices like AWG**	mature electronics, CMOS-compatible photonics	limited light emmision

* QCSE: quantum confined Stark effect ** AWG: arrayed-waveguide grating

2.2.1 Dielectric Crystal and IIIV-Semiconductor Materials

Crystals, such as $LiNbO_3$ and KH_2PO_4 , have strong electrooptic effects and can modulate light at high speed (over 40Gbps) and with low signal chirping and low RF power consumption [37][84]. Particularly, $LiNbO_3$ has unique piezo-electric properties for acoustic modulation [88]. However, $LiNbO_3$ can not be integrated with CMOS electronics directly and its large size on the order of centimeters prohibits the electronic and photonic integration. The high cost of $LiNbO_3$ makes it less attractive than semiconductor materials.

InP and GaAs are the popular IIIV semiconductor platforms for photonic device fabrication [86]. The electrooptic effects, the Franz-Keldysh effect and the quantum confined Stark effect (QCSE) are the dominant effects in GaAs and InP based modulators. They have a direct bandgap and can not only modulate but also emit and detect light. InP-based Distributed Feedback Lasers (DFB) operating at 1310/1550 nm are already widely used in industry. GaAs/AlGaAs Vertical Cavity Surface Emitting Laser (VCSEL) transmitters and GaAs pin photodetector-based receivers are widely applied for 850 nm wavelength products [89]. Furthermore, the integration of electronics and photonics based on GaAs and InP can provide high-speed high-quality electronic photonic integrated circuits. In 2006, Infinera had developed 100Gbps

transceivers based on InP large-scale photonic circuits [90].

2.2.2 Group-IV Materials

Silicon-based materials are CMOS-compatible materials, including silicon and germanium. Silicon oxide and silicon nitride usually act as cladding to confine light. Germanium has an enhanced Franz-Keldysh effect present in epitaxial germanium films on silicon and tensile strained germanium [15]. The quantum confined Stark effect (QCSE) is also found in $\text{Ge}/\text{Si}_{1-x}\text{Ge}_x$ quantum wells [87]. These two effects make it possible to fabricate a high-speed modulators based on germanium.

Silicon is the dominant material of the microelectronics industry. The availability of the well-established and cost-effective fabrication processes renders silicon an attractive choice for optoelectronics, for all applications where infrared light beyond the silicon absorption edge is to be guided, filtered, modulated and switched [17][18]. In addition, optical components that can be manufactured in a CMOS compatible fabrication process, promise full integrability with CMOS microelectronic circuits. Thus, silicon offers a potential platform to integrate microelectronic and optical circuits. The next section will summarize the prior art on silicon-based modulators, most of which are based on silicon using plasma dispersion effect.

2.3 Prior Art of Silicon-Based Optical Modulators

Silicon-based optical modulators attract a lot of attention because of their CMOS-compatible fabrication and high-speed operation. This section reviews the prior art of silicon-based optical modulators in order of time, summarizes and compares their performance, advantages and disadvantages using different electrical and optical structures, as shown in Table 2.2, 2.3 and 2.4.

In the years before 1990, a lot of effort had been put on silicon-based electro-optic modulators, such as the work done by Soref and colleges [35][21][22][23], as shown in Table 2.2. A forward-biased PIN structure was widely used in these publications. However, with the available fabrication technology at that time, the wave-

Table 2.2: Electro-optic modulators (1980–1989)

Year	1986[21] ¹	1987[22] ¹	1988[23] ²	1989[24] ¹
Author/Institution	Soref/AFB ³	Lorenzo/AFB	Friedman/AFB	Hemenway/Standford
Materials	Silicon	Silicon	Silicon	Silicon
Effects	Plasma	Plasma	Plasma	Plasma
Wavelength [μm]	1.3	1.3	1.3	1.3
Electrical Structure	PN-Forward	PIN-Forward	DIFET with Depletion ⁴	PIN-forward
Optical Structure	2×2 X-structure	2×2 X-structure	2×2 X-structure	Reflector
Phase-shift efficiency	N.A.	N.A.	$78V \cdot cm$	N.A.
Length	3mm	400-2000 μm	650 μm	N.A.
Waveguide cross section [μm^2]	10×6 (multi-mode)	15×6 (multi-mode)	3.8×2	3×6
Capacitance	N.A.	N.A.	1.5fF/ μm	N.A.
Effective lifetime [ns]	1 μs	N.A.	10 μs	N.A.
Bandwidth f_{3dB}	N.A.	N.A.	~ 3.2 GHz (RC delay limit)	200MHz
Index Change ΔN	0.9×10^{-3}	$\sim 1.5 \times 10^{-3}$	10^{-3}	N.A.
Modulation depth (Extinction ratio)	N.A.	50%	10%@ $V_g = 1.5$ 90%@ $V_g = 13.5$	10%
Insertion loss [dB]	5-13dB/cm	N.A.	N.A.	N.A.
Total loss [dB]	N.A.	N.A.	N.A.	N.A.
Power [mW]	N.A.	N.A.	N.A.	N.A.
Voltage [V]	N.A.	N.A.	$V_{gate} = 1.5 - 13.5$	N.A.
J [A/cm^2]	N.A.	1-4k	<1k (no heating)	$\sim 5k$

¹ Demonstrated² Design³ AFB:Air Force Base⁴ DIFET: Double-injection Field Effect Transistor

Table 2.3: Electro-optic modulators (1990-2004)

Year	1991[26] ¹	1994[91] ¹	1995[92] ¹	1997[93] ²	1999[94] ²	2000[25] ¹	2004[27] ¹	2002[95]([96]) ¹	2002[84] ¹
Author/ Institution	Treyz/ IBM	Liu/ Xi'an,China	Tang/ Surrey,UK	Cutolo/ Napoli,Italy	Irace/ Napoli,Italy	Dainesi/ Swiss Federal	Barrios/ Cornell	Lee/ Bell Lab	Kondo/ NGK,Japan
Materials	Silicon	Silicon	silicon	Silicon	Silicon	Silicon	Silicon	Polymer	$LiNbO_3$
Effects	Plasma	Plasma	Plasma	Plasma	Plasma	Plasms	Thermooptic	Electrooptic	Electrooptic
Wavelength [μm]	1.3-1.55	1.3	1.5	1.55	1.55	1.3	1.553 μm FWHM=1.54nm	1310 nm	1.55
Electrical Structure	PIN-forward	PIN-forward	PIN-forward	BMFET ³	PIN-forward	PIN-forward	PIN-forward	N.A.	Voltage
Optical Structure	MZ ⁴	2 $\times$ 2 X-structure	Free space, MZ	Waveguide	DBR ⁵	MZ	Bragg reflectors	MZ	MZ
Phase-shift efficiency ($V_\pi \cdot cm$)	N.A.	N.A.	9 V $\cdot cm$	8.3 V $\cdot cm$	N.A.	22.8V $\cdot cm$	N.A.	$V_\pi = 11.3V$	12V $\cdot cm$
Length [μm]	500	200	500	1000	3200	1mm	20 μm	2cm	4cm
Waveguide cross section [μm^2]	10 $\times$ 4.6	6 $\times$ 1.8	8 $\times$ 6.5	6 $\times$ 6	3 $\times$ 36	10 $\times$ 10	1.5 $\times$ 1.5	(15 $\times$ 10)	Mode size $R \simeq 6$
Effective lifetime [ns]	<50	100	N.A.	5.6	N.A.	50ns	N.A.	N.A.	N.A.
Bandwidth f_{3dB}	3MHz	1.5MHz	N.A.	~20MHz	~20MHz	10MHz	N.A.	250GHz	30GHz
Injected Carrier $\Delta N_{e/h}/[cm^3]$	6.5×10^{17}	N.A.	N.A.	N.A.	1.6×10^{17}	2.5×10^{17}	N.A.	N.A.	N.A.
Index Change ΔN	N.A.	N.A.	N.A.	N.A.	N.A.	1.8×10^{-4}	N.A.	N.A.	N.A.
Modulation depth (Extinction rate)	65%	>90%	N.A.	20%	94%	N.A.	3.2dB(53%)	N.A. 20.7dB	N.A. 25dB
Insertion loss [dB]	N.A.	240dB/cm	N.A.	15dB/cm	3.1dB/cm	8.5dB	7dB@DBR	(1.2dB/cm)	10dB
Total loss [dB]	15-20dB/cm	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
power [mW]	N.A.	N.A.	11.9	43, 126	0.3,9.3	210mW	20mW	N.A.	N.A.
Voltage [V]	N.A.	N.A.	$V_\pi = 1.7V$	N.A.	0.64	2.06V	N.A.	6V	N.A.
J [A/cm^2]	1.6	12.5	0.175	N.A.	N.A.	19K	18k	N.A.	N.A.

¹ Demonstrated² Design³ BMFET: Bipolar mode field effect⁴ MZ: Mach-Zehnder⁵ DBR: Distributed Bragg Reflector transistor

Table 2.4: Electro-optic modulators (2004-2007)

Year	2004[12]([97]) ¹	2005[47] ¹	2005[73](2007[98]) ¹	2005[78] ¹	2006[30] ¹	2006[31] ¹	2005[32] ²	2006[99] ¹	2007[14] ¹
Author/ Institution	Liu(Samara-Rubio)/ Intel	Liao/ Intel	Xu/ Cornell	Kuo/ Stanford	Jacobsen/ Denmark	Gu/ Texas	Gardes/ Surrey,UK	Guckenberger/ Luxtera	Liu/ Intel
Materials	Silicon	Silicon	Silicon	Ge/SiGe	Strained-Si	Silicon	Silicon	Silicon	Silicon
Effects	Plasma	Plasma	Plasma	QCSE ³	Pockels	Plasma	Plasma	Plasma	Plasma
Wavelength	1.54 μm	1.55 μm	1.574 μm FWHM=0.04nm	1.5 μm	1.561	1.56	1.55	1.55	1.55
Electrical Structure	MOS	MOS	PIN	PIN-MQW N.A.	Voltage	PIN forward	P^+pnN^+ reverse	PN reverse	PN reverse
Optical Structure	MZ ⁴	MZ	Ring	N.A.	MZ, PCW ⁵	MZ, PCW	MZ	MZ	
$V_\pi \cdot L$	$\sim 8V \cdot \text{cm}$	$3.3V \cdot \text{cm}$		N.A.			$2.5V \cdot \text{cm}$		$4V \cdot \text{cm}$
Length	1-8mm,2.5cm	3.45mm	$R=6\mu\text{m}$	N.A.	20 μm	80 μm	2.5mm	4mm	1-5mm
Waveguide cross section [μm^2]	2.5×2.3	1.6×1.6	0.45×0.25	N.A.	1×0.22	2×0.22	0.415×0.45	0.5×0.25	0.6×0.5
Capacitance	(70pF/cm)	26.4pF/ 3.45mm	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
Effective lifetime	N.A.	N.A.	10	N.A.	N.A.	0.63	3-7ps	N.A.	N.A.
Bandwidth f_{3dB}	1(2.5)GHz	10Gbps	1.5Gbps (12.5Gbps)	50GHz	N.A.	$\sim 250\text{MHz}$	$\sim 20\text{-}40\text{GHz}$	20Gbps	20GHz
Injected Carrier $\Delta N_{e/h}/\text{cm}^3$	N.A.	N.A.	N.A.	N.A.	N.A.	3×10^{17}	N.A.	N.A.	N.A.
Index Change ΔN	N.A.	N.A.	N.A.	N.A.	N.A.	0.001	N.A.	N.A.	N.A.
Modulation depth (Extinction rate)	16dB@DC,1cm (5dB@1GHz,1cm)	4.2dB@0.15 π	97% (9dB)	$\Delta\alpha =$ 2800/cm@3V	N.A.	90%	N.A.	20-80%	N.A.
Insertion loss	6.7dB/cm (5.1dB/cm)	10dB(on-chip) 9dB(coupling)	N.A.	N.A.	N.A.	N.A.	2dB	N.A.	7dB(on-chip)
Total loss	N.A.	N.A.	9dB, including coupling loss	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
power	N.A.	N.A.	$\sim 10 - 100\mu\text{W}$ (4mW)	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
Voltage	N.A.	3.3V	4V	4V	110V	2V	5V	5V	6V
J [A/cm^2]	N.A.	N.A.	0.15	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.

¹ Demonstrated² Design³ QCSE: quantum-confined Stark effect⁴ MZ: Mach-zehnder⁵ PCW: photonic crystal waveguide

uodes usually had an active region longer than 4mm [21] and a cross-section larger than $15\mu m \times 6\mu m$ [22], which limits the maximum speed achievable in modulation. The fastest speed found in the literature was predicted theoretically to be about 3.2GHz [23] in a double-injection field effect transistor (DIFET) and demonstrated to be around 200KHz [24] based on forward-biased PIN diodes.

Selected publications between 1990-2004 on electro-optic modulators are summarized in Table 2.3. Most modulators were designed based on forward-biased PIN structures. The advantage of this kind of electrical structure is that it can provide more carrier injection to enlarge the change of refractive index than other structures, for example, the injected carrier density can reach $\Delta n_{e/h} = 2.5 \times 10^{17}/cm^3$ [25]. However, there are two main disadvantages associated with this kind of structure. Firstly, the modulation speed is usually determined by the lifetime which is on the order of several nano-seconds to several micro-seconds [26]. It can usually provide less than 1GHz modulation speed. Secondly, the refractive index change due to the plasma effect might be cancelled due to the thermo-optic effect. The forward-biased current leads to the heating, the refractive index change induced by the thermo-optic effect is opposite to the plasma effect. In the literature, the highest current density achieved to induce a plasma dispersion effect was found to be $19kA/cm^2$ [25]. Such high current densities are usually undesired because at a current density greater than $18kA/cm^2$, a thermal effect larger than the plasma effect is undesired [27].

Since 2004, more electrical structures have been proposed for silicon electro-optic modulators, such as metal-oxide-semiconductor (MOS) and reverse-biased PIN/PN diodes. A modulator based on a MOS-structure was first proposed and demonstrated in February 2004 by Intel [12]. The advantage of a MOS structure is that it has a very similar process as a transistor because both such modulator and transistor have a MOS-structure. This property may be desirable in future photonic and electronic integration. The disadvantage is that the modulation based on MOS structure is limited by RC delay. Also a modulator with smaller capacitance has a larger bandwidth but it needs a higher RF voltage to achieve the same amount of injected charge $Q = CV$. There is a tradeoff between speed and voltage [97]. So far the demonstrated

widest bandwidth is 10Gbps while the necessary voltage is several volts which can't be provided by standard CMOS circuits [47].

Reverse-biased modulators have attracted a lot of attention since 2004 [32][99][14]. Fast modulation speed can be achieved for a reverse-biased modulator because a high electric field, such as 10^5V/cm , can be applied on reverse-biased PIN structures, which drives carriers to travel close at saturation velocity $V_s = 10^7\text{cm/s}$. The modulation speed is inverse to the transit time $\tau_t = W/v$. Here, W and v are the travelling distance and the carrier drift velocity, respectively. It was predicted that the reverse-biased modulation can achieve a high speed of $20\text{-}40\text{GHz}$ [32]. So far, the fastest modulation speed demonstrated is 20GHz [14]. However, the reverse-biased modulator has its intrinsic disadvantages. The movable carrier density is usually low since it is mainly contributed by donors or acceptors, on the order of $10^{17}/\text{cm}^3$, and a larger doping may introduce more optical loss. This carrier change can only give a refractive index change about $\Delta \simeq 10^{-4}$ resulting in a large figure of merit (FOM) on the order of $V_\pi \simeq 5\text{V} \cdot \text{cm}$ [32]. This large FOM leads to a necessary length of several millimeter of the phase shifter such that a travelling-wave electrode design is necessary in order to achieve high-speed modulation [14].

Most previous work has been using the plasma dispersion effect. Recently, the field effects in silicon-based materials are studied. The quantum confined Stark effect (QCSE) was discovered in Ge/SiGe quantum wells [78]. Theoretically a modulator based on QCSE can operate over 50GHz [29], which is higher than the modulation speed from the plasma dispersion effect. The linear electrooptic effect (Pockels effect) was demonstrated in strained silicon [30]. The Pockels effect in silicon was generated by straining the silicon to break the cubic symmetry. The Pockels effect potentially can be employed for very high speed modulation.

Several ways to enhance the modulation efficiency and to reduce the power dissipation have been investigated, such as using resonators and slow wave structures. A resonator modulator is preferred to increase modulation efficiency in low power dissipation and narrow bandwidth applications. The ring resonator modulators can reduce the power dissipation by several orders of magnitude. A ring modulator has

been demonstrated with 10-100 μW power consumption [73]. A Fabry-perot resonator is another method to deploy resonance and enhance the modulation efficiency [27]. Moreover, slow wave structures made of resonators, photonic crystals or other structures can be used to enhance modulation efficiency [28]. The photonic crystal slow wave structure has been demonstrated in Mach-Zehnder interferometers [30][31].

In summary, the previous electro-optic modulators tended to use the forward-biased PIN structure to achieve a large modulation depth and apply the reverse-biased PIN/PN structure to obtain high modulation speed. The electric field effects in silicon-based materials, such as the quantum-well Stark effect, are yet well-investigated to be used in modulating light. The insertion loss was usually large around 5-20dB/cm [21] in the modulators based on ridge waveguides from 1980' to 1990' and was reduced to $< 1dB/cm$ [32] for modulators based on buried waveguides after 2000. Lower index contrast waveguides, such as ridge waveguide, were widely used previously and higher index contrast systems have been used recently, such as a small buried waveguide with a cross-section of $500nm \times 250nm$, to achieve efficient modulation at high speed and large scale integration.

2.4 Theory of Optical Modulators

The optical modulator studied in this thesis employs a PIN diode structure to inject carriers and a Mach-Zehnder Interferometer (MZI) to convert a phase shift into an amplitude modulation of light. In this section, the relevant theory of a PIN diode structure and Mach-Zehnder interferometers are discussed. In particular, the analytical PN theory is reviewed because of the similarity of PN and PIN diodes. The DC and AC characteristics of PIN diodes are investigated theoretically, which can be used as a benchmark for numerical modeling and SPICE circuit simulations of PIN diode modulators. In addition, the transmission characteristics of the Mach-Zehnder interferometer is analyzed.

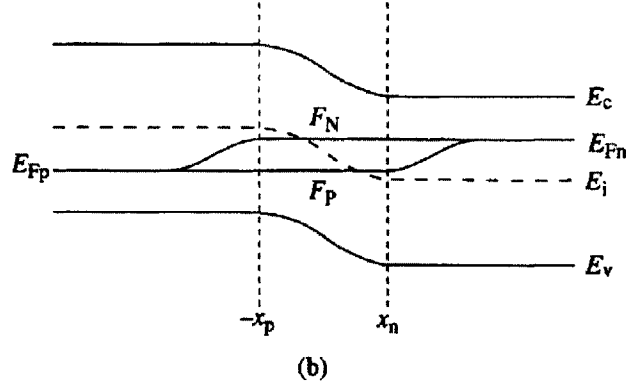
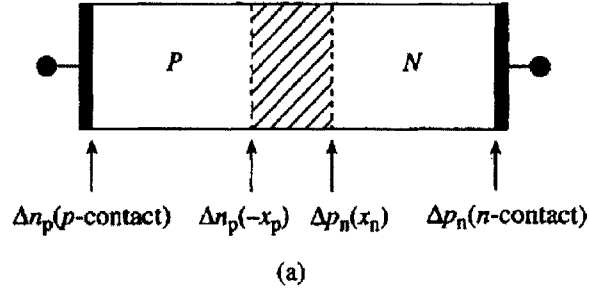


Figure 2-6: (a) PN junction with the depletion region from $-x_p$ to x_n , (b) the band structure of a forward-biased diode [33].

2.4.1 PN Diode Theory

PN Junction DC Characteristics

To obtain the current density as a function of the applied voltage, one needs to solve the continuity equation with the boundary conditions at the contacts and the depletion edges $-x_p$ and x_n , shown in Fig.2-6(a). The electron continuity equation in the P-region is

$$\begin{aligned} \frac{\partial n}{\partial t} &= G_n - \frac{n}{\tau_n} + D_n \frac{\partial^2 n}{\partial x^2} + \frac{\partial(qn\varepsilon\mu_n)}{\partial x} \\ &\simeq G_n - \frac{n}{\tau_n} + D_n \frac{\partial^2 n}{\partial x^2} \end{aligned} \quad (2.18)$$

where n is the free electron density, D_n is the electron diffusion coefficient, μ_n is the electron mobility, G_n is the generation rate, ε is the electric field across the junction,

and τ is the electron lifetime, respectively. The drift term $\partial(qn\varepsilon\mu_n)/\partial x$ for minority carriers induced by the electric field is neglected in the high doping regions for low level injection [33]. The hole continuity equation is similar to equation (2.18). A DC injected carrier distribution can be obtained from equation (2.18) by setting $\partial n/\partial t = 0$ and $G_n = 0$

$$\Delta n(x) = \Delta n(-x_p)e^{(x+x_p)/L_n} \quad (2.19)$$

for the P region and

$$\Delta p(x) = \Delta p(x_n)e^{-(x+x_n)/L_p} \quad (2.20)$$

for the N region. Here, $L_n = \sqrt{\tau_n D_n}$ and $L_p = \sqrt{\tau_p D_p}$ are the diffusion lengths for electrons in the P region and holes in the N region, respectively.

The boundary conditions at the depletion edges $-x_p$ and x_n depend on the external applied voltage V_A . At the depletion edges, the free electron density n and hole density p satisfy the relation [33]:

$$np = n_i^2 e^{V_A/(KT/q)} \quad (2.21)$$

where V_A is the applied voltage, K is the Boltzmann constant, T is the temperature, q is the electron charge, and n_i is the intrinsic carrier density, respectively. This relation has been referred as the "law of the junction". At the edge $x = -x_p$, the majority holes don't vary much for low level injection which means that the total free hole density p is equal to the acceptor density N_A , i.e, $p = N_A$. Therefore, the injected electron density is achieved at $x = -x_p$,

$$\Delta n(-x_p) = n(-x_p) - \frac{n_i^2}{N_A} = \frac{n_i^2}{N_A} (e^{V_A/(KT/q)} - 1) \quad (2.22)$$

and similarly, we have

$$\Delta p(x_n) = p(x_n) - \frac{n_i^2}{N_D} = \frac{n_i^2}{N_D} (e^{V_A/(KT/q)} - 1) \quad (2.23)$$

where N_D is the donator density in the N region. Putting the boundary conditions (2.22) and (2.23) into the carrier distribution equations (2.19) and (2.20) separately, the diffusion current densities can be obtained

$$J_n = qD_n \frac{\partial \Delta n}{\partial x} \Big|_{x=-x_p} = \frac{qn_i^2 D_n}{N_A L_n} (e^{V_A/(KT/q)} - 1) \quad (2.24)$$

$$J_p = -qD_p \frac{\partial \Delta p}{\partial x} \Big|_{x=x_n} = \frac{qn_i^2 D_p}{N_D L_p} (e^{V_A/(KT/q)} - 1) \quad (2.25)$$

The total current is derived by neglecting the recombination current in the depletion region,

$$I_{DC} = I_n + I_p = qA \left(\frac{n_i^2 D_n}{N_A L_n} + \frac{n_i^2 D_p}{N_D L_p} \right) (e^{V_A/(KT/q)} - 1) \quad (2.26)$$

where A is the cross-section.

PN Junction AC Characteristics

The PN junction AC characteristics is studied by applying the small-signal analysis on the PN junction. The applied voltage can be assumed to be $V_A = V_0 + \tilde{V}_1$, where V_0 is the DC component and $\tilde{V}_1$ is a small perturbation on the order of $KT/q = 25.9mV$ at $T = 300K$. In the analysis, we only keep the linear terms of perturbations. If the perturbation term $\tilde{V}_1$ is a sinusoidal voltage the carrier variation only has the first harmonic term and all the higher harmonic terms are negligible. Therefore, the linearized continuity equation of the carrier density variation (2.18) is written as

$$-\frac{\Delta n}{\tau_n/(j\omega\tau_n + 1)} + D_n \frac{\partial^2 \Delta n}{\partial x^2} = 0 \quad (2.27)$$

by setting $G_n = 0$. Equation (2.27) has a similar form as the DC continuity equation except that the effective lifetime becomes $\tau_{eff} = \tau_n/(j\omega\tau_n + 1)$. Putting $V_A = V_0 + \tilde{V}_1$ into the DC current boundary condition (2.22) at $x = -x_p$, the boundary condition becomes,

$$\Delta n(-x_p) = \Delta n_{DC}(-x_p) + \Delta n_{AC}(-x_p) = \frac{n_i^2}{N_A} (e^{(V_0 + \tilde{V}_1)/(KT/q)} - 1) \quad (2.28)$$

where

$$\Delta n_{DC}(-x_p) = \frac{n_i^2}{N_A} (e^{V_0/(KT/q)} - 1) \quad (2.29)$$

$$\Delta n_{AC}(-x_p) = \frac{n_i^2}{N_A} e^{V_0/(KT/q)} (e^{\tilde{V}_1/(KT/q)} - 1) = \frac{n_i^2}{N_A} \frac{\tilde{V}_1}{KT/q} e^{V_0/(KT/q)} \quad (2.30)$$

Thus, putting the boundary condition (2.30) and the hole distribution (2.20) into the diffusion current density equation (2.24), we conclude

$$\tilde{i}_n = A\tilde{J}_n = qAD_n \frac{\partial \Delta n_{AC}}{\partial x} \Big|_{x=-x_p} = qA \frac{n_i^2 D_n}{N_A L_n} \frac{\tilde{V}_1}{KT/q} e^{V_0/(KT/q)} \quad (2.31)$$

Substituting the effective diffusion length $L_n = \sqrt{D_n \tau_n / (j\omega\tau_n + 1)}$ into the above equation, the electron admittance is derived as

$$Y_D = \frac{\tilde{i}_n}{\tilde{V}_1} = G_0 \sqrt{1 + j\omega\tau_n} \quad (2.32)$$

where $G_0 = I_0/u_t$, $I_0 = qA \frac{n_i^2 D_n}{N_A \sqrt{\tau_n D_n}} e^{V_0/(KT/q)}$ and $= KT/q$. The admittance can be separated into real and imaginary parts to obtain the diffusion conductance and capacitance,

$$G_D = \frac{G_0}{\sqrt{2}} (\sqrt{1 + (\omega\tau_n)^2} + 1)^{1/2} \quad (2.33)$$

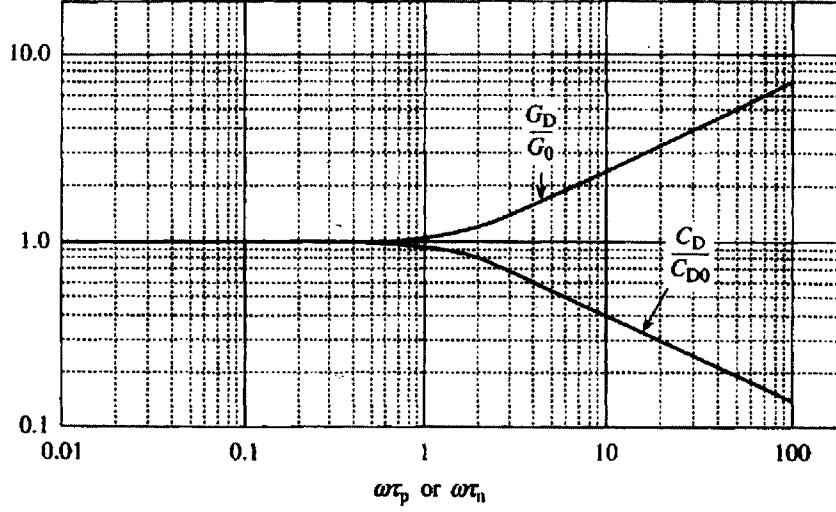


Figure 2-7: Normalized PN diode diffusion capacitance and conductance as a function of $\omega\tau_p$ (p^+n diode) or $\omega\tau_n$ (n^+p diode) [33].

$$C_D = \frac{G_0}{\omega\sqrt{2}}(\sqrt{1 + (\omega\tau_n)^2} - 1)^{1/2} \quad (2.34)$$

Fig.2-7 shows the normalized diffusion conductance and capacitance. At low frequencies, $G_D = G_0$ and $C_D = G_0\tau/2$. At high frequencies, the conductance is proportional to $\sqrt{\omega}$ and the capacitance is inversely proportional to $\sqrt{\omega}$.

2.4.2 PIN Diode Theory

Similar to PN junction theory in Sec.2.4.1, the DC and AC carrier distribution of PIN diode can be derived by solving the continuity equations with the boundary conditions at the two edges $x = \pm d$, as shown in Fig.2-8. The 1-dimensional continuity equations for electrons and holes are,

$$\frac{\partial n}{\partial t} = \frac{1}{q} \frac{\partial J_n}{\partial x} - \frac{n}{\tau_n} = D_n \frac{\partial}{\partial x} \left(\frac{q}{kT} (\epsilon n) + \frac{\partial n}{\partial x} \right) - \frac{n}{\tau_n} \quad (2.35)$$

$$\frac{\partial p}{\partial t} = -\frac{1}{q} \frac{\partial J_p}{\partial x} - \frac{p}{\tau_p} = -D_p \frac{\partial}{\partial x} \left(\frac{q}{kT} (\epsilon p) - \frac{\partial p}{\partial x} \right) - \frac{p}{\tau_p} \quad (2.36)$$

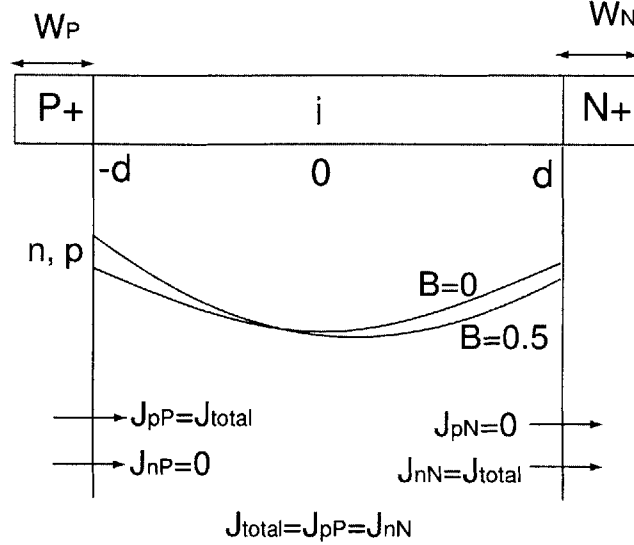


Figure 2-8: Diagram of PIN diode structure, carrier distribution at intrinsic region and the current density at the edges $x = \pm d$ when assuming that the injection efficiency at both ends is unity. The small subscripts n/p indicate electrons and holes, and the capital subscripts N/P indicate the current flows from the P/N-region.

where the Einstein relation $\mu = \frac{q}{kT}D$ is used, τ_p and τ_n are the intrinsic region hole lifetime and the minority electron lifetime, D_p and D_n are the hole diffusion coefficient and the electron diffusion coefficient, and ε is the electric field, respectively. Assuming the free electron density and the free hole density are equal, i.e., $n \simeq p$ and $\partial n \simeq \partial p$, εp and εn terms are eliminated by calculating $(2.35)/D_n + (2.36)/D_p$,

$$\frac{1}{D_n} \frac{\partial n}{\partial t} + \frac{1}{D_p} \frac{\partial p}{\partial t} = \frac{\partial^2 n}{\partial x^2} + \frac{\partial^2 p}{\partial x^2} - \left(\frac{n}{\tau_n D_n} + \frac{p}{\tau_p D_p} \right) \quad (2.37)$$

We define ambipolar diffusion coefficient $D_a = \frac{2D_n D_p}{D_n + D_p}$ and $\tau_a = \frac{\tau_n \tau_p (D_n + D_p)}{D_n \tau_n + D_p \tau_p}$ and rewrite equation (2.37) as

$$\frac{\partial n}{\partial t} = D_a \frac{\partial^2 n}{\partial x^2} - \frac{n}{\tau_a} \quad (2.38)$$

Equation (2.38) is called the ambipolar equation.

At the present, we assume the injection efficiency at both ends of a PIN diode is unity, in which case all the forward current comes from the current J_{pP} due to the

holes injected into i-region from p^+ -region and the current J_{nN} due to the electrons injected into the i-region from the n^+ -region, as shown in Fig.2-8. The injected holes and electrons recombine in the i-region. The current J_{pN} due to the minority holes injected from the i-region into the n^+ -region and the current J_{nP} due to the minority electrons injected from the i-region into the p^+ -region are neglected. At the boundaries $x = +d$, the current density is given by

$$J_{+d} = J_{nN} = q\mu_n \varepsilon n + qD_n \frac{\partial n}{\partial x} \quad (2.39)$$

If the injection efficiency is assumed to be unity, i.e., $J_{pN} = 0$, at $x = +d$, ε is obtained

$$\varepsilon_{+d} = \frac{kT}{q} \frac{1}{n_{+d}} \frac{\partial n}{\partial x} \Big|_{x=+d} \quad (2.40)$$

Therefore, the current at $x = +d$ becomes

$$J_{+d} = 2qD_n \frac{dn}{dx} \Big|_{x=+d} \quad (2.41)$$

Similarly,

$$J_{-d} = -2qD_p \frac{dp}{dx} \Big|_{x=-d} \quad (2.42)$$

where quasi-neutral condition $n \simeq p$ is used. Solving the ambipolar equation (2.38) with the boundary conditions (2.41) and (2.42), the DC solution for the free carrier distribution can be found [34],

$$n_0(x) = \frac{\tau_a J_0}{2qL_a} \left[\frac{\cosh(x/L_a)}{\sinh(d/L_a)} - B \frac{\sinh(x/L_a)}{\cosh(d/L_a)} \right] \quad (2.43)$$

where $B = \frac{\mu_n - \mu_p}{\mu_n + \mu_p}$, $L_a = \sqrt{D_a \tau_a}$, $J_0 = J_{+d} = J_{-d}$ is the current injected from external circuits. The AC characteristics with the unit injection efficiency and the DC/AC characteristics with injection efficiency less than 1 are given in Appendix A.

2.4.3 Transmission Characteristics of Mach-Zehnder Interferometer

A Mach-Zehnder interferometer uses one beam splitter to split an incident beam into two beams and uses the other beamsplitter to recombine the two beams to one output port, as shown in Fig.2-9(a). The output signal is modulated by a difference in phase shift between the two beams. The transmission of the Mach-Zehnder interferometer can be calculated as follows.

The incident electric field can be written in terms of position x and time t ,

$$E_i(x, t) = E_0 e^{-j(kx - \omega t)} \quad (2.44)$$

where E_0 , k and ω are the electric field amplitude, the wavenumber and the frequency, respectively. If the light field is split to each arm in half, the electric field amplitudes at the end of each arm can be written as,

$$E'_1 = \frac{1}{\sqrt{2}} E_0 e^{-j\phi_1 - \alpha L_1} \quad (2.45)$$

$$E'_2 = \frac{1}{\sqrt{2}} E_0 e^{-j\phi_2 - \alpha L_2 - j\phi_B} \quad (2.46)$$

where ϕ and αL are the phase shift and optical absorption, and ϕ_B is the phase shift between the two arms to bias the operation point, respectively. The total field amplitudes of the MZ-interferometer output, as shown in Fig.2-9(a), is expressed as

$$\begin{aligned} E &= \frac{1}{\sqrt{2}} E_0 e^{-j\phi_{DC} - \alpha L_{DC}} (e^{-j\Delta\phi - \Delta\alpha L} + e^{j\Delta\phi + \Delta\alpha L - j\phi_B}) \\ &= \frac{1}{\sqrt{2}} E_0 e^{-j\phi_{DC} - \alpha L_{DC} - j\frac{\phi_B}{2}} (e^{-j\Delta\phi - \Delta\alpha L + j\frac{\phi_B}{2}} + e^{j\Delta\phi + \Delta\alpha L - j\frac{\phi_B}{2}}) \end{aligned} \quad (2.47)$$

where $\phi_{DC} \equiv (\phi_1 + \phi_2)/2$, $\alpha L_{DC} \equiv (\alpha L_1 + \alpha L_2)/2$, $\Delta\phi \equiv (\phi_1 - \phi_2)/2$ and $\Delta\alpha L \equiv (\alpha L_1 - \alpha L_2)/2$. If the push-pull method is used to modulate the MZ modulator [8],

the two terms ϕ_{DC} and αL_{DC} are constant. The normalized power transmission is calculated by

$$\begin{aligned}
T = \frac{1}{2E_0^2} E \bullet E^* &= \frac{1}{4} e^{-2\alpha L_{DC}} [e^{-2\Delta\alpha L} + e^{2\Delta\alpha L} + 2 \cos(2\Delta\phi - \phi_B)] \\
&= \frac{1}{2} e^{-2\alpha L_{DC}} [\cosh(2\Delta\alpha L) + \cos(2\Delta\phi - \phi_B)] \\
&\simeq \frac{1}{2} e^{-2\alpha L_{DC}} [\cosh(\frac{\Delta\phi}{4}) + \cos(2\Delta\phi - \phi_B)] \quad (2.48)
\end{aligned}$$

where ϕ_B is the bias phase shift between two arms and the substitution $\Delta\phi = \frac{2\pi}{\lambda} \Delta n L$ and $2\alpha L = \frac{\Delta\phi\lambda}{2\pi\cdot\mu m} \simeq \Delta\phi/4$ are used to simplify the expression. This substitution is approximately true only if electrons are taken into account in the plasma effect at wavelength $\lambda = 1.55\mu m$ [35].

We can get the first and second derivative of the output light intensity

$$\begin{aligned}
\frac{\partial T}{\partial \Delta\phi} \big|_{\Delta\phi=0} &= \frac{1}{2} e^{-2\alpha L_{DC}} [\frac{1}{4} \sinh(\frac{\Delta\phi}{4}) - 2 \sin(2\Delta\phi - \phi_B)] \\
&= e^{-2\alpha L_{DC}} \sin(\phi_B) \quad (2.49)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial^2 T}{\partial \Delta\phi^2} \big|_{\Delta\phi=0} &= \frac{1}{2} e^{-2\alpha L_{DC}} [\frac{1}{16} \cosh(\frac{\Delta\phi}{4}) - 4 \cos(2\Delta\phi - \phi_B)] \\
&= \frac{1}{2} e^{-2\alpha L_{DC}} [\frac{1}{16} - 4 \cos(\phi_B)] = 0 \quad (2.50)
\end{aligned}$$

In order to eliminate the second order harmonic, the second derivative in equation (2.50) has to be zero. Therefore, we have

$$\phi_B = \cos^{-1}(\frac{1}{4 \times 16}) \simeq 89.1^\circ \quad (2.51)$$

Figure 2-9(b) plots the transmission characteristics with the bias of $\phi_B = 89.1^\circ$.

The minimum DC optical loss can be $e^{-2\alpha L_{DC}} \geq e^{-2\Delta\alpha L_{min}} \simeq e^{-\Delta\phi_{min}/4}$ for the push-pull method. For example, a phase shift $\Delta\phi = 0.5\pi$ gives the minimum DC optical loss to be $-10 \log_{10}[e^{-0.5\pi/4}] = 1.7dB$.

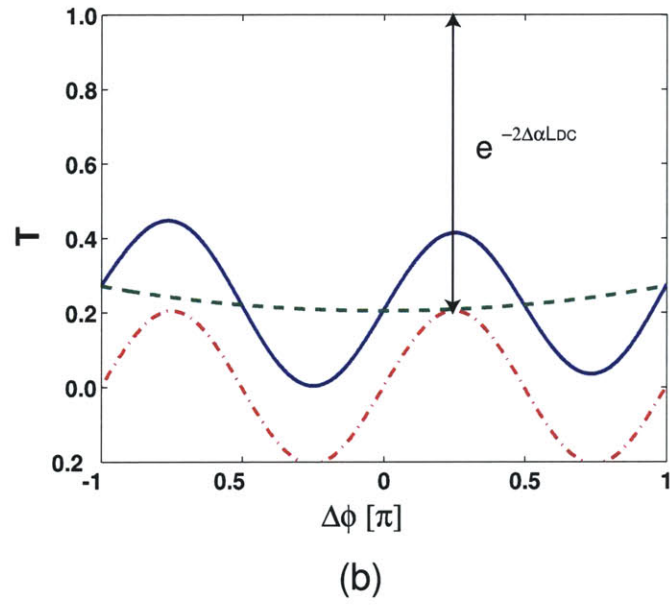
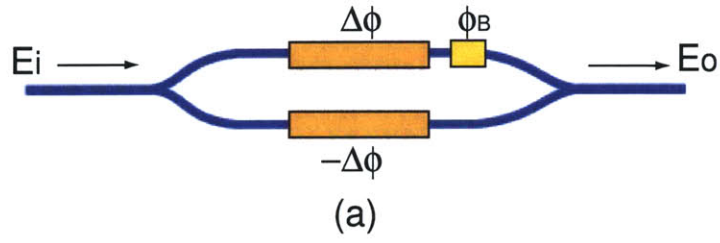


Figure 2-9: Transmission of Mach-Zehnder interferometer with push-pull method and $\phi_B = 89.1^\circ$: $\frac{1}{2}e^{-2\alpha L_{DC}} \cos(2\Delta\phi - \phi_B)$ (red solid-dot), $\frac{1}{2}e^{-2\alpha L_{DC}} \cosh(\Delta\phi/4)$ (green dash), $\frac{1}{2}e^{-2\alpha L_{DC}} [\cosh(\Delta\phi/4) + \cos(2\Delta\phi - \phi_B)]$ (blue solid).

Chapter 3

High-Speed Silicon Modulators for High-Index-Contrast (HIC) Waveguides

One of the most important electro-optic devices for future electronic-photonic integrated circuits based on silicon technology is a CMOS-compatible high-speed optical modulator that converts electronic signals into the optical domain. The plasma and thermal effects in silicon have long been known as efficient means to achieve such modulation [36]. Early on PIN-structures based on the plasma effect have been proposed and realized for silicon-based optoelectronic phase and amplitude modulators with switching speeds up to 10MHz [25]. The speed of the frequency response in those devices is limited by carrier recombination. Very recently a modulator based on a reverse-biased PIN/PN structure has been reported operating at a modulation speed of 20GHz [14]. However, this device needs a large applied voltage of around 5V and has a figure of merit (FOM) $V_{\pi} \cdot L$ larger than $1V \cdot cm$. In this chapter, we investigate high-speed electro-optic modulators based on high-index-contrast silicon buried waveguides to achieve minimum optical loss, low RF power consumption and small voltage-length products while increasing the modulation bandwidth.

This chapter is organized as follows. In Sec.3.1, the modulators based on four different structures, the forward-biased PIN diode with lifetime reduction, the forward-

biased diode without lifetime reduction, the reverse-biased PIN/PN diode and the metal-oxide-semiconductor (MOS), are presented. The electrical properties of modulators are modeled with the 2-dimensional semiconductor simulator MEDICI and the experimental results are analyzed analytically and numerically. In Sec.3.2, the PIN diode is modeled with SPICE equivalent circuits so that the PIN SPICE model can be integrated with electronic driving circuits for circuit simulations. In Sec.3.3, the fabrication and measurement setup are illustrated and discussed.

3.1 Investigation of Modulation

Recently, a lot of applications need to process high-speed signals with high performance. For example, the analog-digital converter mentioned in chapter 1 requires a bandwidth up to 100Gbps. The speed of those electro-optic applications is limited by the speed of optical modulators which convert electrical signals to optical signals. The modulation speed of optical modulators usually depends on at least three mechanisms. Firstly, the carrier lifetime determines the upper speed of an electro-optic modulator based on the plasma dispersion effect, such as a forward-biased PIN diode modulator. The 3dB corner frequency is given by $f_{3dB} = 1/2\pi\tau$. For example, a carrier lifetime of 1ns gives a frequency of 169MHz. Secondly, the carrier drift velocity is the dominant parameter in those electro-optic modulators based on drift effects, such as the PIN/PN reverse-biased modulator. The carrier drift velocity determines the bandwidth of a drift-effect-dependant electro-optic modulator according to $f_{3dB} = 2.4/(2\pi\tau_{tr})$ with $\tau_{tr} = W/v$ [37]. Here, W and v are the width of i-region and the carrier drift velocity, respectively. The modulation speed can be increased by decreasing the length or increasing the drift velocity. If the carriers drift over a distance $W = 500nm$ at the saturation velocity $v = 10^7cm/s$ in silicon, the bandwidth is $f_{3dB} = 2.4/2\pi\tau_{tr} \simeq 80GHz$. Finally, the RC delay of driving circuits and optical modulators can limit the upper modulation speed.

It's important to achieve high modulation efficiency for high-performance applications. The modulation efficiency of electro-optic modulators can be quantized with

the figure of merit (FOM), which is the product of the voltage and length to achieve π phase shift, i.e., $V_\pi \cdot L$. A larger capacitance of a modulator usually gives larger modulation efficiency because more carriers can be obtained to introduce larger refractive index change at the same driving voltage. In addition, an efficient way to increase modulation efficiency is to enhance the overlap between carrier change and optical mode. This can be achieved by optimizing geometry, as will be discussed in Sec.3.1.1.

In this section, silicon electro-optic modulators are discussed based on four structures: the modulator based on forward-biased PIN diodes with lifetime reduction, the modulator based on forward-biased diodes without lifetime reduction, the modulator PIN/PN reverse-biased and the modulator based on the metal-oxide-semiconductor (MOS). A schematic layout of the proposed modulators is shown in Fig.3-1. Fig.3-1(a) shows a conventional Mach-Zehnder arrangement and Fig.3-1(b) shows the most important sub-component of the modulator, the electrically driven phase modulator section. Three types of doping profile in the phase shifter will be thoroughly studied to optimize optical modulators.

The fabrication and testing of the designed silicon modulators have been done through a collaboration with MIT-Lincoln Laboratory. The measurement data is analyzed analytically and numerically in Sec.3.1.2. The method of fabrication and measurement setup is illustrated in Sec.3.3.

3.1.1 Modulator Based on Forward-Biased PIN Diodes with Lifetime Reduction

In this section, we explore the modulator based on forward-biased PIN diodes with lifetime reduction. The waveguide geometry and doping profile are investigated to minimize the optical loss and figure of merit. The bandwidth is broadened by reducing the carrier drift distance and increasing the drift velocity. A drift velocity can be increased by a high electric field applied to the PIN diode. Two conditions need to be satisfied so that a high electric field can be applied to the diode. Firstly, the contact

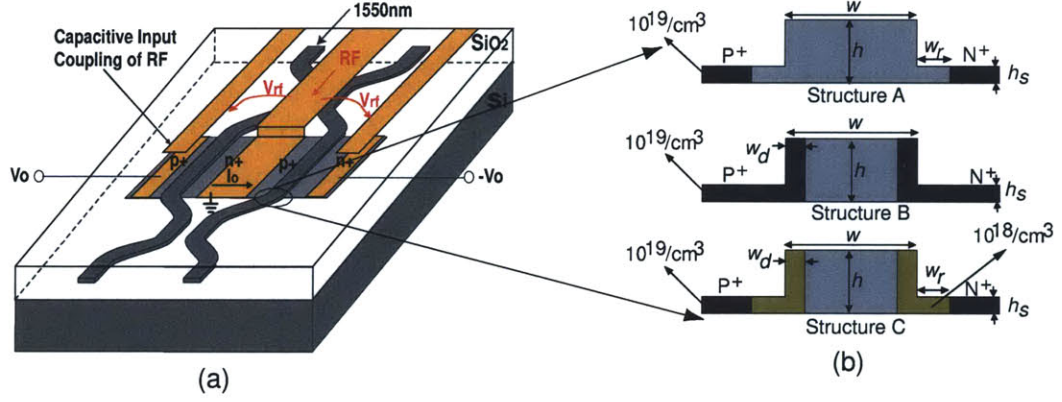


Figure 3-1: (a) Mach-Zehnder modulator with forward biased PIN-phase modulators, (b) different doping profiles for the Si-buried waveguide pin-phase modulators.

resistivity has to be small so as to make most voltage drop across the PIN diode. The reported contact resistivity is about $10^{-7} \Omega \cdot \text{cm}^2$ [38] which is small enough to minimize the voltage drop across the series resistance and enlarge the voltage drop across the i-region of the PIN-modulator. Secondly, the current needs to keep in a low level and thus not much heating is generated when a high electric field is applied to the PIN diode. This can be realized by reducing carrier lifetime. Carrier lifetime can be controlled to the order of tens pico-second using doping or ion implantation [39].

Numerical Modeling

Three kinds of pin-diode structures are numerically investigated to achieve optimum modulator performance, as shown in Fig.3-1(b). The modulation bandwidth is increased by moving the highly doped region P/N to the waveguide as close as possible to the optical waveguide by decreasing the length of the undoped region w_r in structure A, or even moving the higher doped regions inside the buried waveguide as in structure B. However, structure B leads to a high optical loss of over 60dB/cm due to the overlap between the high doping region and the optical mode. Much lower loss is achieved with structure C where the sidewalls of the waveguide are weakly doped with $10^{18}/\text{cm}^3$ and high doping $10^{19}/\text{cm}^3$ is w_r away from the waveguide. The

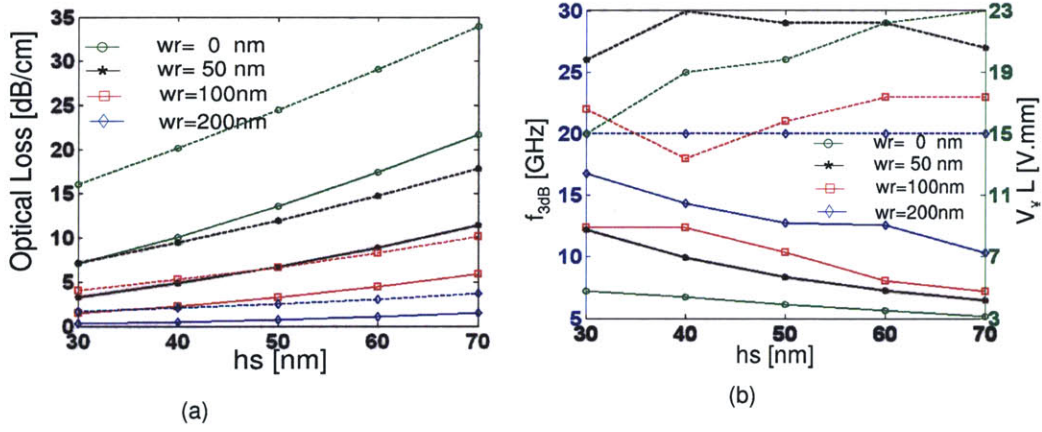


Figure 3-2: (a) Optical loss due to doping (solid) and with carrier injection (dashed), (b) f_{3dB} (solid) and $V_\pi L$ (dashed) of structure A with different width w_r , $\tau = 10ps$, $V_{DC} = 3V$.

sidewall doping improves equalization of the carrier distribution across the i-region and, therefore, decreases the FOM $V_\pi \cdot L$.

Fig.3-2 and 3-3 show the performance of structure A and structure C. The optical loss due to doping (solid curves) and that due to doping and carrier injection (dash curves) is plotted in 3-2(a) for structure A with different height h_s and width w_r . The optical loss is around 0-20dB/cm due to doping and 3-35dB/cm due to doping and free carriers injected by the 3V DC forward bias. The optical loss of structure C is slightly larger than that of structure A, as shown in Fig.3-3(a) and 3-3(b). Therefore, the lower doping doesn't add much additional optical loss. However, the 3dB corner frequencies for structure C ($w_d=50nm$) are about 20% larger than structure A and C ($w_d=0nm$), around 29-35GHz (see Fig.3-3(d)), from which it can be concluded that the short intrinsic length in structure C ($w_d = 50nm$) plays an important role in broadening the modulation bandwidth. Finally, the FOM of structure A is larger than structure C, around 3-12 $V \cdot mm$, shown in Fig.3-2(b), 3-3(c) and 3-3(d). A comparison of Fig.3-3(c) and 3-3(d) shows that structure C ($w_d=0nm$) has about 4% higher FOM than structure C ($w_d=50nm$), because the sidewall doping reduces the area where the carrier density can be changed by the current flow. In summary, structure C ($w_d=0nm$) has the smallest FOM and structure C ($w_d=50nm$) shows

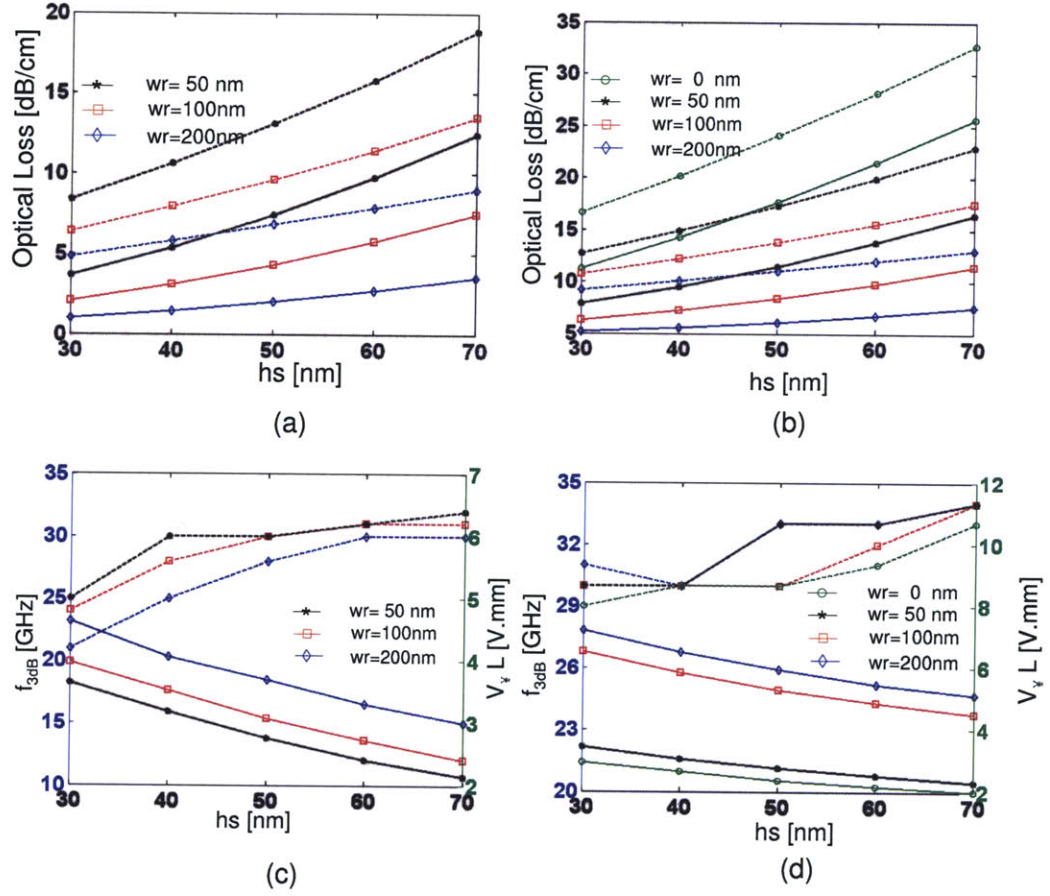


Figure 3-3: Optical loss due to doping (solid) and with carrier injection (dashed) of structure C (a) with $w_d = 0$ nm and (b) with $w_d = 50$ nm, f_{3dB} (solid) and $V_{\pi}L$ (dashed) of structure C (c) with $w_d = 0$ nm and (d) with $w_d = 50$ nm. ($\tau = 10$ ps, $V_{DC} = 2.5$ V)

the largest modulation bandwidth. Based on the discussion above, structure C was chosen to complete a full high-speed modulator design. In the following calculations we choose dimensions $h_s = 50$ nm, $w_d = 50$ nm, $w_r = 200$ nm, $w = 450$ nm, and $h = 250$ nm. The high and low doping levels are $10^{19}/\text{cm}^3$ and $10^{18}/\text{cm}^3$, respectively.

Detailed two-dimensional simulations have been carried out for a modulator implementing structure C. The current density flowing through the intrinsic zone as a function of the applied voltage is shown in Fig.3-4(a) for three different carrier lifetimes. The current density saturates at voltages greater than 5V. We can not bias the voltage larger than 5V. Therefore, $V_{DC} = 2$ V is chosen as the bias operation

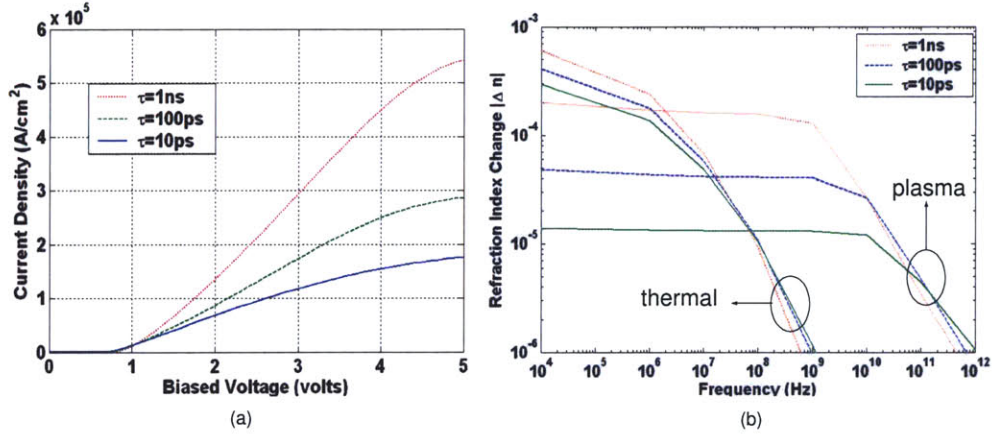


Figure 3-4: (a) Modulator I-V characteristics; (b) effective index change due to plasma and thermal effects for $\tau = 10ps$ (solid), $100ps$ (dashed), $1ns$ (dotted) in structure C. ($w = 450nm$, $h = 250nm$, $h_s = 50nm$, $w_r = 200nm$, $w_d = 50nm$)

point for small-signal analysis. We can investigate the bandwidth based on small-signal analysis. Fig.3-4(b) shows the amplitude of the effective index change due to both the thermal effect and the plasma dispersion effect for a small sinusoidal driving voltage amplitude of $0.1V$. The simulations gave $f_{3dB} = 32GHz$, $15GHz$ and $1.5GHz$ for $\tau = 10ps$, $100ps$ and $1ns$, respectively, as shown in Fig.3-4(b). According to the speed dependence on the lifetime $f_{3dB} = 1/4\pi\tau$, the 3dB corner frequencies due to recombination are $8GHz$, $0.8GHz$ and $0.08GHz$ for $\tau = 10ps$, $100ps$ and $1ns$, respectively [40]. The discrepancy between the lifetime dependent speed and the simulated speed indicates that the high modulation bandwidth in simulation is enabled by the high drift velocity of the electrons close to the saturation velocity. The modulation speed of a pin-diode depends critically on both the carrier lifetime and drift velocity, i.e., how fast carriers can be removed from the intrinsic zone.

For frequencies $f > 100$ MHz, the plasma effect dominates. At low frequencies the thermal effect dominates because the thermal time constant is about $300ns$ estimated from Fig.3-4(b), which is much longer than the carrier lifetimes used in the simulations. This is of major importance, because both effects do have opposite sign and, therefore, there is a frequency range, between $1 - 100$ MHz, where the response is zero. For this design $6mm$ length is necessary for a π phase shift with a voltage of

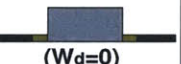
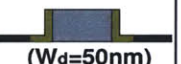
	Structure A 	Structure B 	Structure C ($W_d=0$) 	Structure C ($W_d=50nm$) 
Optical loss (dB/cm) at ON-state	3	60	6	10
f_{3dB} (GHz)	20	NA	27	34
$V_\pi L$ (V.mm)	9	NA	3.8	6

Figure 3-5: Modeled performance of the modulators based on different structures.

1V because of $V_\pi L = 6Vmm$ as indicated in Fig.3-3(d). The DC-power and the dissipated microwave power of a 6mm long section are $P_{DC} = 3.6W$ and $P_{AC} = 500mW$ for a phase shift of π .

Fig.3-5 summarizes the performance of structures shown in Fig.3-1(b). Structure A has the lowest optical loss because the high doping is over $200nm$ away from the waveguide. Structure B has the highest optical loss resulting from the high doped sidewall. Structure C with $w_d = 50nm$ can work in a wider bandwidth $f_{3dB} = 34$ GHz. Structure C with $w_d = 0$ gives the largest modulation efficiency with a figure of merit $V_\pi \cdot L = 3.8$, which has been confirmed by the experiment as shown in Fig.3-12 in Sec.3.1.2.

Experimental Exploration

Waveguides are processed using three different methods: only silicon ion implanted, only annealed, and both silicon ion implanted and annealed. The implantation was done with silicon ion at energy 190keV and dose $10^{13}/cm^2$. The annealing was done under $450^\circ C$ temperature and lasted 20 seconds. In Fig.3-6(a), the I-V curves are plotted for these three types of waveguides. The ion implanted sample without annealed has larger current than the other two samples. In the low injection forward-biased regime (the bias voltage $V < 0$ in Fig.3-6(a)), the recombination current dominates the forward-biased current. The ion implantation introduces defects so that the silicon structure is damaged, which reduces the carrier lifetime [41]. A two-color pump-probe lifetime measurement shows that a sample with dose $10^{14}/cm^2$ without annealing has

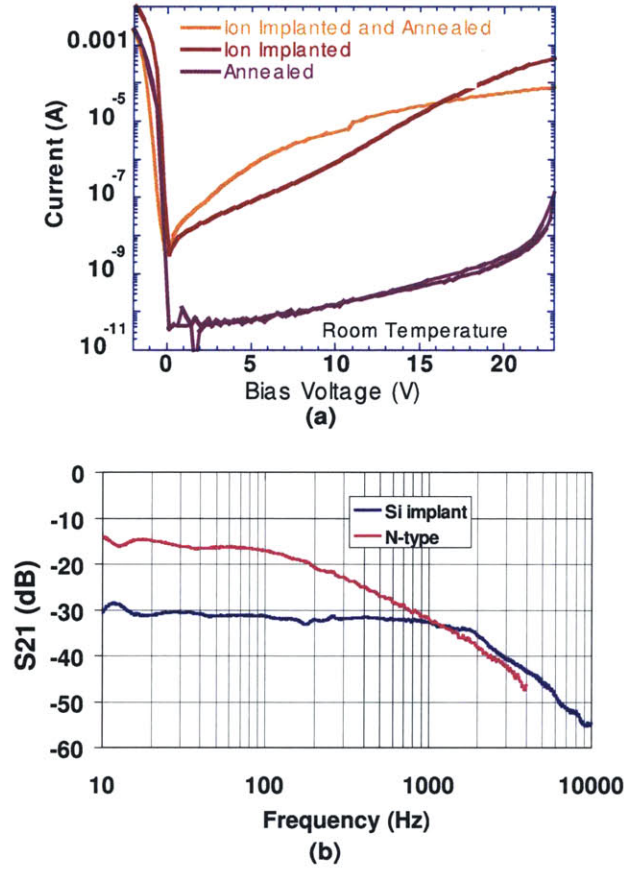


Figure 3-6: (a) Measured I-V curves of PIN diodes with three different processes, (b) comparison between the modulators with silicon ion implantation (blue line) and light n-type doping (red line).

0.6ps carrier lifetime. The annealing efficiently repairs the damaged region in the ion implanted waveguide. Therefore, the two samples, only annealed and both ion implanted and annealed, have similar IV characteristics in the forward-biased modulators. In the reverse-biased region (the bias voltage $V > 0$ in Fig.3-6(a)), the current of the ion implanted sample is 2-3 orders of magnitude larger than the annealed sample, because the ion implantation could excite free carriers to mid-gap states and conduction band [41].

Fig.3-6(b) illustrates the frequency response of two kinds of modulators based on forward-biased PIN diodes. One with silicon ion implantation and the other with lightly n-type doping in the waveguide region. The n-type doped sample has a corner

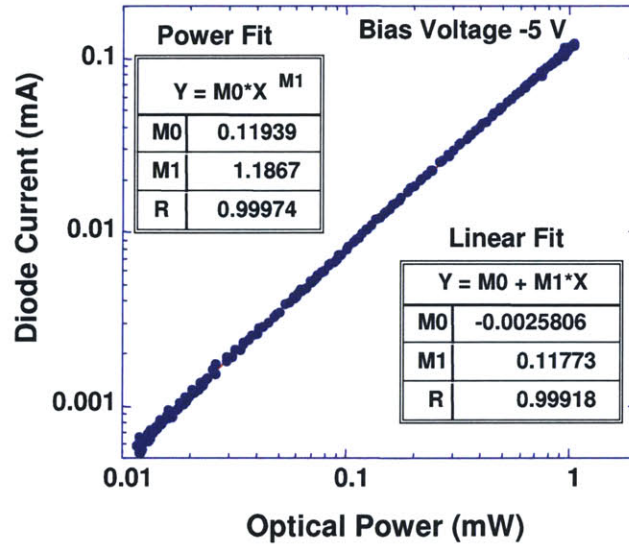


Figure 3-7: Diode current in terms of input optical power. The waveguide is silicon ion implanted.

frequency of $f_{3dB} = 100$ MHz which corresponds a lifetime of 1ns. The silicon ion implanted sample has an extended f_{3dB} of 1-2GHz which is still much smaller than the 34GHz bandwidth obtained in the numerical design in Fig.3-5, where the lifetime was assumed to be around 10ps . This could result from the slow RC delay due to the high contact resistance and the external circuits.

The optical loss is measured up to $100 - 200 \text{ dB/cm}$ in the waveguides with Si ion implantation [41]. It's believed that the implanted Si ions introduce mid-gap energy states. The electrons can absorb light and transit from the valence band to the midgap energy state or from the midgap state to the conduction band. Furthermore, Fig.3-7 illustrates that the photo current is linearly proportional to the input optical power, which indicates that there is only one photon involved in the electron transition from valance to conduction band and only half of the excitation energy is from 1550nm photon. However, it's not well-understood which process provides the other half energy and how electrons are excited to the conduction band. It's believed that the electric field at the vicinity of defects makes it happen. Therefore, more efforts are necessary to reduce these excitation and reduce the optical loss.

In addition to the silicon ion implantation, an alternative process to reduce carrier

lifetime is doping silicon with gold/platinum [42]. The doping of gold and platinum introduces defects and generates midgap states so that the carrier lifetime is reduced. A fast variable optical attenuator (VOA) has been proposed based on the gold/platinum doped silicon waveguide [43]. Switching speed is determined by the recombination lifetime.

In a short summary, a high-speed electronic carrier-injection modulator based on a forward biased PIN diode has been proposed and analyzed numerically. Simulations showed that with a graded doping profile operating frequencies as large as 34GHz can be reached while maintaining a low insertion loss of 3dB and a low figure of merit $V_\pi L = 0.6Vcm$. The experiment has given 1-2GHz 3dB corner frequency.

3.1.2 Modulator Based on Forward-Biased PIN Diodes without Lifetime Reduction

In this section, the DC characteristics of the forward-biased PIN diode modulator is investigated experimentally and numerically. The measured I-V curves are used to extract a series resistance.

The RF power consumption scaling as a function of frequency is studied analytically based on the well-known PN diode theory, and then is investigated numerically and experimentally. Furthermore, the strategies of reducing power consumption are proposed based on the RF power scaling.

The modulation efficiency of the forward-biased PIN diode modulator at low and high frequencies is investigated in the third subsection. In particular, the modulation efficiency of forward-biased PIN diodes with different doping profiles is studied numerically, and further verified by experiments.

Finally, the AC characteristics of the PIN diode modulator is analyzed. The high modulation efficiency at the high-speed operation opens an opportunity to use a pre-compensation scheme to flatten the frequency response up to 10GHz with low RF power and figure of merit $V_\pi \cdot L$. The numerical simulations and experimental data confirm the feasibility of the pre-compensation scheme.

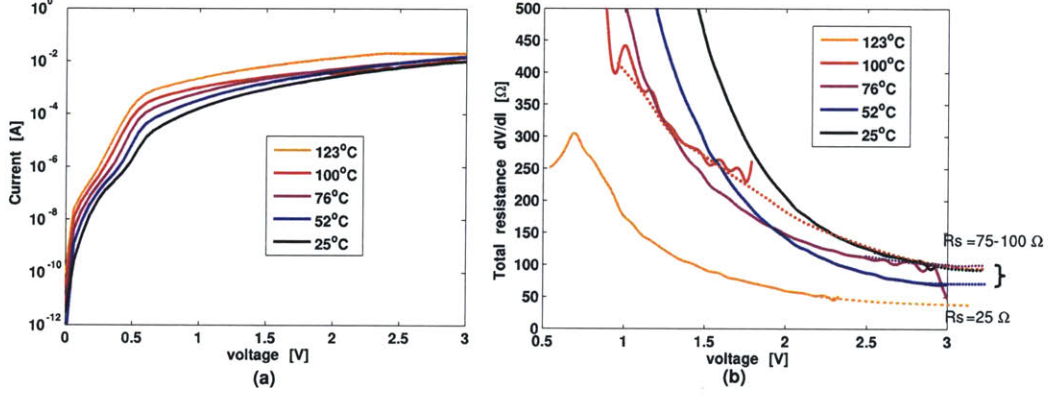


Figure 3-8: (a) Measured I-V curves of PIN diodes at different temperatures, (b) extracted total resistance and series resistance R_s from measured I-V curves. (6mm long, 20 seconds annealing in vacuum 450°C , the waveguide is not silicon ion implanted and has no sidewall doping.)

DC Characteristics of Forward-biased PIN Diode Modulator

The modulator based on the forward-biased PIN diode has been fabricated and its DC characteristics and frequency response have been measured. The I-V curves of the PIN diodes are measured when the device is heated to different temperature, as plotted in Fig.3-8(a). The current increases as the temperature rises for the same applied voltage, since there exist more intrinsic free carriers at higher temperature according to $n_i = \sqrt{N_c N_v} e^{-E_g/kT}$. Here, N_c , N_v , E_g , k and T are the effective density of states in the conduction band, the effective density of states in the valence band, the Fermi energy of intrinsic material, the band-gap energy, the Planck's constant and the absolute temperature, respectively [44]. At voltage $V > 0.8\text{V}$, the I-V curves are flattened, indicating that a large series resistor exists in the diode. Fig.3-8(b) illustrates the total resistance as a function of the voltage for different temperature. The total resistance is obtained by differentiating the voltage over the current, i.e., $R_{tot} = dV/dI$. The total extracted resistance decreases as the current increases and approaches the residual resistance which is the series resistance R_s . The series

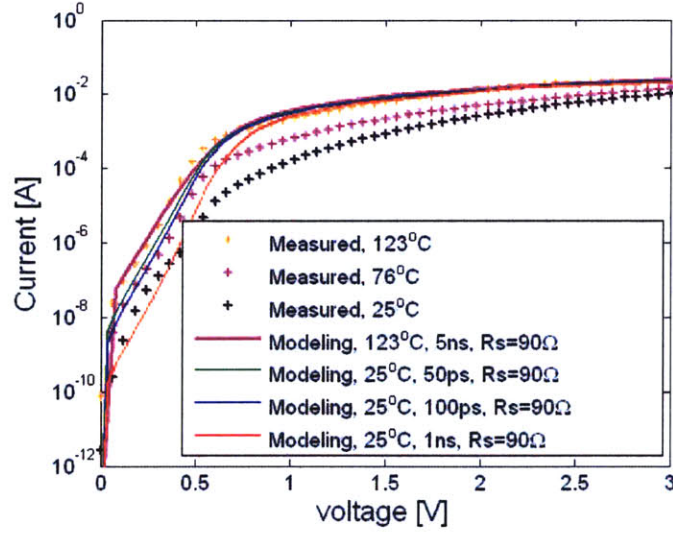


Figure 3-9: Comparison of I-V curves between measurement and modeling. (6mm long, 20 seconds annealing in vacuum 450°C , the waveguide is not silicon ion implanted and has no sidewall doping.)

resistance mostly results from the contact resistance and is therefore independent on the diode current. The extracted value for series resistance is in the range of $75\text{--}100\Omega$ for different operating temperatures from 25°C to 10°C . The smallest series resistance 25Ω was achieved at the highest operating temperature 123°C .

The 2-dimensional numerical semiconductor simulator MEDIIIC [45] is used to fit the measured I-V curves. The results are shown in Fig.3-9. The numerical modeling explores the effects on the I-V curves due to temperature, minority carrier lifetime τ and series resistance R_s . Firstly, both numerical modeling and experimental data indicate that at higher temperature the current rises because of the increased equilibrium free carriers densities. Secondly, the minority carrier lifetime plays an important role in the low injection regime. Smaller lifetimes cause a higher recombination current which dominates the total current in the low injection regime. Finally, the current at high voltage level largely depends on the series resistance. The modeled I-V curves in Fig.3-9 illustrates that for different lifetimes and temperature the same series resistance induces the same current at voltage larger than 1V. Overall, the numerical I-V curve at 123°C with $\tau = 5\text{ns}$ and $R_s = 90\Omega$ matches very well with the one from

measurement at temperature 123°C.

RF Power Consumption Scaling

Based on the theory of PN and PIN diodes in Chapter 1, we can derive RF power consumption. In the low frequency regime $\omega\tau \ll 1$, the diffusion conductance $G_D \approx G_0 = \frac{qA\sqrt{D_n}}{\sqrt{\tau}} \frac{n_i^2}{N_A} \cdot M$, the diffusion capacitance $C_D \approx \frac{G_0}{2}\tau = \frac{qA\sqrt{D_n\tau}}{2} \frac{n_i^2}{N_A} \cdot M$, $M = \frac{e^{V_0/(KT/q)}}{KT/q}$ according to equations (2.33) and (2.34), the RF power is

$$P_{RF,D} = \frac{1}{2}V^2G_D = \frac{1}{2}V^2G_0 = \frac{1}{2}V^2 \frac{qA\sqrt{D_n}}{\sqrt{\tau}} \frac{n_i^2}{N_A} M \quad (3.1)$$

The diffusion capacitor C_D is half of the total capacitance $C_{total} = \frac{\partial Q}{\partial V} = \frac{\partial(I\tau)}{\partial V} = G_0\tau$, which indicates that only half of the charges can be swept out and the other half recombine in the waveguide.

In the high frequency regime $\omega\tau \gg 1$, the diffusion conductance $G_D \approx \frac{G_0}{\sqrt{2}}\sqrt{\omega\tau} = \frac{qA\sqrt{D_n}}{\sqrt{2}} \frac{n_i^2}{N_A} \sqrt{\omega} \cdot M$, and the diffusion capacitance $C_D \approx \frac{G_0}{\sqrt{2}}\sqrt{\frac{\tau}{\omega}} = \frac{qA\sqrt{D_n}}{\sqrt{2}} \frac{n_i^2}{N_A} \sqrt{\frac{1}{\omega}} \cdot M$, then the RF power is

$$P_{RF,D} = \frac{1}{2}V^2G_D = \frac{1}{2}V^2 \frac{G_0}{\sqrt{2}} \sqrt{\omega\tau} = \frac{1}{2\sqrt{2}}V^2 qA\sqrt{D_n} \frac{n_i^2}{N_A} \sqrt{\omega} M \quad (3.2)$$

In the above equations, G_D , C_D and P_{RF} have no correlation with the lifetime τ . This is because the driving frequency is faster than the carrier recombination speed. All carriers are swept out before they recombine, i.e., the total charges are stored in the diffusion capacitor. Therefore, the total capacitance is equal to the diffusion capacitance, i.e., $C_{total} = C_D = G_D/\omega$. Furthermore, in the high frequency regime $\omega\tau \gg 1$, $\frac{G_D}{\omega C_D} = 1$ indicates that half of the current goes to the capacitance, the other half goes to the conductance.

In a more realistic situation, there is additional power consumption besides the power consumption based on the ideal PN junction theory in equation (3.1) and (3.2), such as power consumption induced by the series resistance R_s . In our calculation, the possible series capacitance (such as the residual capacitance because of imperfect

metal contacts) and the junction capacitance C_j are not included for simplification because C_j dominates in the reverse-biased and low injection forward-biased operation.

For low frequencies $\omega\tau \ll 1$, the voltage drop across a PN junction is $V_D = \frac{Q}{C_D} = \frac{2Q}{G_0\tau}$. Q is the charge stored in the diffusion capacitor. The total RF power consists of the power dissipated in the PN junction $P_{RF,D}$ and in the series resistance P_{RF,R_s} ,

$$P_{RF} = P_{RF,D} + P_{RF,R_s} \quad (3.3)$$

where,

$$P_{RF,D} = \frac{1}{2}V_D^2G_D = \frac{1}{2}V_D^2G_0 = \frac{Q^2}{\tau^2} \cdot \frac{2}{G_0} \quad (3.4)$$

$$P_{RF,R_s} = \frac{1}{2}|I_D|^2R_s = \frac{1}{2}|V_D(j\omega C_D + G_D)|^2R_s = \frac{1}{2}V_D^2G_0 \cdot G_0R_s = \frac{Q^2}{\tau^2} \cdot 2R_s \quad (3.5)$$

For high frequencies $\omega\tau \gg 1$, $V_D = \frac{Q}{C_D} = \frac{\sqrt{2}Q}{G_0} \sqrt{\frac{\omega}{\tau}}$, then $P_{RF,D}$ and P_{RF,R_s} are

$$P_{RF,D} = \frac{1}{2}V_D^2G_D = \frac{Q^2\omega^{\frac{3}{2}}}{\sqrt{2\tau}G_0} = \frac{N_A Q^2}{\sqrt{2qA\sqrt{D_n}n_i^2}}\omega^{\frac{3}{2}} \quad (3.6)$$

$$P_{RF,R_s} = \frac{1}{2}|I_D|^2R_s = \frac{1}{2}|V_D(j\omega C_D + G_D)|^2R_s = \frac{1}{2}V_D^2G_D \cdot 2G_DR_s = Q^2R_s\omega^2 \quad (3.7)$$

The ratios of the high frequency power to the low frequency power for diodes and R_s are, respectively,

$$\frac{P_{RF,D}^{high.freq}}{P_{RF,D}^{low.freq}} = \frac{(\omega\tau)^{\frac{3}{2}}}{2\sqrt{2}} \quad (3.8)$$

$$\frac{P_{RF,R_s}^{high.freq}}{P_{RF,R_s}^{low.freq}} = \frac{(\omega\tau)^2}{2} \quad (3.9)$$

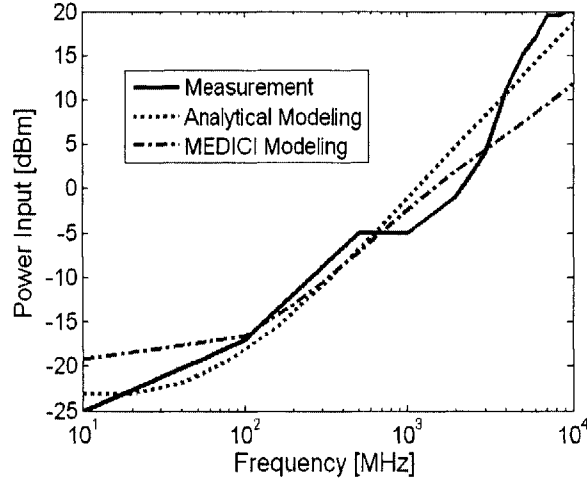


Figure 3-10: Power consumption of a 500- μm -long Mach-Zehnder interferometer in forward bias with 25% modulation depth: measurement (solid), result from analytic modeling (dash) and MEDICI numerical modeling (dash-dotted).

The ratio of the RF power consumed by series resistor to the RF power consumed by diode is

$$\frac{P_{RF.Rs}^{high.freq}}{P_{RF.D}^{high.freq}} = \sqrt{2}G_0R_s(\omega\tau)^{0.5} \quad (3.10)$$

In the low frequency regime $\omega\tau \ll 1$, equations (3.4) and (3.5) give $P_{RF.D}/P_{RF.Rs} = 1/(G_0R_s)$. In the high frequency regime $\omega\tau \gg 1$, equations (3.8) and (3.9) give $P_{RF.D} \propto \omega^{3/2}$, and $P_{RF.Rs} \propto \omega^2$ RF power dissipated in R_s increases faster than in PN junction in terms of frequency. More RF power is consumed at high frequencies $\omega\tau \gg 1$ because the ratio $P_{RF.Rs}/P_{RF.D} = \sqrt{2}G_0R_s(\omega\tau)^{0.5}$ can be much larger than one, where $G_0 \sim I/(kT/q) \simeq 1\text{mA}/25\text{mV} = 0.04\Omega$ and $R_s \sim 50\Omega$, as indicated in Fig.3-8. Since the series resistor may consume more RF power than the diode at high frequencies, reducing the series resistance is crucial for the reduction of RF power consumption.

The power scaling model in the PN junction is used to fit the experimental results of a PiN diode optical modulator. The RF power input is plotted as a function of frequency in Fig.3-10. In the analytical modeling, the parameters are chosen to be:

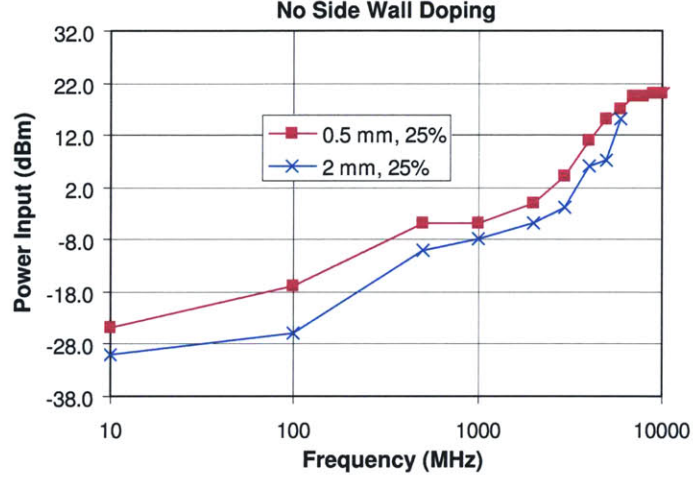


Figure 3-11: Measured RF power input in terms of driving frequency for phase shifters with length 0.5mm (square line) and 2mm (cross line). The measurement was done with the modulation depth to be 25% for both modulators.

lifetime $\tau = 5ns$, series resistance $R_s = 57\Omega$, junction capacitance $6.54pF$, DC capacitance $C_0 = 3.27pF$, DC conductance $G_0 = 0.00131S$. Most of these data is extracted from IV-curve measurement in Fig.3-8. The measurement and modeling both give a power increase by 55dBm for achieving the same modulation from frequency 10MHz to 10GHz. The numerical simulations with MEDICI give only 30dBm increase.

Based on the analytical power scaling theory above, we propose three strategies to reduce the RF power. The first strategy is to reduce the series resistance. This method is efficient at high frequencies because most of the RF power is consumed by the series resistance, i.e., $P_{RF.Rs}/P_{RF.D} = \sqrt{2}G_0R_s(\omega\tau)^{0.5} > 1$ at high frequencies. Improving the metal contacts and reducing the contact resistance are crucial for reducing RF power consumption.

The second strategy is to use a resonator with an enhancement factor n . Compared with a modulator without the resonator, the same phase change can be achieved with Q/n charge, then V is n times smaller by keeping C_D and G_D the same. Therefore, $P_{RF} = \frac{1}{2}|V|^2G_D = \frac{1}{2}(Q/C_D)^2G_D$ is n^2 times smaller.

Thirdly, increasing the length of the diode can decrease the power consumption. The RF power consumption in a PN diode can be expressed as $P_{RF.D} = \frac{1}{2}|V|^2G_D =$

$\frac{1}{2}(Q/C_D)^2 G_D$. The capacitance C_D and the conductance G_D increase by a factor of n if the diode length is increased n times. Therefore, the RF power in the PN diode is reduced n times while keeping the charge Q constant. The measured RF power of two modulators with different length has confirmed this prediction, as shown in Fig.3-11. The modulator with a length of 0.5mm needs about 2-4 times more RF power than the one with a length of 2mm for frequencies below 10GHz.

Modulation Efficiency

Before discussing the modulation efficiency, we investigate the necessary amount of charge to achieve a π phase shift. The total charge to achieve a phase change $\Delta\phi = \frac{2\pi}{\lambda} \Delta n L = \pi$ in a phase shifter (the refractive index change Δn is proportional to the carrier change $\Delta n = C \cdot \Delta n_e$) can be estimated as

$$Q_\pi = \Delta n_e q \cdot LA = \frac{1}{2} \frac{\lambda A}{C} q = 10^8 q \quad (3.11)$$

where the wavelength $\lambda = 1.55\mu m$, a length L and cross-section $A = 0.45\mu m \times 0.25\mu m$ are used, $q = 1.6 \times 10^{19} C$ is the electron charge, and $C = 8.8 \times 10^{-22} cm^3$ is the plasma effect coefficient for electrons. Here, only electrons are considered for simplification. One-third of the above Q_π is necessary if equal number of holes and electrons are injected because holes induce a 2 times larger refractive index change at a given carrier injection of around $\Delta N_{e/h} = 10^{17}/cm^3$ according to equation (2.16). It can be seen that 10^8 electrons are necessary to obtain a π phase change for a cross-section $A = 0.25\mu m \times 0.45\mu m$. If $V = 1V$ is used, the capacitance $C = Q_\pi/V = 10pF$ is necessary. The current can be estimated as $I_\pi = Q_\pi/\tau = 16mA$ for a carrier lifetime of $\tau = 1ns$.

The modulation efficiency can be quantified by a figure of merit. The figure of merit is defined as the product of voltage and modulator length, $V_\pi \cdot L$. Given the diffusion capacitance to be $G_0\tau/2$ at low frequencies $\omega\tau \ll 1$ and $\frac{G_0}{\sqrt{2}}\sqrt{\frac{\tau}{\omega}}$ at high frequencies $\omega\tau \gg 1$, according to equation (2.34), combined with $Q_\pi = V_\pi \cdot C_D$ and

$G_0 = qA_D \frac{n_i^2 D_n}{N_A \sqrt{\tau_n D_n}} \frac{e^{V_0/(KT/q)}}{KT/q}$, the figure of merit can be evaluated as

$$V_\pi \cdot L = \frac{\sqrt{2} Q_\pi N_A}{qh \sqrt{D_n n_i^2}} \cdot \frac{\sqrt{2}}{\sqrt{\tau}} \cdot \frac{KT/q}{e^{V_0/(KT/q)}} \quad (3.12)$$

for $\omega\tau \ll 1$ and

$$V_\pi \cdot L = \frac{\sqrt{2} Q_\pi N_A}{qh \sqrt{D_n n_i^2}} \cdot \frac{1}{\sqrt{\omega}} \cdot \frac{KT/q}{e^{V_0/(KT/q)}} \quad (3.13)$$

for $\omega\tau \gg 1$ with the n^+p diode cross-section $A_D = L \times h$, L and h are the n^+p diode width and thickness, respectively. From equations (3.12) and (3.13), it can be seen that the figure of merit $V_\pi \cdot L$ is decreased by reducing the high doping N_A or increasing the thickness h , which is equivalent to an increase in the capacitance. This explains that structure C, shown in Fig.3-1(b), has lower $V_\pi \cdot L$ than structure A because structure C has larger h than structure A, which will further be discussed later in this subsection. Note that the junction capacitance is not considered because the free carriers injected into the junction capacitor is negligible compared with the carriers injected into the diffusion capacitor for forward bias.

The DC $V_\pi \cdot L$ is readily estimated from the measured I-V curves in Fig.3-8. The total charge in the PIN diode is $Q \simeq I_{DC} \cdot \tau = 10mA \cdot 1ns = 10pC$ at the bias point of $I_{DC} = 10mA$ and $V_{DC} = 2V$, where the carrier lifetime is on the order of $\tau = 500ps - 1ns$ in these fabricated devices. The length is expressed as $L_\pi = \frac{\lambda}{2\Delta n}$ for a phase shift of $\Delta\phi = 2\pi\Delta n L/\lambda = \pi$ and therefore the DC figure of merit can be extracted,

$$V_\pi \cdot L = V_{DC} \frac{\lambda}{2\Delta n} = V_{DC} \frac{\lambda}{2C\Delta n_{e/h}} = V_{DC} \frac{\lambda}{2C \frac{Q/e}{500\mu m \cdot 500nm \cdot 240nm}} = 0.037V \cdot cm \quad (3.14)$$

where plasma dispersion effect coefficient is $C = 4 \times 10^{-21}$. This is close to the measured value $V_\pi \cdot L = 0.02V \cdot cm$.

At a speed of 10GHz, the RF power was measured and the figure of merit can be extracted. Only 100mW is necessary to achieve a 25% modulation depth and it is

Table 3.1: Modulation efficiency at 10GHz

	measurement	modeling
$R_s = 0\Omega$		$P_{RF} = 8.8mW$ $V_\pi \cdot L = 0.18V \cdot cm$
$R_s = 10\Omega$	$P_{RF} = 100mW$ $V_\pi \cdot L = 0.28V \cdot cm$	$P_{RF} = 15mW$ $V_\pi \cdot L = 0.31V \cdot cm$
$R_s = 50\Omega$	$P_{RF} = 100mW$ $V_\pi \cdot L = 0.84V \cdot cm$	$P_{RF} = 77mW$ $V_\pi \cdot L = 0.81V \cdot cm$

Note: the RF power is for 25% modulation depth.

believed that most RF power is consumed by the series resistance, as stated in the discussion of the RF power consumption scaling earlier in this section. The results are illustrated in column 2 of Table 3.1. If the series resistance of the diode is assumed to be 10Ω , the measured RF power $P_{RF} = V^2/2R = 100mW$ leads to $V = 1.4V$ for 25% modulation depth which corresponds to approximately 0.25π phase shift. Therefore, the figure of merit is $V_\pi \cdot L = 1.4V \times 500\mu m \times \pi/0.25\pi = 0.28V \cdot cm$. If assuming $R_s = 50\Omega$, the figure of merit $V_\pi \cdot L = 0.84V \cdot cm$ can be extracted.

The MEDICI numerical large-signal transient analysis is used to calculate the RF power and $V_\pi \cdot L$ of the PIN diode with sidewall doping at a speed of 10GHz. In the modeling, the DC bias voltage $0.75V$ is applied to the PIN diode, the minority carrier lifetime is $\tau = 1ns$, the phase shift length is set to $500\mu m$ and the RF voltage is tuned so that the modulation depth is around 25%. If the series resistance is set to $R_s = 10\Omega$ and the total RF voltage $V_{pp} = 1.7V$ is employed, the change of average electron density and hole density $\Delta n_{e/h} = 10^{17}/cm^3$ and RF power $P_{RF} = 15mW$ are obtained. This amount of carrier density change induces a phase shift of 0.2751π and thus a figure of merit of $V_\pi \cdot L = 1.7V \times 500\mu m \times \pi/0.2751\pi = 0.31V \cdot cm$. By setting $R_s = 50\Omega$, we extract $V_\pi \cdot L = 0.81V \cdot cm$ and $P_{RF} = 40mW$ from simulations for 25% modulation depth. Setting $R_s = 0$, we find $V_\pi \cdot L = 0.18V \cdot cm$ and $P_{RF} = 8.8mW$ for 25% modulation depth. It can be seen that reducing series resistance is important for decreasing the RF power and $V_\pi \cdot L$. It's believed that the series resistance has enough room for reduction, as well as $V_\pi \cdot L$ because the series resistance in the fabricated

Structures	Modulation depth @ 5 GHz	RF Power (mW)
 (Structure A)	13%	100
 (Structure C, wd=50nm)	43 %	100
 (Structure C, wd=0)	59 %	100

Figure 3-12: Measured modulation depth of different structures by applying the same RF power.

devices is larger than $10\Omega \times 500\mu m \times 10\mu m = 5 \times 10^{-4}\Omega \cdot cm^2$, about three orders of magnitude larger than the series resistance $10^{-7}\Omega \cdot cm^2$ reported in [38].

The MEDICI numerical large-signal transient analysis has also been used to calculate the RF power and $V_\pi \cdot L$ of the PIN diode without sidewall doping. The modeling parameters are the same as the modeling on PIN diodes with sidewall doping. The change of the average electron density and hole density is $\Delta n_{e/h} = 0.3 \times 10^{17}/cm^3$, which induces a phase shift of 0.1π and a figure of merit of $0.8V \cdot cm$ if assuming $R_s = 10\Omega$. It can be seen that the modulation efficiency of a PIN diode with sidewall doping is around 3 times better than the one without sidewall doping.

The modulation efficiency for different structures is studied by measuring the modulation depth for the same RF power, as shown in Fig.3-12. The modulation efficiency of structure C with $w_d = 50nm$ is slightly larger than structure C with $w_d = 0$ because the former structure has more overlap between carrier change with optical mode at the sidewalls. The modulation efficiency of structure C with $w_d = 50nm$ is about 5 times better than structure A because the former structure has larger capacitance. This observation confirms the numerical modeling in the above paragraph and in Fig.3-5.

In a short summary, the modulator based on the forward-biased PIN diode has high modulation efficiency, needs very low drive voltage and has extremely low power consumption in the low frequency regime. The measurement shows that the DC bias

is around 1-2 V with 5-10 mA current. Around 75% modulation depth is obtained at low freq with RF voltage $V_{RF} = 0.1V$ and $P_{RF} = 1mW$. A very low figure of merit $V_{\pi} \cdot L = 0.02V \cdot cm$ illustrates the high modulation efficiency. At a frequency of 10GHz, a RF power consumption of 100mW for 25% modulation depth and a figure of merit of $V_{\pi} \cdot L = 0.28V \cdot cm$ are achieved.

Frequency Response with and without Pre-Compensation Scheme

The frequency response of modulators based on forward-biased PIN diodes without pre-compensation scheme are investigated analytically, numerically and experimentally. If keeping the applied RF voltage constant for all frequencies, a analytical derivation leads to the frequency response S_{21} of a Mach-zehnder interferometer [33]

$$|S_{21}| \propto Q^2 = |V_{RF,D}(C_j + C_D)|^2 = \frac{|V_{RF}(C_j + C_D)|^2}{(R_s G_D + 1)^2 + (R_s \omega(C_j + C_D))^2} \quad (3.15)$$

where $V_{RF,D}$ is the RF voltage applied to the PIN diode, V_{RF} is the total voltage applied to the PIN diode and the series resistance, and Q is the charge stored in the diode capacitor. Here, the output frequency response of a Mach-Zehnder interferometer is biased at $\pi/2$ where the second harmonic term diminishes, and thus is proportional to the square of refractive index change. The refractive index change is approximately proportional to the injected charge Q .

In Fig.3-13(a) the Mach-Zehnder transmission S_{21} is plotted as a function of the frequency for modulators based on PIN diodes. By keeping the voltage input constant, the transmission characteristics S_{21} of the forward-biased modulator is obtained analytically, numerically and experimentally. The numerical modeling simulates a modulator with the minority carrier lifetime $\tau = 1ns$ and the series resistor $R_s = 200\Omega$. The experimental device was annealed for 20 seconds in vacuum 450°C. All fabricated devices studied in Fig.3-13 have no sidewall doping and silicon ion implantation. The measured $|S_{21}|$ rolls off in terms of frequency a little slower than the analytical and numerical response.

In the forward bias regime, the electrical bandwidth f_{3dB} is ~ 200 MHz. In this

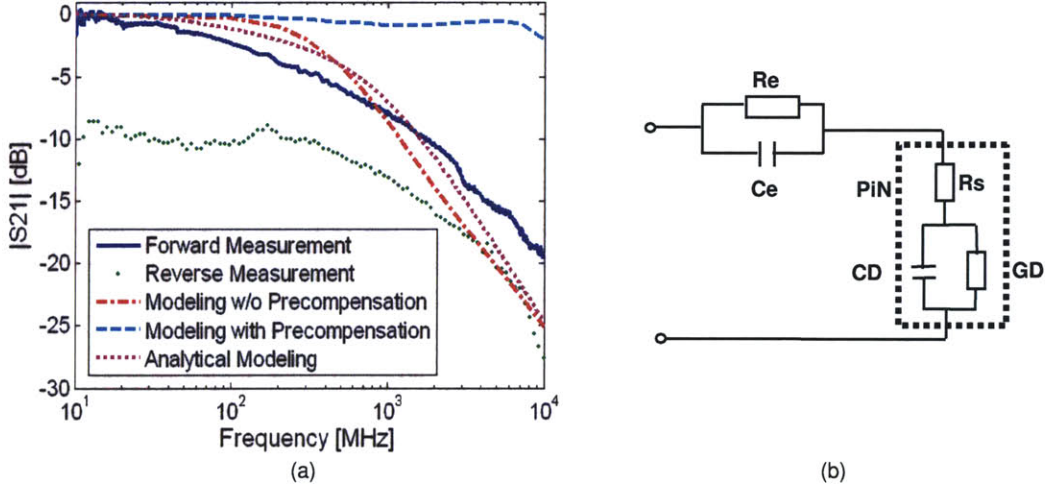


Figure 3-13: (a) Frequency response of Mach-Zehnder modulators: measurement under forward biasing (solid); MEDICI simulation without pre-compensation circuit (dash-dotted); MEDICI modeling with pre-compensation circuit (dashed); measurement of reverse biased device (dotted); analytical modeling under forward biasing (short dashed), (b) PIN diode circuit model with a simple RC pre-compensation circuit. All forward-biased diodes are $250\text{-}\mu\text{m}$ -long and reverse-biased diode is 6-mm long.

mode of operation the device bandwidth is carrier lifetime dependent. The lifetime of this device is expected to be $\tau = 1\text{ns}$, so that $f_{3dB} = 1/2\pi\tau = 160\text{ MHz}$, which is consistent with the measurement. The optimum bias voltage was $1\text{-}2\text{ V}$ with a current of $\sim 10\text{ mA}$ in the forward bias regime. The measured rf-power consumption for a $500\text{-}\mu\text{m}$ -long device for 25% modulation depth is shown in Fig.3-10 together with the result obtained from the analytical modeling. At a low frequency of 10 MHz , only $1\mu\text{W}$ of RF drive power is necessary to achieve 25% modulation depth. At a frequency of 10GHz , 100mW of RF drive power is needed. Fig.3-14(a) illustrates the detected optical output wave in terms of time. Fig.3-14(b) shows the modulation depth and the corresponding RF power input and operation frequencies. The relatively low electrical power dissipation $\sim 100\text{mW}$ at 10GHz enables to use a pre-compensation technique to increase the modulator bandwidth. This is demonstrated by simulations. A high pass filter consisting of a parallel resistor and capacitor is series with the PIN-diode, as shown in Fig.3-13(b), where $R_e = 4000\Omega$, $C_e = 90\text{fF}$, $R_s = 200\Omega$,

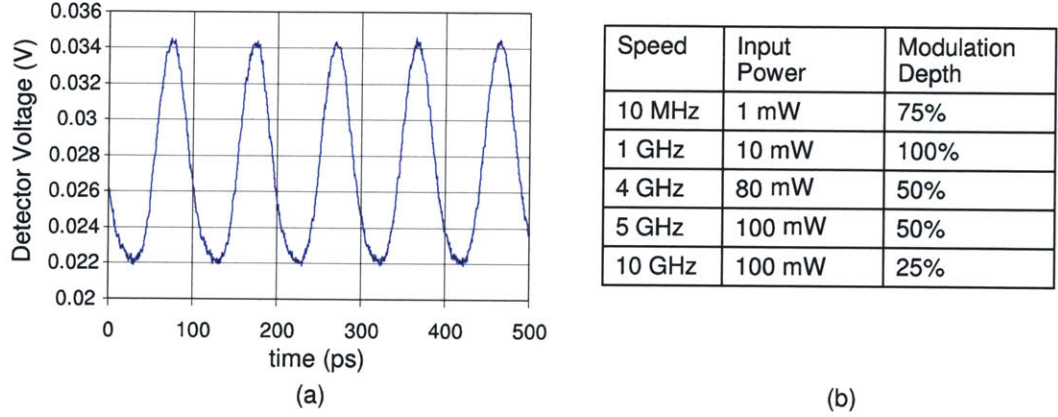


Figure 3-14: (a) Measured modulation at 10GHz without pre-compensation. The modulation depth is 25%, (b) measured modulation depth with different RF power and frequencies. The length is $500\mu m$.

$\tau = 1ns$, $V_{DC} = 10.5V$, $I_{DC} = 2.5mA$ and $L = 250\mu m$. The high pass filter in the modulator driver circuit extends the bandwidth of the modulator from 100 MHz to 10GHz as shown in Fig.3-13(a) with the dashed curves. A similar high pass filter was used to extend the frequency response to a higher bandwidth experimentally. The frequency response of the PiN diode modulator without a filter indicates a bandwidth of ~ 200 MHz, as shown in Fig.3-15. Using a high pass filter, the frequency response level is raised in the high frequency range 1-5GHz. A better designed high pass filter is necessary to flatten the frequency response between 100MHz and 1GHz. The preliminary experimental results prove the feasibility of the pre-compensation scheme.

3.1.3 Modulator Based on Reverse-Biased Diodes

Numerical Modeling

A silicon electro-optic modulator based on reverse bias can potentially achieve the high-speed modulation because a high electric field, such as $10^5 V/cm$, can be applied to the reverse-biased PIN structure such that carriers travel close to the saturation velocity $V_s = 10^7 cm/s$ in silicon. The modulation speed is inverse to the transit time $\tau_t = W/v$. Here, W and v are the travelling distance and the carrier drift

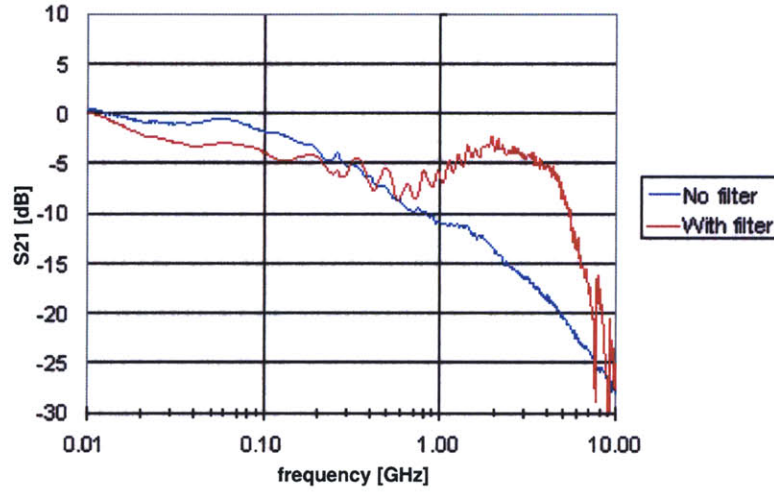


Figure 3-15: Measured frequency response of the Mach-Zehnder modulator. The plots are normalized to be 0dB at the lowest frequency [46].

velocity, respectively. For example, the transit time is around $\tau_t = W/v_s = 10ps$ for a $W = 500nm$ carrier travelling distance at the saturation velocity. This corresponds to a 3dB corner frequency $f_{3dB} = 2.4/2\pi\tau_s = 80\text{ GHz}$ [37].

A silicon modulator based on reverse-biased PIN diodes has been designed. It employs structure C as illustrated in Fig.3-1. The ribs are highly doped with $10^{19}/cm^3$ to reduce the contact resistance. The rib sections close to the waveguide and the waveguide sidewalls are doped with $10^{18}/cm^3$ to reduce the intrinsic region but without introducing much optical loss. An additional loss of 3dB/cm is induced by the doping of $10^{18}/cm^3$. The frequency response of the modulation from the plasma effect demonstrates a 3dB corner frequency of 80GHz as shown in Fig.3-16(a), very close to the theoretical expectation. The refractive index change due to the thermal effect is over 40dB lower than that due to the plasma effect. The 80GHz bandwidth is the intrinsic bandwidth. The overall bandwidth will depend on the total RC delay of the driving circuits and the phase matching between RF-drive voltage and optical group velocity.

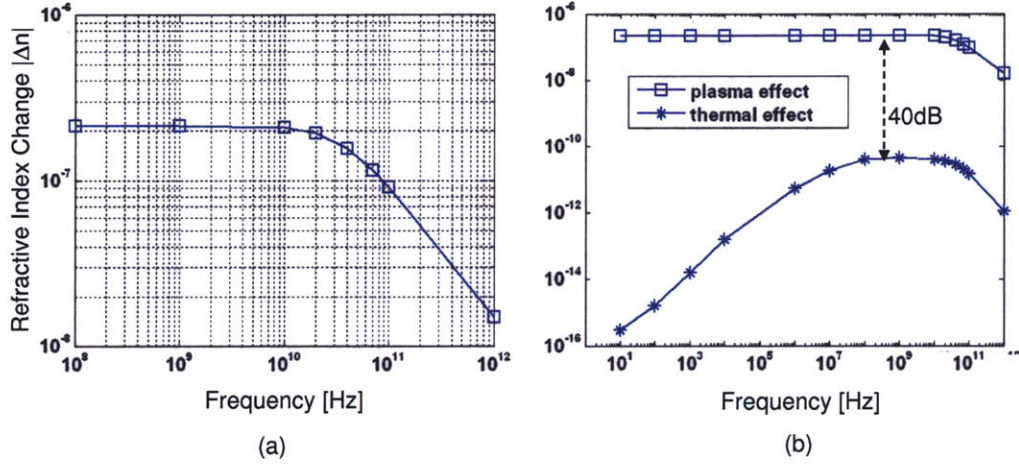


Figure 3-16: (a) Refractive index change due to plasma effect, (b) refractive index change due to plasma and thermal effects. The series resistance $R_s = 4\Omega$, the length is $L = 6mm$.

Fabrication and Measurement

The electrical to optical frequency response S_{21} of a Mach-Zehnder modulator with 250- μm -long PIN-sections in forward bias operation and a 6-mm-long device in reverse bias operation has been measured (see Fig.3-13(a)). Under reverse bias the modulator is expected to have broad bandwidth because it is drift-velocity dependent; $f_{3dB} = 2.4/2\tau_{drift} = 80$ GHz assuming the carriers drift at the saturation velocity $10^7 cm/s$. However, the measured f_{3dB} is only ~ 1 -2GHz. This indicates firstly that the current experimental setup is RC delay limited. Second, due to fabrication problems, the metal contacts showed Schottky diode characteristics rather than ohmic characteristics. The induced large RC delay reduces the modulation bandwidth. Therefore, the measured bandwidth is less than estimated. The response of the device under reverse bias is much weaker compared to that under forward bias, thus it requires a longer arm length of 6.0 mm. The optimum bias voltage was ~ -9 V with a current less than $2 \mu A$ for reverse bias operation.

Figure of Merit (FOM) $V_\pi \cdot L$

The depletion region width of a N^+P junction can be calculated as [33]

$$W = \sqrt{\frac{2\epsilon}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right) (V_{bi} - V_A)} \simeq \sqrt{\frac{2\epsilon}{q} \frac{1}{N_A} (V_{bi} - V_A)} \quad (3.16)$$

where N_A , N_D , V_{bi} and V_A are the acceptor density, the donor density, the build-in voltage and the applied voltage, respectively. $N_A \ll N_D$ is assumed for simplification. The phase change induced by the majority carriers has to be π to calculate $V_\pi \cdot L$

$$\Delta\phi = \frac{2\pi}{\lambda} L \Delta n = \frac{2\pi}{\lambda} L [C N_A F] = \pi \quad (3.17)$$

where L , C and F are the diode length, the plasma dispersion effect coefficient and the overlap factor between the carrier change and the optical mode, respectively. The DC figure of merit $V_\pi \cdot L$ of a PN-reverse modulator can be extracted from (3.16) and (3.17),

$$V_\pi \cdot L = (V_{bi} - V_A)_\pi \cdot L = \frac{W^2 \lambda e}{2CF \cdot 2\epsilon} \quad (3.18)$$

For $W = 450nm$, the DC figure of merit is $V_\pi \cdot L = \frac{11.58}{F} V \cdot cm$ if donors are used to induced refractive index change, and $V_\pi \cdot L = \frac{5.79}{F} V \cdot cm$ if accepters are used because holes can induce twice larger index change than electrons. The figure of merit is $13V \cdot cm$ for the former and $6.5V \cdot cm$ for the latter, respectively, since the factor F is around 0.9 for high-index-contrast waveguide. A figure of merit of $V_\pi \cdot L = 4V \cdot cm$ at low frequencies has been demonstrated experimentally in Ref.[14].

From the figure of merit expression (3.18), the width W and the overlap factor F are the only two parameters which can be optimized to reduce the figure of merit. The Factor F has no much room to optimize because it's 0.8–0.9 already while the maximum F can only be 1. Based on the modulator discussed in reference [14], the effective width is $W \simeq 530nm$ and the figure of merit is $V_\pi \cdot L = 4V \cdot cm$ at frequency $f = 10$ GHz. If the width is reduced to $W = 250nm$, the figure of merit

can be reduced to around $V_\pi \cdot L = 1V \cdot cm$. $V_\pi \cdot L$ as low as $1V \cdot cm$ is desired because a modulator with $1V \cdot cm$ FOM can achieve a phase shift of $\Delta\phi = 0.015\pi$ (extinction ratio ER=9.7dB) with a modulator length of $L = 1mm$ and a voltage of $V_{RF} = 1.5V$ with push-pull method. Such extinction ratio is enough for most optical communication applications, and the relatively short length and CMOS-compatible voltage are desired in CMOS large-scale integration. However, reducing width W to around $250nm$ may be challenging in fabrication and cause additional optical loss because high doped regions may be too close to optical modes.

3.1.4 Modulator Based on Metal-Oxide-Semiconductor (MOS)

Figure of Merit (FOM) $V_\pi \cdot L$

The injected electrons Δn_e and holes Δn_h are related to the drive voltage in metal-oxide-semiconductor (MOS) structure by [48]:

$$\Delta n_e = \Delta n_h = \frac{\epsilon_0 \epsilon_r}{et_{ox}t} [V_D - V_{FB}] \quad (3.19)$$

where ϵ_0 and ϵ_r are the vacuum permittivity and the low-frequency relative permittivity of the oxide, e is the electron charge, V_D is the drive voltage, V_{FB} is the flat band voltage of the MOS structure, t_{ox} and t are the gate oxide thickness and the effective accumulated charge thickness, respectively. In this analysis, a flat band voltage of $V_{FB} = 1.25V$ and an effective accumulated charge thickness of $t = 10nm$ are assumed [48]. The phase change induced by carriers has to be π to calculate $V_\pi \cdot L$

$$\Delta\phi = \frac{2\pi}{\lambda} L \Delta n = \frac{2\pi}{\lambda} L [C \Delta n_{e/h} F] = \pi \quad (3.20)$$

where Δn , $C = 10^{-21}$ and L are the refractive index change, the plasma effect coefficient and the active phase shifter length, respectively. Combine (3.19) and (3.20), the estimated DC figure of merit (FOM) $V_\pi \cdot L$ is

$$V_\pi \cdot L \simeq [V_D - V_{FB}]_\pi \cdot L = \frac{\lambda \cdot et_{ox}t}{2CF\epsilon_0\epsilon_r} = \frac{0.13V \cdot cm}{F} \quad (3.21)$$

Assuming the wavelength $\lambda = 1.55\mu m$, the oxide thickness $t_{ox} = 20nm$, the effective charge layer thickness $t = 10nm$, and the overlap factor F to be 0.01–0.1 in the MOS structure, we have the DC figure of merit of $V_{\pi} \cdot L = 1.3 - 13V \cdot cm$ which is consistent with the demonstrated $V_{\pi} \cdot L = 3.3V \cdot cm$ in [47]. An optimized design has demonstrated that the MOS structure can achieve $0.68V \cdot cm$ [48].

Performance in High-speed Operation

A MOS structure can be modeled as a series RC circuit. The AC current and the AC voltage across the capacitor of the series RC circuit can be written as

$$I = \frac{V_{tot}\omega C}{\sqrt{(\omega RC)^2 + 1}} \quad (3.22)$$

$$V_{C.RF} = \frac{V_{tot}}{\sqrt{(\omega RC)^2 + 1}} \quad (3.23)$$

where V_{tot} is the total applied voltage amplitude. In reference [47], the MOS structure is divided to 11 sections. Each section has a resistance of $R = 6.5\Omega$, a capacitance of $C = 26.5pF$ and a length of $L = 315\mu m$ at a data rate of $6Gbps$. A phase change of 0.15π was obtained with a total voltage of $V_{tot1} = 1.4V$ in the modulator with a length of $L = 11 \times 315\mu m = 3.45mm$ and a data rate of $6Gbps$ which roughly corresponds to $3GHz$.

For a frequency of $10GHz$, we compute the necessary total voltage to achieve a phase shift of π . The total voltage for a phase shift of 0.15π at $10GHz$ can be calculated by using the same R and C at $3GHz$,

$$\begin{aligned} V_{C.RF}(3GHz) &= \frac{V_{tot1}(0.15\pi)}{\sqrt{(2\pi \cdot 3GHz \cdot RC)^2 + 1}} \\ &= V_{C.RF}(10GHz) = \frac{V_{tot2}(0.15\pi)}{\sqrt{(2\pi \cdot 10GHz \cdot RC)^2 + 1}} \end{aligned} \quad (3.24)$$

which leads to $V_{tot2}(0.15\pi) = 1.88V$ at $10GHz$. The total voltage for a phase shift

of π is $V_{tot2}(\pi) = V_{tot2}(0.15\pi) \times \frac{\pi}{0.15\pi} = 12.5V$ because the voltage is approximately proportional to the phase shift. A figure of merit of $V_\pi \cdot L = 12.5V \times 3.45mm = 4.3V \cdot cm$ is achieved. The RF power is computed as

$$P_{RF} = \frac{R}{2} I^2 = \frac{R}{2} [V_{tot2}(\pi) \cdot \frac{\omega C}{\sqrt{(j\omega RC)^2 + 1}}]^2 \times \frac{3.45mm}{315\mu m} = 64.5W \quad (3.25)$$

for $\Delta\phi = \pi$, $L = 3.45mm$ and $f = 10$ GHz. The estimated 64.5W RF power is too high for most CMOS technology. The large resistor inside the MOS structure consumes most of the electrical power.

From equation (3.21), the parameters which can be optimized to reduce the $V_\pi \cdot L$ are the oxide thickness t_{ox} , the charge effective thickness t and the overlap factor F . Reducing t_{ox} can reduce $V_\pi \cdot L$ but also reduce the operation speed because the speed of MOS modulator is limited by the RC delay. The factor F can be increased by optimizing the geometry of the structure, but F can not be enlarged much because the effective carrier thickness t , on the order of 10nm, is too thin and the overlap between optical mode and carrier change. In Ref.[48], an optimized design with $L=1.25mm$, N doping $3 \times 10^{17}/cm^3$ and $t_{ox} = 6nm$ achieved a bandwidth of $f_{3dB} = 13$ GHz, an optical loss of around 20dB/cm and a total loss of 1.3dB. The figure of merit is $V_\pi \cdot L = 0.75V \cdot cm$ for DC characteristics and $V_\pi \cdot L = 1V \cdot cm$ for $f_{3dB} = 13$ GHz.

3.1.5 Comparison of Forward-Biased PIN, Reverse-Biased Diodes and MOS Structures

Based on the above detailed analysis on these three kinds of structures (forward-biased PIN, reverse-biased PIN/PN and metal-oxide-insulator(MOS)), their performance is summarized in Table 3.2. The forward-biased PIN structure with lifetime reduction reaches 34GHz and achieves a low figure of merit $V_\pi \cdot L = 0.6V \cdot cm$, while requires more DC power and generates much heat at low frequencies. The forward-biased PIN structure without lifetime reduction is able to achieve the lowest figure of merit $V_\pi \cdot L = 0.28V \cdot cm$ and work up to 10GHz with the aid of pre-compensation. The

Table 3.2: Comparison of forward-biased, reverse-biased PIN/PN diodes and MOS structures

Structure	P_{DC} [W]	P_{RF} @10GHz, $\Delta\phi = \pi$ [W]	$V_{\pi} \cdot L$ @10GHz [V · cm]	Thermal		Modulation speed [GHz]	Voltage [V]	Speed limited by	Insertion loss (dB)
				ΔT_{DC} [K]	ΔT_{AC} [K]				
Forward ¹	3.6W	~1W (L=6mm $\Delta V = 1V$)	0.6	50-100	Dominant $0 < f < 100MHz$	34	1-3	V_{DC} and driving circuits	3dB(injection and contact) (L=6mm)
Forward ²	~ 1mW	<1.6W ($L = 500\mu m$ $\Delta V < 5.6V$)	<0.28 (measured) ³ <0.31 (modeling)	small	small	10	~ 1V	driving circuits	10dB/cm (on-chip) (measured)
Reverse[14]	0	1.89W (L=5mm $\Delta V = 8.9V$)	4	0	small	>20	~5	driving circuits	7dB(on-chip) (L=3mm)
MOS[47]	0	64.5W (L=3.45mm, $\Delta V = 12.5V$)	4.3	0	small	10Gps	3-5V	driving circuits	10dB(on-chip) 9dB(coupling) (L=3.45mm)

¹ Forward-biased PIN structure with lifetime reduction² Forward-biased PIN structure without lifetime reduction³ Assume series resistance $R_s = 10\Omega$.

reverse-biased structure can reach very high speed, even up to 40-80GHz in theory but it is difficult to reduce $V_\pi \cdot L$ lower than $1V \cdot cm$ and the voltage is usually higher than the voltage provided by the CMOS technology. The last structure, metal-oxide-insulator(MOS), is hard to work up to 10GHz because large RF power is necessary.

3.2 Large-Signal Modeling of PIN Diodes Using SPICE

3.2.1 SPICE Model Implementation

In this section, the PIN diode is modeled with SPICE equivalent circuits so that the PIN SPICE model can be integrated with electronic driving circuits for circuit simulations.

The carrier distribution in the intrinsic region of PIN diodes is governed by the ambipolar diffusion equation

$$\frac{\partial^2 p}{\partial x^2} = \frac{p}{L_a^2} + \frac{1}{D_a} \frac{\partial p}{\partial t} \quad (3.26)$$

where $D_a = 2D_n D_p / (D_n + D_p)$ is the ambipolar coefficient, $L_a = \sqrt{D_a \tau_a}$ is the ambipolar diffusion length, and τ_a is carrier lifetime. The DC carrier distribution in the quasi-neutral intrinsic region can be derived as in Appendix A

$$n(x) = p(x) = p_0 \sinh(w/L_a) \left[\frac{\cosh((x - w/2)/L_a)}{\sinh(w/2L_a)} - B \frac{\sinh((x - w/2)/L_a)}{\cosh(w/2L_a)} \right] \quad (3.27)$$

where $B = (\mu_n - \mu_p) / (\mu_n + \mu_p)$. For silicon, $\mu_n / \mu_p = 3$. The carrier distribution diagram is shown in Fig.3-17.

The characteristics of P^+n and nN^+ junctions and the carrier distribution in equation (3.27) and in Fig.3-17 have been modeled with a three-stage equivalent circuit [49]. The correlation between the current $i_1(t)$ and the injected charge $Q_0(t)$ can be simulated with this circuit, as shown in equation (B.18). The detailed model implementation is described in Appendix B.

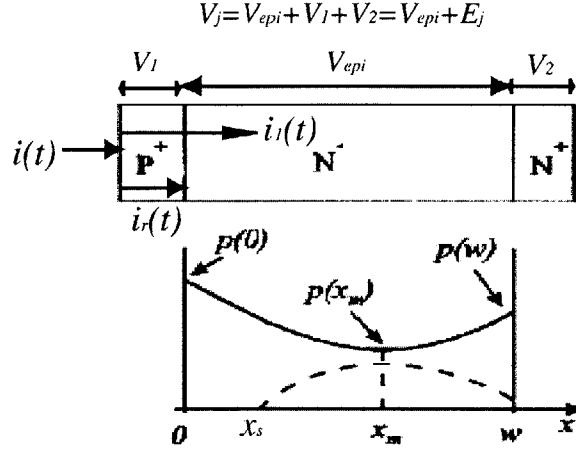


Figure 3-17: Carrier distribution in the base region during forward conduction (solid line) and reverse-recovery (dashed) [49].

The PIN SPICE model in Ref. [49] only works for DC or small-signal analysis, but not for large-signal analysis. In large-signal analysis, the injected carrier distribution is very different. The carrier distribution of DC/small-signal analysis has hyperbolic function shape as in Fig.3-18(a). The free hole density and electron density are approximately equal and form the quasi-neutral base-region. However, the large-signal input will drive the carrier distribution away from the quasi-neutral condition and makes it look like the distribution as shown in Fig.3-18(b). Because free hole density is not equal to free electron density, the equivalent circuit in Appendix B can not be used directly to model the correlation between input current and injected carriers.

Without the quasi-neutral condition, the current density J_p with carrier distribution p at the boundary $x = 0$ is written as,

$$J_p|_{x=0} = -qD_p \frac{\partial p}{\partial x} + qpE\mu_p \simeq -qD_p \frac{\partial(F_1 \cdot p)}{\partial x} \quad (3.28)$$

Under the quasi-neutral condition, the current density $J_{p'}$ with carrier distribution p' at the boundary $x = 0$ is written as,

$$J_{p'}|_{x=0} = -qD_p \frac{\partial(p')}{\partial x} \quad (3.29)$$

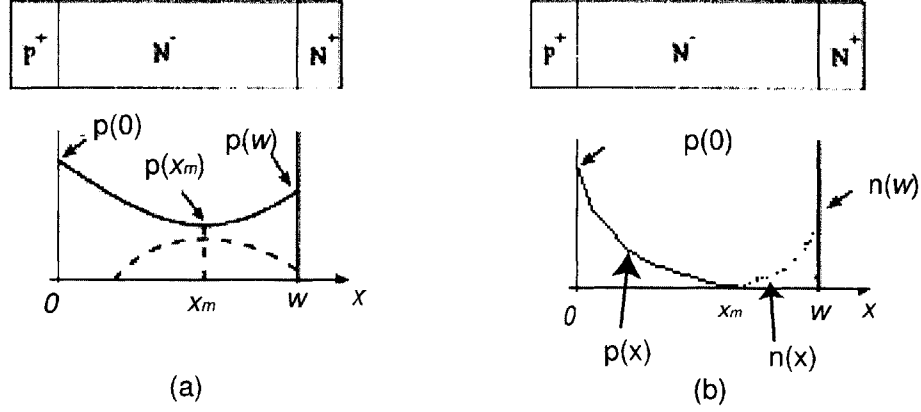


Figure 3-18: Carrier distribution in base-region for (a) DC/small-signal analysis and (b) large-signal analysis [49].

Here, we introduce a new factor F_1 to correlate the current density J_p with $J_{p'}$ at the boundary $x = 0$,

$$J_p(x = 0) = F_1 \cdot J_{p'}(x = 0) \quad (3.30)$$

The total injected carriers by these two currents are equal,

$$Q = A \int J_p(x = 0) dt = A \int F_1 \cdot J_{p'}(x = 0) dt \quad (3.31)$$

Therefore, the correlation between the current J_p and the injected density p under large-signal signal situation can be modeled with the current $J_{p'}$ and the carrier density p' with quasi-neutral condition. The transfer function between J_p and p of a PIN diode can be simulated with the same equivalent circuit for $J_{p'}$ and p' . The factor F_1 indicates how far the carrier distribution is away from quasi-neutral condition. It depends on the operating frequency and amplitude of the applied voltage or current source. It can be extracted from large-signal numerical modeling or experiments.

Table 3.3: SPICE model parameters of PIN diode

Parameters	Unit	Comments
I_S	A	Saturation current
N		Ideality factor
I_{KF}	A	High-injection knee current
P_{HI}	V	Build-in voltage
T_0	s	Transit time
I_E	A	Emitter recombination knee current
V_M	V	High-injection voltage drop on the base
R_{LIM}	Ω	Carrier-scattering series resistance
R_{EPI}	Ω	Base region resistance
L_{AM}		Forward-recovery coefficient
τ	s	Lifetime
V_{PT}	V	Reverse-recovery coefficient
R_{SC}	Ω	Reverse-recovery coefficient
*F1		Initial condition coefficient
*F2		Reverse-recovery coefficient

3.2.2 SPICE Model Results

The parameters I_S , N , I_{KF} , P_{HI} , T_0 , I_E , V_M , R_{LIM} , R_{EPI} , L_{AM} , τ , V_{PT} , R_{SC} , F_1 and F_2 are parameters for the diode model in Appendix C, which are indicated in Table.3.3. The parameters I_S , N , I_{KF} , I_E , V_M , R_{LIM} and R_{EPI} can be extracted from I-V curves. The parameters T_0 , τ , V_{PT} and R_{SC} are extracted from reverse-recovery simulations. The forward-recovery coefficient L_{AM} is extracted from forward-recovery simulations. The parameter P_{HI} doesn't contribute much in this model and can be set to 1V. The initial condition factor F_1 is obtained by a large-signal analysis simulation. The factor F_2 is extracted from reverse-recovery. The factor F_2 is added to the model used in reference [49] because HSPICE can't converge in simulating the circuits with the nonlinear element current source G_{rs1} [49]. The SPICE subcircuits of the PIN diode model are discussed in Appendix C.

The circuit simulation results of the PIN SPICE model has been compared with

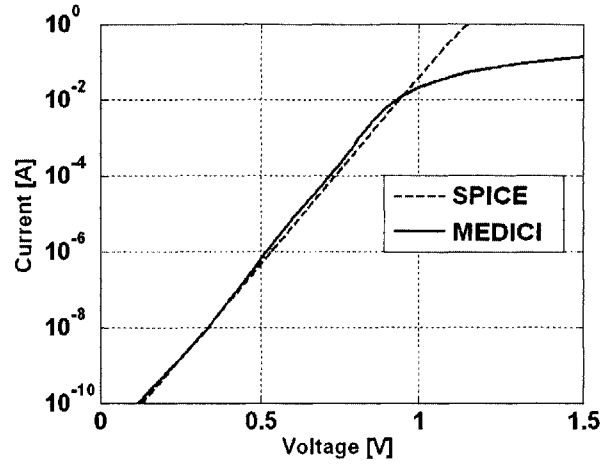


Figure 3-19: DC characteristics of MEDICI simulation (solid) and SPICE modeling (dashed).

the results from the 2D simulator MEDICI [45]. The simulated diode has structure A, shown in Fig.3-1(b). It has height $\times$ width = $240nm \times 520nm$ and two 50nm thick ribs on both sides highly doped to $10^{19}/cm^3$ for n-type and p-type, respectively. The high doping regions are 200nm away from the optical waveguide to minimize optical loss.

The DC and small-signal analysis have been conducted using the sub-circuit in Appendix C with the factors $F_1 = 1$ and $F_2 = 1$. The DC current-voltage characteristics is shown in Fig.3-19. For $V < 1V$, the SPICE model results match MEDICI numerical simulation results very well. For $V > 1V$, the heating mechanism in MEDICI simulation may be the reason to cause the I-V roll off faster than SPICE model because the heating leads to mobility reduction and diode current saturation. The heating mechanism was not included in the SPICE model. The small-signal analysis was done with a small voltage source of $V_{RF} = 0.01$. Using small-signal analysis, the capacitance and resistance properties at different frequencies and DC bias are calculated. Fig.3-20(a)(b) illustrate the total resistive and reactive impedance versus frequency at bias $V_{DC} = 0.5$. The results from MEDICI match those from SPICE well up to 5-10GHz. Fig.3-20(c) shows the calculated diffusion capacitance by sweeping the forward bias. The diffusion capacitance from MEDICI is close to those from

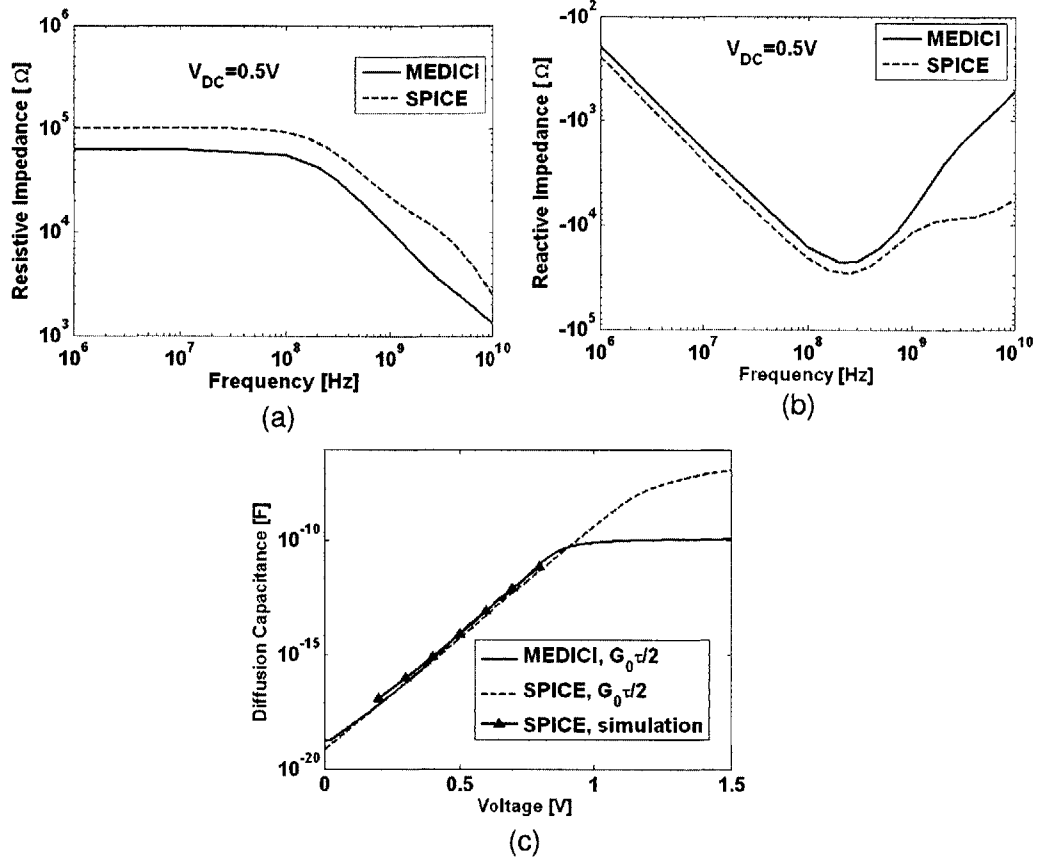


Figure 3-20: (a) Total resistance and (b) total reactive impedance versus frequency at forward bias $V_{DC} = 0.5V$, (c) diffusion capacitance versus forward bias at frequency $f = 100Hz$.

SPICE for $V < 1V$.

For digital applications, a pulse voltage source is used to simulate logic 1 and 0. The current and total carriers have been computed with MEDICI [45] and HSPICE [50] for random voltage square pulses. A detailed comparison can be found in Appendix B. The transient carrier change injected into the PIN diode can be obtained from the ladder subcircuit as shown in Fig.3-21. By reformatting, the subcircuit Fig.3-21(a) is equivalent to the subcircuit in Fig.3-21(b). The charges $q1$ and $q2$ in Fig.3-21(b) are the charges in the PIN intrinsic region as shown in Fig.3-21(c). The

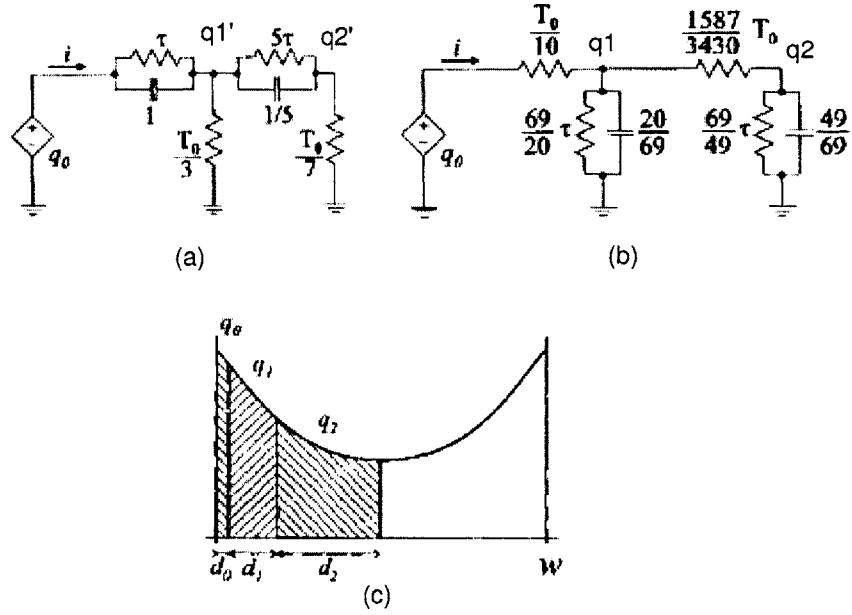


Figure 3-21: (a) 4th order PIN subcircuit, (b) another equivalent 4th order PIN subcircuit, (c) carrier distribution in PIN intrinsic region [49].

total charges are calculated as

$$Q = [q_1 \times \gamma + q_2 \times (1 - \gamma)] \times F_1/2 \quad (3.32)$$

where F_1 is the factor to evaluate the carrier distribution difference between steady-state and transient as shown in Fig.3-18(a). The factor 2 indicates the fact that the transient carriers are only about half of those in steady-state, see Fig.3-18(b). $\gamma = d_1/(w/2)$ is the ratio of d_1 to the half intrinsic width. Fig.3-22(a) shows the transient current from MEDICI numerical simulation (solid curve) and HSPICE simulation (dashed curve). Fig.3-22(b) illustrates the transient carriers computed with MEDICI modeling (solid curve) and transient carriers simulated with HSPICE and calculated from equation (3.32) (dashed curve). Note that the voltage in the subcircuits in Appendix C is equivalent to charge divided by carrier lifetime.

In Fig.3-22, the transient carriers increase to some large level when there are continuous "1" signals. The optical output is not a flat top signal as the input electrical signal. Since a long continuous "1" signal may be applied to the PIN diode

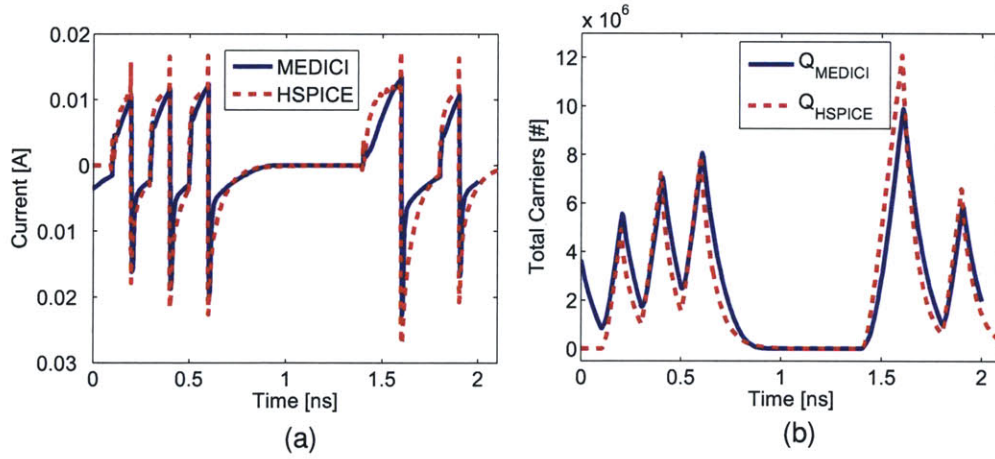


Figure 3-22: (a) Transient MEDICI current (solid) and HSPICE current (dashed), (b) transient MEDICI total carriers (solid) and HSPICE total carriers current (dashed). The total carriers include holes and electrons. The random signal is [0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 1, 0] with one bit period to be 100ps.

in some idle links, techniques are necessary to obtain constant optical output for the same electronic signal input.

3.3 Fabrication and Measurement Setup

3.3.1 Device Fabrication

The silicon modulator has been fabricated at Lincoln Laboratory with CMOS process. Fig.3-23 illustrates several key steps. Fig.3-23(a)(b) show how to make the high doping region to reduce the contact resistance. It's estimated that 10-20nm fabrication error on the distance between the high doping region and the waveguide does not seriously affect the device performance. However, a 5-10nm fabrication error can induce performance distortion if the sidewall doping is necessary, such as structure C shown in Fig.3-1(b), because the sidewall doping is designed to be around 50nm and close to the optical mode center. Fig.3-23(c) shows how to make the low doping (such as n-type $10^{15}/cm^3$) for forward-biased operation, medium doping (such as p-type $10^{17}/cm^3$) for reverse-biased operation and silicon ion implantation to introduce defects and control carrier lifetime. Fig.3-23(d) illustrates the final structure after

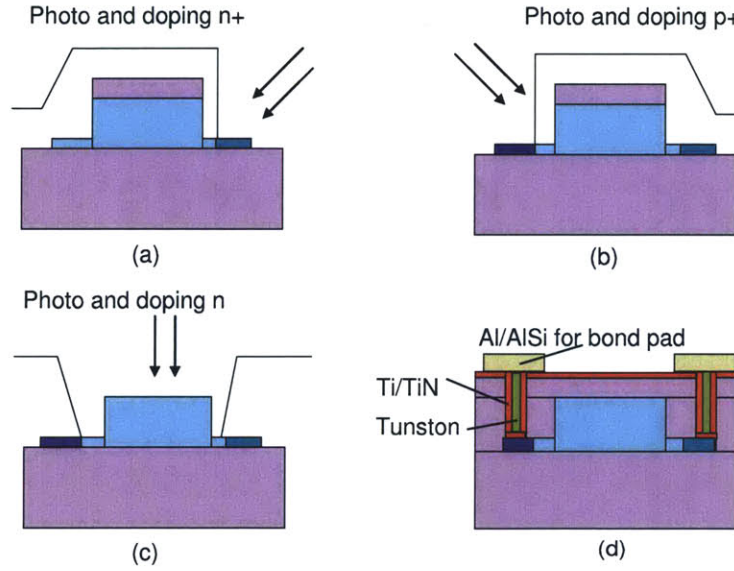


Figure 3-23: (a) high n-type doping, (b) high p-type doping, (c) low n-doping or silicon ion implantation, (d) final structure after process.

processing with metal.

The Mach-Zehnder modulator is shown in Fig.3-24(a). Y-junction is employed to split the light from the input and to combine it at the output. The SEM images of the fabricated waveguide are shown in Fig.3-24(b) and (c). The height of the highly doped contact regions is 38.7nm. The height of the waveguide is 276.2nm. The width at the top and bottom of the waveguide is 517.8nm and 606.3nm, respectively. This waveguide can generate the second TE mode, but this mode can not affect the waveguide's performance much because its effective index is a lot lower than those of the first TE mode and the first TM mode.

3.3.2 Experimental Setup

The fabrication and testing of the designed silicon modulators has been carried out though a collaboration at MIT-Lincoln Laboratory. The experimental setup is shown in Fig.3-25 and Fig.3-26.

A 1550nm laser generated by a tunable laser diode (Santec TSL-210V) with maximum output power $\sim 15\text{mW}$ is amplified in an optical fiber amplifier. The amplifier

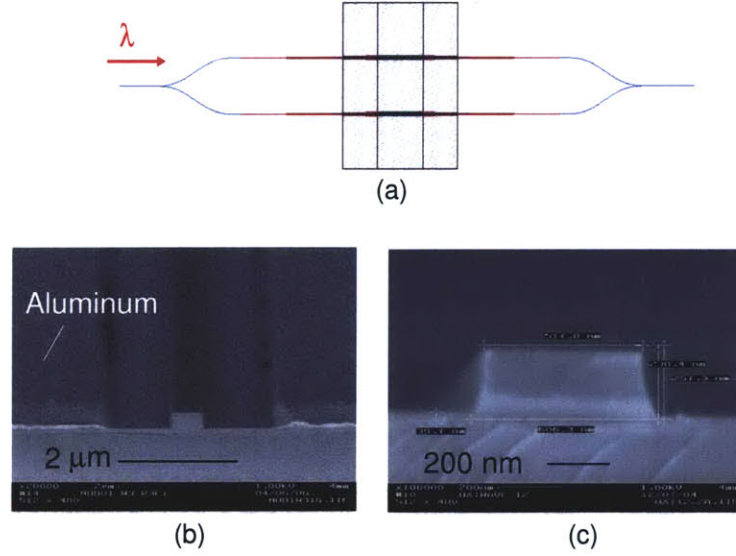


Figure 3-24: (a) Mach-Zehnder interferometer, (b) SEM image of waveguide with metal contact, (c) SEM image of waveguide cross-section.

is required because of the optical loss due to the tested devices and coupling losses. A fiber polarization controller is used to match the polarization of the input laser to the quasi-TE single-mode waveguides. An optical lensed fiber matches the input fiber mode to the on-chip waveguide mode. The lensed fiber has a fiber core diameter of $9\ \mu\text{m}$, a cladding diameter of $125\ \mu\text{m}$ and a diameter of $250\ \mu\text{m}$.

The output beam from the on-chip waveguide is collected by a lensed fiber, goes through a fiber preamplifier, a 2nm tunable bandpass filter and a fiber amplifier and finally is detected by an InGaAs photo-detector. The pre-amplifier raises the overall power level and the filter suppresses unwanted light before coupling the light into the main amplifier.

The RF power is generated by a network analyzer, amplified by a RF power amplifier which has a bandwidth of 30kHz-20GHz and a gain of 18dBm. The analyzer power can be up to 5dBm, but is typically kept to 0dBm because some nonlinearity is observed when power goes higher. There are only three pads in the modulators and the electrical probes are shown in Fig.3-26(b). The DC bias is applied across the left arm and goes through the right arm to the ground. The RF probe is applied to

the middle pad together with a DC bias. The optical output can be maximized by adjusting the two DC bias voltages.

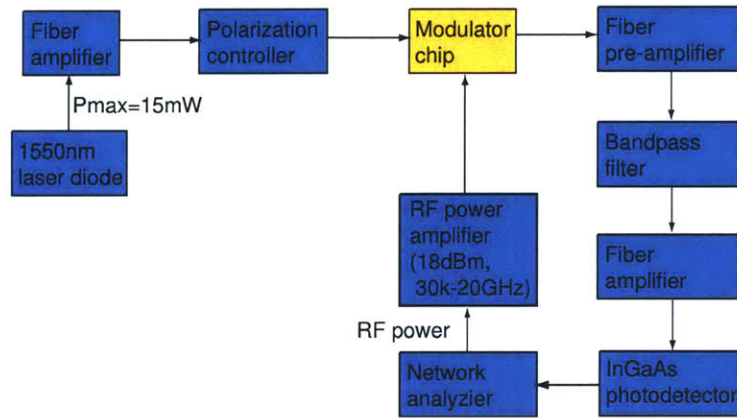
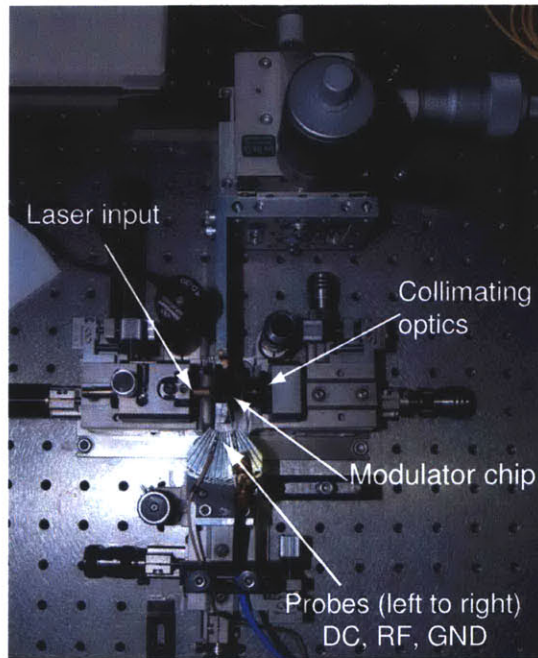
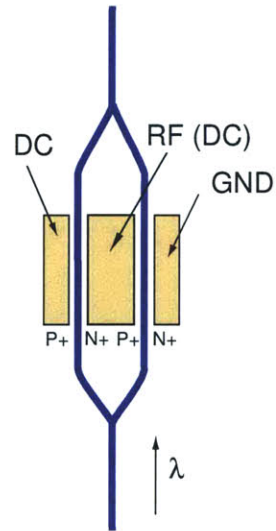


Figure 3-25: Schematic diagram of the experimental setup for modulator RF measurement.



(a)



(b)

Figure 3-26: (a) Experimental setup, (b) Mach-Zehnder modulator with probes.

Chapter 4

Electro-Optic Modulators and All-Optical Switches Based on Carrier-Injection in Si/SiO_2 Split-Ridge Waveguides (SRW's)

In chapter 3, a buried waveguide was employed in high-speed silicon electro-optic modulators. The advantage of the buried waveguide is the tight optical confinement due to a small mode size, which makes it applicable for large scale integration. However, the compact size is often outweighed by strong device heating caused by the low thermal conductivity in the surrounding silicon dioxide, and the fabrication tolerances of the buried single-mode waveguide are fairly small and easily deteriorate the performance in a fabricated device. For example, the heat induced refractive index change in the forward-biased PIN modulator with lifetime reduction is larger than that induced by plasma effect at low frequencies, shown in Fig.3-4. The waveguide structure must be optimized to minimize thermal effects and maximize fabrication tolerance, and the optical field must well overlap with the generated carriers. The two most common waveguide structures, buried and ridge waveguides, are not well suited for these requirements. Therefore, we propose a Si/SiO_2 split-ridge waveguides (SRW),

which offers excellent heat sinking, good carrier confinement and a high tolerance toward fabrication errors. The electro-optic modulator and all-optical switching are investigated based on SRW waveguide and show the advantages of low heating and high overlap between optical mode and carrier change.

In Sec. 4.1, the SRW is explored and its properties of good heat sinking, large fabrication tolerance and broadband coupling are discussed. In Sec.4.2, an electro-optic modulator is realized based on the SRW waveguide. In Sec.4.3, two architectures for all-optical carrier injection switches using two or three waveguides are proposed. These switches exhibit large fabrication tolerances, low power consumption and reduced undesired heating.

The commercial simulator FIMMWAVE [51] is used to solve optical modes and model wave propagation and coupling. The program is a fully vectorial waveguide solver based on the film mode matching method. It can find arbitrary order modes and mixed polarisation and is an eigenvalue solver. The 2-dimensional semiconductor simulator MEDICI is employed to simulate the electrical properties. MEDICI simulates the physical mechanisms governed by the carrier continuity equations, the Poisson equation, and the heating equation.

4.1 Waveguide and Coupler Design

4.1.1 Split-Ridge Waveguide (SRW)

Waveguides are the most fundamental components of every integrated optical device. They ascertain confinement of light and contain the active medium whose index is changed in the case considered here, and simultaneously act as a heat sink for the energy deployed. High-index-contrast (HIC) waveguides with a silicon core and SiO_2 cladding offer the advantage of strong light and carrier confinement. Therefore, a short interaction length in optical switching applications using the plasma effect in silicon [35] is possible.

Two popular waveguide geometries on the Si/SiO_2 platform are the buried and

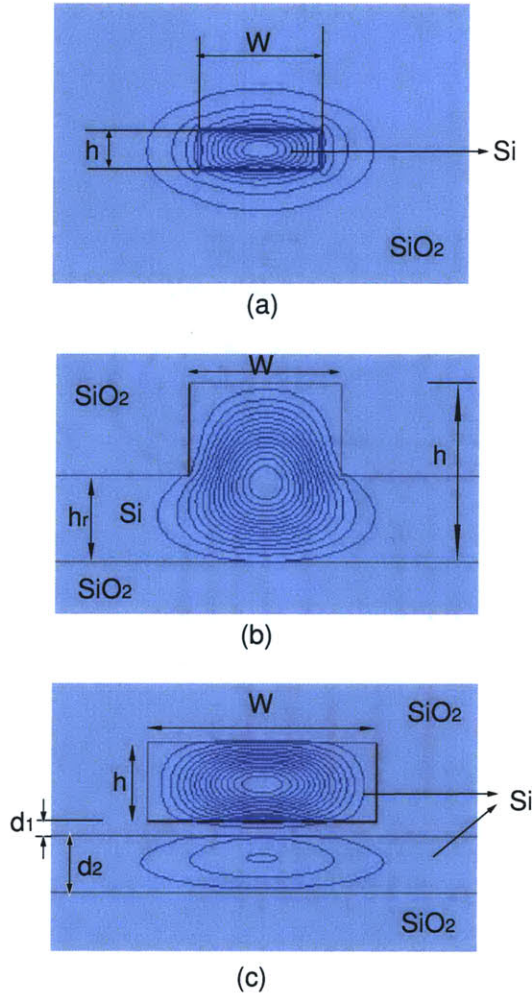


Figure 4-1: Structure layout and single-mode profile for: (a) buried waveguide, (b) ridge waveguide, (c) split-ridge waveguide (SRW)

ridge waveguides as shown in Fig.4-1(a) and (b). Typical dimensions of a single mode buried waveguide in this material platform are a width of $0.5 \mu m$ paired with a height of $0.2 \mu m$ [18]. For waveguides, the height can be rather well controlled by chemical vapor deposition of the individual layers or with the SIMOX [52] or Smartcut process [53]. However, the lateral dimensions are determined by the accuracy of the photolithography employed and can exhibit large deviations from the design goal. The fabrication tolerances of the buried single-mode waveguide are fairly small and easily deteriorate the performance in a fabricated device. The advantage of the buried

waveguide is the tight optical confinement due to a small mode size, which is often outweighed by strong device heating caused by the low thermal conductivity in the surrounding silicon dioxide. In contrast, ridge waveguides exhibit much larger mode sizes and also show a much larger fabrication tolerance of several micrometers [54][55]. However, their open structure leads to low carrier confinement because the carriers can diffuse away $62\ \mu\text{m}$ within 100ns in bulk silicon. Most carriers may diffuse away upon carrier injection, which results in a low carrier concentration. Here, a Si/SiO_2 SRW is investigated to overcome these limitations and to combine the advantages of buried and ridge waveguide in a single structure.

A SRW can be considered a buried waveguide on top of a closely spaced thin Si layer, see Fig.4-1(c). The whole structure is embedded in SiO_2 . The thin layer draws part of the mode down from the buried waveguide. The amount of light guided in the thin Si-slab depends on the distance between the slab and the buried waveguide as well as the thickness of the slab. Hence, its mode spreads out more than in a buried waveguide but less than in a ridge waveguide, resulting in a lower center of the mode profile than that in the buried waveguide. Single-mode operation is obtained by adjusting the vertical dimensions d_1 and d_2 . Only first TE mode can be highly confined, the first TM mode is weakly confined and no other modes can be guided. The Si-slab makes the cladding effective index larger than in the buried waveguide so that the index contrast is weaker than in a buried waveguide. Therefore, the SRW remains single-mode for larger dimensions than a buried waveguide.

The SRW structure has a number of advantages over the buried and ridge waveguides. First of all, the insertion of the Si-slab allows to choose a larger width of the waveguide to reduce sensitivity to fabrication errors while still maintaining single-mode operation by carefully adjusting the layer thickness. For example, at a communications wavelength of around $1.5\ \mu\text{m}$, a waveguide width of $1\ \mu\text{m}$, twice as large as for a comparable buried waveguide, can be chosen. In addition, the thickness of the isolating SiO_2 layer is much smaller than in buried waveguides, resulting in a significant improvement in heat sinking. To achieve good heat sinking, the silicon slab thickness should be larger than 350nm and the heat sink is less than $5\ \mu\text{m}$ away

from the waveguide so that the thermal resistance of silicon slab is smaller than the thin SiO_2 layer. An electronic carrier-injection modulator design has made full use of this feature [40]. In our current proposed device, an SiO_2 thickness d_1 of less than 100 nm is sufficient, compared with 1- 2 μm that would be required in a comparable buried waveguide, yielding a 10 to 20-fold improvement in thermal conductivity. Finally, the SRW allows for good confinement of carriers in the buried section, while they would easily escape in ridge waveguides due to the open structure. Thus, the desired switching effect can be achieved at lower switching power.

4.1.2 Broadband Coupler with Split-Ridge Waveguide (SRW)

The SRW described above also enables the design of couplers that are less sensitive to fabrication variations and wavelength. This enables the construction of all-optical switches using broadband SRW-couplers.

Fabrication Tolerances

Small length and thickness variations are unavoidable in waveguide fabrication. We can differentiate between absolute width variations, i.e., errors that result in size changes for all devices on a wafer, and relative variations, such as different thickness of adjacent waveguides in an individual waveguide coupler. The coupling efficiency in waveguide couplers tends to be particularly sensitive to relative errors of the adjacent waveguides. For example, the coupling of a single-mode coupler at the communications wavelength based on buried waveguides, with a cross-section of $400\text{ nm} \times 200\text{ nm}$, is fundamentally degraded by a deviation in feature size as small as a few nanometers in the horizontal direction. The vertical dimension can be controlled much more precisely. Therefore, variations in the vertical dimension are not considered. Even for state-of-the-art E-beam fabrication techniques [11], the relative variations in waveguide width can be controlled only to the order of 1 nm which can cause a serious variation in the coupling coefficient. To obtain a worst-case estimate we assume a relative variation of 2.5 nm in the following calculations. A good design must be

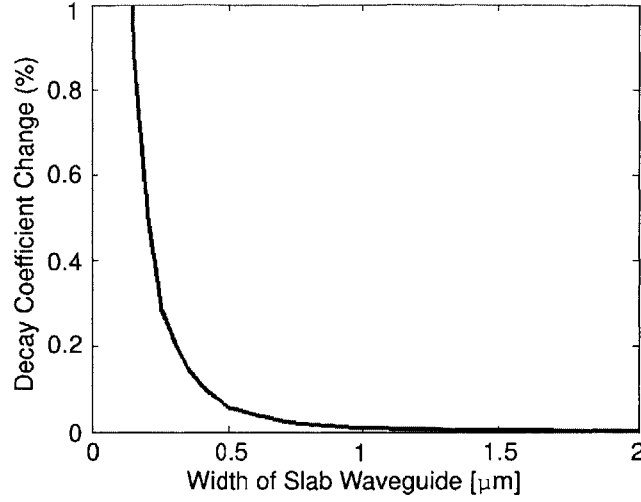


Figure 4-2: Relative change of cladding decay coefficient $\Delta\alpha/\alpha_0$ as a function of slab waveguide width with a waveguide width variation of 2.5 nm at the wavelength of $\lambda = 1.55\mu\text{m}$

sufficiently robust against fabrication errors.

In the following, the fabrication tolerance of the SRW-coupler is investigated based on a simplified analytical 1-dimensional model and a numerical 3-dimensional model. In the former case, the SRW is modeled as a 1D slab waveguide. This approximation is chosen because the weak coupling in the devices is proportional to $e^{-\alpha L}$, dominated by the product of the coupler length L and the field decay coefficient (α) in the horizontal direction while the vertical field decay coefficient is of minor importance. The coupler length's effect on tolerance will not be studied in this thesis, more information can be found in [56]. The horizontal cladding decay α of the symmetrical TM mode in a slab waveguide is governed by the transcendental equation [56]

$$\tan(k_x \omega / 2) = \frac{\epsilon_{co}}{\epsilon_{cl}} \frac{\alpha}{k_x} \quad (4.1)$$

where k_x , ω , ϵ_{co} and ϵ_{cl} are the horizontal wavenumber in the core region, the core width, the core and cladding dielectric constants, respectively. The relative change of field decay in the slab waveguide cladding $\frac{\Delta\alpha}{\alpha_0} = \frac{\Delta\alpha(w_0 + \Delta w) - \Delta\alpha(w_0)}{\alpha(w_0)}$ is calculated as a

function of slab width in Fig.4-2. For waveguide widths below $0.5 \mu m$, small variations in thickness lead to a large relative change of field decay $\Delta\alpha/\alpha_0$. In contrast, for a width of the waveguide exceeding $1 \mu m$ the relative change of field decay experiences merely a negligible drop. Thus, a coupler made out of waveguides wider than $1 \mu m$ will have a considerably less change in coupling strength. Fig.4-2 illustrates the significant improvement in sensitivity against fabrication tolerances of SRW couplers due to its larger waveguide width.

The insensitivity to fabrication errors has also been numerically verified in a 3-dimensional coupler with single-mode SRWs at a communications wavelength of $1.55 \mu m$ by performing simulations with the software FIMMWAVE [51]. For a 100% coupler with the waveguide dimensions $w \times h = 1 \mu m \times 0.55 \mu m$, the layer thickness $d_1 = 100 \text{ nm}$ and $d_2 = 350 \text{ nm}$ (Fig.4-1c), a 600 nm distance between the two quasi-TE single-mode waveguides and $365 \mu m$ coupling length, our simulations show that a relative fabrication error of 2.5 nm leads to a tolerable drop in output power of 0.11 dB . In contrast, in a quasi-TE single-mode coupler based on buried waveguides with the dimensions of $w \times h = 450 \text{ nm} \times 250 \text{ nm}$, the lateral distance of $d = 400 \text{ nm}$, and the coupling length of $l = 180 \mu m$, the same fabrication error leads to a 8 dB power penalty. The computation was done by varying the waveguide width while keeping the waveguide center-to-center distance constant. These numbers illustrate the robustness of the SRW based coupler with respect to fabrication tolerances.

The insensitivity to fabrication tolerances also leads to wavelength insensitivity for a directional coupler because both changes in dimensions and wavelength can substantially vary the waveguide mode size and the coupling strength. The wavelength-sensitivity of SRW-couplers will be discussed later in this section.

Bandwidth

Another key requirement for all-optical switches is the broadband behavior which is required to cover the entire C-band of optical communications with very low OFF-state insertion loss over the full bandwidth. For example, the construction of a switch matrix with $N = 64$ channels may need a matrix of N^2 couplers, as shown in Fig.4-3.

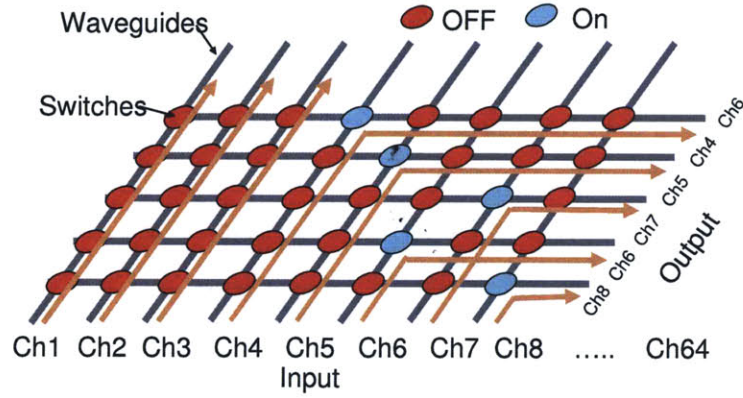


Figure 4-3: 64 x 64 all optical router.

In such a device, an individual channel encounters the OFF-state, i.e., the switch without carrier-injection, up to maximum $2N-2$ times, whereas the ON-state, i.e., the switch with carrier-injection, is encountered only once, when the signal is routed to a different channel. For our design, we found that the OFF-state of each individual coupler is almost frequency independent and does exhibit very low loss on the order of 1% (see Fig.4-4), neglecting additional waveguide losses. Thus, the channel passing through the entire matrix of 64 switches would encounter an overall insertion loss of only 5.5 dB.

The broadband operation of the SRW-coupler results from the large width of the waveguide. The SRW coupler with $1\ \mu\text{m}$ width offers a significantly broader bandwidth than that based on the buried waveguide with a typical width of less than $0.5\ \mu\text{m}$. We verified this hypothesis with the software package FIMMWAVE [51]. Our calculations reveal that the SRW-coupler exhibits over 99% transmission over the entire C-band (Fig.4-4), while the coupler based on buried waveguides has much narrower band and its power transmission drops below 94% at both ends of the C-band. For a complete switch matrix such a high transmission uniformity of a SRW-coupler would be significant: with a seemingly small loss of about 1% in each buried waveguide coupler at 1530 nm, the total losses at this wavelength accumulate to maximally 5.5 dB in a 64-channel switch matrix, while the signal at the center wavelength of 1550 nm remains un-attenuated. A 64-channel switch matrix based on

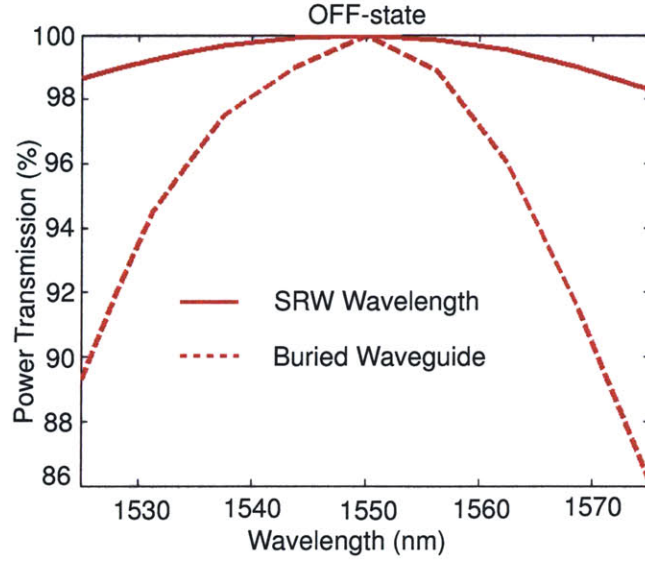


Figure 4-4: Bandwidth for two-waveguide couplers based on buried (dashed) and SRW (solid) waveguides

buried waveguide coupler has 34dB maximum loss at the wavelength 1530nm. This problem is circumvented in the SRW-coupler due to its low variation in transmission over the entire C-band.

4.2 Electro-Optic Modulators Based on SRW

In this section, we discuss a Si/SiO₂ high-index contrast waveguide modulator based on a split ridge-waveguide that operates under forward biased conditions and shows corner frequencies of up to 24 GHz, which has a smaller figure of merit than previous studied PIN/PN reverse-biased structures [14].

4.2.1 Device Structure

A schematic layout of the proposed modulator is shown in Fig.4-5. Fig.4-5(a) shows a conventional Mach-Zehnder arrangement, and Fig.4-5(b) shows the most important sub-component of the modulator, the electrically driven phase modulator section. We use a split-ridge waveguide, i.e., the high-index ridge is separated from the high-index

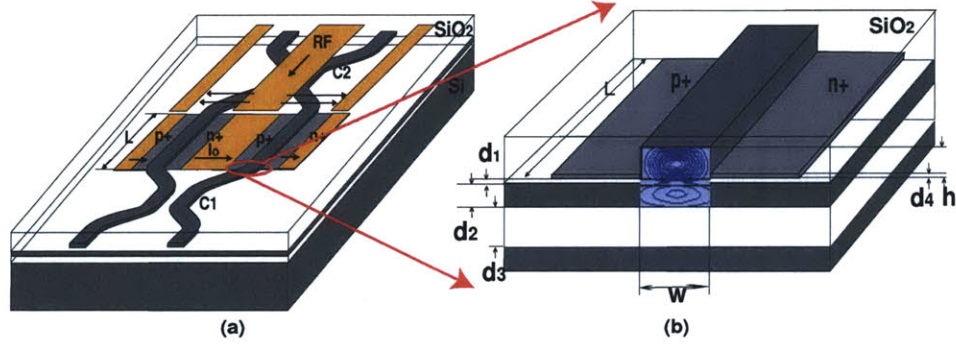


Figure 4-5: (a) Externally-biased Mach-Zehnder modulator with coplanar RF-feeder, (b) split-ridge waveguide pin-phase modulator with single mode intensity profile.

slab via a thin low index layer, which combines the following advantages of a buried waveguide and a ridge waveguide (see Fig.4-5(b)). By adjusting thickness d_1 and d_2 , we can obtain single-mode operation. For example with $d_1 = 100nm$, $d_2 = 350nm$, $h = 550nm$ and $w = 1\mu m$, we obtain from simulations with FIMMWAVE that there is only one guided TE and one guided TM mode, only the TE mode is studied in this thesis. The optical mode is 94% confined in the ridge section of the waveguide which is isolated so that injected carriers cannot escape in the transverse directions. One way to fabricate this structure is to use double SOI wafers which can be generated by SIMOX or Smart-cut process [52][53]. The SiO_2 -isolating layer, which splits the ridge-waveguide, is rather thin ($d_1 = 100nm$), so that there is good heat sinking to the silicon slab portion of the waveguide acting as a heat spreader. The optical mode is well confined within the ridge in the horizontal direction, which greatly reduces loss due to highly doped contact regions and sidewall roughness. Assuming that also a 50nm thick layer on both sides of the ridge waveguide section is doped to ensure homogenous carrier distribution, even doping levels as high as $n_D = 10^{19}cm^{-3}$ lead only to an absorption coefficient of $\alpha_C = 2.8dB/cm$. The PIN-diodes in both arms of the Mach-Zehnder modulator are biased in series and a static current is driven through both arms, leading to equal phase shifts, absorption and heating in both arms. This ensures that the thermo-optic effects in both arms are balanced to a large degree. The RF-signal is capacitively coupled to the diodes out of phase via the even mode of a coplanar transmission line, see Fig.4-5(a).

The electrical and thermal characteristics of the phase modulator sections are evaluated using the semiconductor device simulator MEDICI [45]. Subsequently, the optical characteristics of the phase modulator sections, i.e., effective optical index and absorption changes, and the resulting Mach-Zehnder modulator characteristics are determined from these computations by exporting the electrical and thermal characteristics and building proper averages over the optical mode profile that is calculated separately with the software tool FIMMWAVE [51]. The carrier lifetime is dynamically adjusted including the surface, the Shockley-Read-Hall(SRH), and the Auger recombination. The carrier lifetime in the intrinsic section can be varied over a wide range using doping or ion implantation [39]. Simulations with three values for the electron/hole lifetimes have been performed: $\tau=10$, 100, and 1000ps. The contact resistivity is assumed to be small, about $10^{-7}\Omega \cdot cm^2$ [38]. Therefore, most of the voltage drop is located across the i-region of the pin-modulator. For short carrier lifetimes, the electric field can be increased to a level where the current flow gets close to saturation and the carriers drift at the saturation velocity $v_s = 10^7 cm/s$ in silicon. The carriers injected into the i-region induce a change in the refractive index and the optical absorption. The effective index and absorption of the guided mode are obtained by averaging over the spatially varying index and absorption using the numerically determined optical mode profile of the split-ridge waveguide, see Fig.4-5(b).

4.2.2 Device Performance

Detailed two-dimensional simulations have been carried out for a device with dimensions: $d_1 = 100nm$, $d_2 = 350nm$, $d_3 = 2\mu m$, $d_4 = 100nm$, $h = 550nm$ and $w = 1\mu m$ assuming longitudinal homogeneity. The static I-V characteristics of the phase modulator section in terms of the average current density flowing through the intrinsic zone as a function of applied voltage are shown in Fig.4-6(a) for the three different carrier lifetimes. In addition for different operating points, the maximum device temperature occurring in the center of the ridge waveguide section is shown in Fig.4-6(a). For voltages greater than 6V the current strongly saturates. Acceptable internal operat-

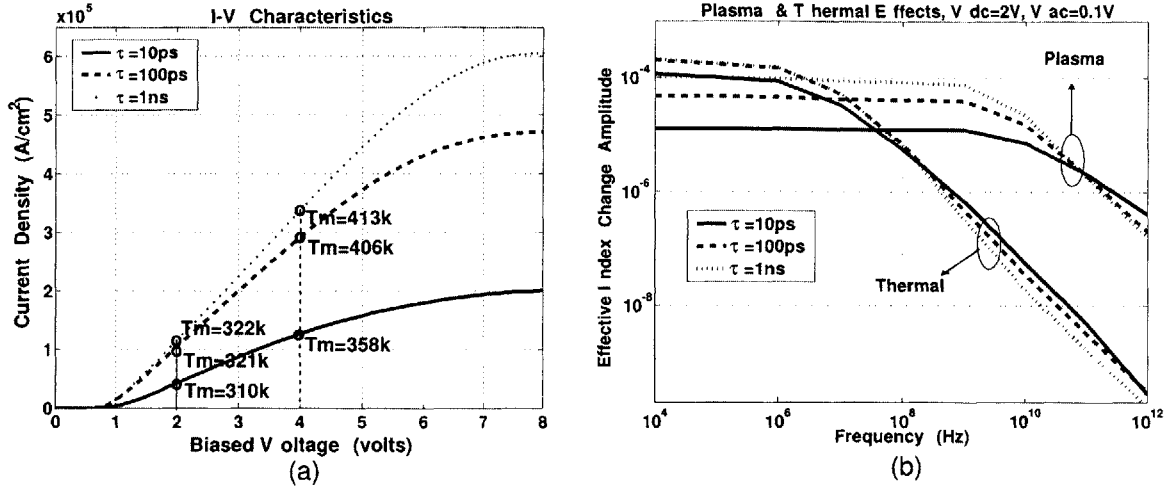


Figure 4-6: (a) Modulator I-V-characteristics and maximum device temperature T_m at different operating points and carrier lifetimes $\tau=10\text{ps}$ (solid), 100ps (dashed), 1ns (dotted); (b) effective index change due to plasma and thermal effects for $\tau=10\text{ps}$ (solid), 100ps (dashed), 1ns (dotted).

ing temperatures between 300K and 400K are computed for this device for operating voltages of 2V and 4V. The heat is efficiently removed from the device via the metal contacts and the thin slab-Si layer (see Fig.4-5b). Roughly 65% of the heat escapes over the metal contacts and 35% over the Si-slab for the current dimensions. Thus the Si-slab layer does not only reduce the waveguide loss by better mode confinement in the ridge but also greatly improves the heat sinking.

The AC characteristics of the device are determined by computing the small signal modulation transfer function between the effective index change in the phase modulator section and the applied voltage. The modulation speed of a pin-diode depends critically on the carrier recombination time and the drift velocity. The faster of the two processes, recombination or drift, determines the 3dB corner frequency of the AC-response. For a structure with a carrier lifetime equal to or shorter than 1ns, and assuming a thermally grown oxide passivation with a surface recombination velocity of 100 cm/sec., the SRH-recombination dominates the carrier lifetime. The SRH lifetime depends on the density of recombination centers due to the doping or ion bombardment. For high carrier injection, the SRH recombination time is in the order of

$\tau_{SHR} = 2\tau$, where τ itself depends on the carrier concentration [45]. Correspondingly, the 3dB corner frequencies due to recombination are $f_{3dB}^{recom} = 1/2\pi\tau_{SHR} = 1/4\pi\tau = 8$ GHz, 0.8 GHz, 0.08GHz for $\tau=10$ ps, 100ps, and 1ns, respectively. In addition to recombination, carrier drift under the influence of a strong applied field also determines the maximum modulation speed. The simulations result in carrier velocities $v \simeq 8 \times 10^6 cm/s$, $2.4 \times 10^6 cm/s$ and $2.4 \times 10^6 cm/s$ for $\tau=10$ ps, 100 ps, and 1 ns, which corresponds to maximum modulation frequencies $f_{3dB} = 2.4/2\tau_{drift} = 30$ GHz, 9 GHz, 9 GHz with $\tau_{drift} = w/v$ [37]. We obtain the 3dB frequency from Fig.4-6(b) to be 24 GHz, 4 GHz, and 3 GHz, which agrees well with the estimation of 30 GHz, 9 GHz and 9 GHz, respectively.

Fig.4-6(b) shows both the amplitude of the effective index change due to the thermal effect and due to the plasma dispersion effect for an assumed sinusoidal drive voltage modulation amplitude of 0.1V. For frequencies $f > 100MHz$, the plasma effect dominates the thermal effect. At low frequencies thermal effects dominate because the thermal time constant is about 30ns and, therefore, much longer than the carrier lifetimes used in the simulations. This is of major importance because both effects do have opposite sign and thus there is a frequency range, i.e., 1 – 100MHz, where the response is zero. If a modulator for a combined low and high frequency range is desired, this problem can be fixed by using two modulators with widely different carrier lifetimes in series, which address the low and high frequency ranges of the input signal separately. A similar modulator as in Fig.4-5 with a carrier lifetime of 30 ns could be used to modulate the low frequency band from 0 – 100 MHz.

To bias the modulator section, we dissipate a DC-power $P_{DC} = V_{DC} \cdot I_{DC}$, where the DC-current I_{DC} is the current density from Fig.4-6(a) multiplied by the cross section of the device through the ridge. Operating at a bias voltage, V_{DC} , we obtain for the DC-power of a 1mm long section according to Fig.4-5 (b), with the area $A = 0.55\mu m \times 1mm = 5.5 \times 10^{-6} cm^2$, the DC power $P_{DC} = 1/2 V_{AC} I_{AC} \cos(\phi) = 1/2 \cdot 1V \cdot 2.78 \times 10^4 A/cm^2 = 550mW$. At this bias point and a modulation frequency of $f=10GHz$, with amplitude(half peak-to-peak) $V_{AC} = 1V$, the simulation gives for the amplitude of the AC current density $J_{AC} = 2.78 \times 10^4 A/cm^2$, resulting in a

dissipated microwave power of $P_{AC} = 1/2 V_{AC} I_{AC} \cos(\phi) = 1/2 \cdot 2.78 \times 10^4 A/cm^2 \cdot 5.5 \times 10^{-6} cm^2 \cdot \cos(18^\circ) = 76 mW$. The corresponding amplitude of the effective index change is $\Delta n = -1.5254 \times 10^{-4}$. To achieve a phase shift of π at a wavelength of $1.55 \mu m$ a 5 mm long section would be necessary, resulting in a figure of merit, $FOM=0.5 Vcm$, of the phase modulator section. Note that for a MZ-modulator a differential phase shift of π is necessary between the two arms to achieve full on off modulation. This is possible for a 1 V amplitude signal and 2.5 mm long phase modulator sections in each arm resulting in a total DC-Power consumption of 2.5W and an estimated microwave power of 380 mW.

Both the losses and the phase shift depend approximately linearly on the injected carriers [36] and, therefore, we obtain for the relationship of total losses accumulated in the structure versus phase shift $\alpha \cdot L \approx \phi/4$, which leads to the following transmission characteristic of the modulator as in chapter 2

$$T(\phi) = \frac{1}{2} \cdot e^{-(\alpha_C + \alpha_{DC})L} [\cosh(\phi/4) - \cos(2\phi - \phi_B)] \quad (4.2)$$

for $|\phi| < \pi/2$, where ϕ_B is the bias phase shift. For the current example of a 2.5 mm long structure the total static losses due to contact doping and the DC-bias current would be about

$$10 \log_{10}[e^{(\alpha_C + \alpha_{DC})L}] = 4 dB \quad (4.3)$$

not including possible bend and scattering losses due to sidewall roughness.

Furthermore, the thermal improvement by the silicon slab can be seen by comparing the 10ps lifetime AC characteristics between the modulators based on buried waveguides and SRW. A 3dB corner frequency of the refractive index change due to the thermal effect is around 500kHz in the buried waveguide modulator (see Fig.3-4), while it is around 1MHz for the SRW modulator (see Fig.4-6). The higher corner frequency in SRW structure indicates SRW has shorter thermal time constant, i.e., the generated heat in the waveguide can diffuse to the heat sink faster. It is the SRW silicon slab that introduces another heat sinking and reduces thermal effect.

In summary, a high-speed electronic carrier-injection modulator based on a high-index-contrast split-ridge Si-waveguide has been proposed and investigated. Simulations show that with a short carrier lifetime in the intrinsic region, $\tau=10\text{ps}$, by lifetime doping or ion bombardment and operation in the saturation region where carrier travel at saturation velocity close to $v_s \approx 10^7\text{cm/s}$, modulators with a frequency range of 1-24GHz should be possible. At these operation points reasonable values for DC and AC power consumption, operating temperatures and an intrinsic loss of only 4 dB can be achieved.

4.3 All-Optical Switches Based on SRW

In Sec.4.1, we demonstrated the advantages of using SRW's to achieve a broadband coupler. In this section we analyze two different architectures for all-optical switches based on SRW-couplers. The key attraction of all-optical switches is the potential for high switching speeds and low insertion loss. The high speed results from the absence of RC low-pass filters of the electrical drive circuitry, and the absence of significant optical losses at the signal wavelength typically occurs in switches due to the absorption of evanescent field components in the vicinity of the drive electrodes. All-optical switches have been realized at high power levels in large silicon structures [57]. However, the high power regime and large dimensions of the silicon switches demonstrated to date render them inappropriate for on-chip integration. Furthermore, recently the possibility of all-optical switching in the silicon material platform was demonstrated by injection of 500 ps pulses in a microring resonator structure resulting in an ultrafast switch [17]. Here, the design of all-optical $\Delta\beta$ -switches is investigated. Switching is triggered by carrier generation caused by absorbing 800 nm light, that is coupled into the switch with a dedicated SiN waveguide (Fig.4-7). Our simulations (Sec.4.3.2 and 4.3.3) and carrier lifetime measurements (Sec.5.3) show that such all-optical switches enable switching with cw control light at low optical power level.

In Sec.4.3.1, the switching principle and device structures are depicted. In Sec.4.3.2

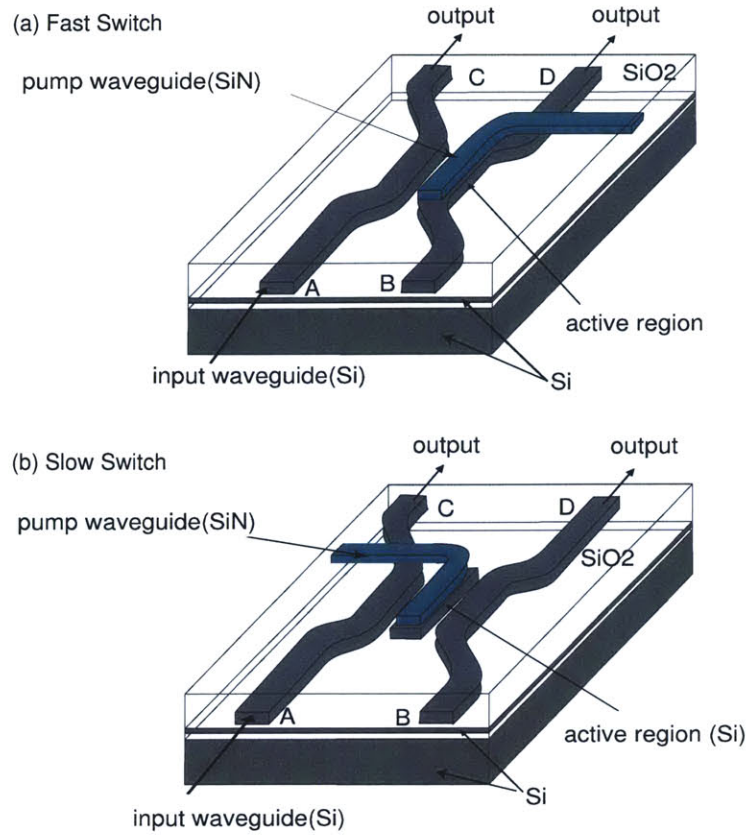


Figure 4-7: Sketch of the proposed switches (a) two-waveguide Switch, (b) three-waveguide Switch.

and 4.3.3, two architectures for all-optical carrier-injection switches using two or three waveguides are proposed. These switches exhibit large fabrication tolerances, low power consumption and reduced undesired heating. In Sec.4.3.4, the mechanism determining switching speed is discussed.

4.3.1 Switching Principle and Device Structures

The structures of the carrier injection switches are depicted in Fig.4-7. The switches are based on the SRW-coupler discussed in Sec.4.1.1 and a SiN injection waveguide for transport of the 800 nm modulation light. The modulation light is assumed to come from an off-chip source. Two different coupler architectures have been explored with two and three waveguide elements for the signal light, respectively (Fig.4-7).

The coupler consisting of two waveguides (Fig.4-7(a)) is suitable for high switching speeds, exceeding 1 GHz, since carriers diffuse out of the active region along the signal waveguides. In contrast, the switch depicted in Fig.4-7(b) comprises an additional middle section in which the carriers are generated. The effective carrier lifetime in this structure is longer, since the surrounding isolating material prevents the escape of the carriers as discussed in Sec.4.1.1, such that switching can be maintained with very low optical power if the surfaces are sufficiently well passivated. The length of the active regions is chosen to allow for efficient coupling of the 800 nm control light from the injection waveguide. Numerical simulations show that, for a distance between control and signal waveguides of about 10-50 nm and for a length of the active zone of about 100 μm , over 90% of the control light is absorbed in the signal waveguide.

In the "OFF-State" of the switch no carriers are injected into the active region and the input waveguide couples fully to the output waveguide B-D. In this state the system is ideally lossless and the switch is transparent. The broadband transfer characteristics from the input to output are shown in Fig.4-4. In the case of the two-waveguide switch, coupling occurs directly to the output waveguide B-D (Fig.4-7a). In the case of the three-waveguide switch, coupling occurs via a short intermediate coupling section into which the carriers are injected (Fig.4-7b). In the "ON-State" of the switch, control light entering the SiN-waveguide is absorbed in the active waveguide and changes the index, which detunes the active waveguide. The coupling of light to the output waveguide B-D is blocked and almost all light remains in the input waveguide A-C. The maximum loss in this state can be estimated by assuming that all the input light that is still coupled into the active region gets absorbed by the induced free carrier absorption. The next two sections discuss the performance of the two/three-waveguide switches in detail.

4.3.2 Two-Waveguide All-Optical Switch

We propose an all-optical switching structure capable of high-speed modulation. We start our analysis by investigating the power transfer characteristics as a function

of the bulk carrier lifetime, the surface recombination, the coupler dimensions, and other parameters. For this purpose, we calculate the switching power required to switch permanently from one channel to the other. Only the steady state "ON" and "OFF" conditions are considered and transients are neglected. Finally, we discuss the requirements for carrier lifetime to reach a certain switching speed.

The power transfer characteristics of the device depend on the changes of the refractive index and absorption coefficient between ON and OFF states, as well as the recombination time constant τ_{eff} . Using coupled mode theory, the power output from port D in Fig.4-7(a) can be written as [56]:

$$P_D = P_0 \frac{\kappa^2}{\delta^2 + \kappa^2} \sin^2(\sqrt{\delta^2 + \kappa^2} L) \quad (4.4)$$

where L , κ , $P_A = P_0$ and $\delta = (\beta_1 - \beta_2)/2$ are the coupler length, the coupling coefficient, the input power into waveguide A and the wavevector mismatch between the two waveguides, respectively. In the OFF-state with $\delta = 0$, the half beat length of the coupler is $L = \pi/2\kappa$ to obtain full power transfer to port D. In the ON-state $\delta \neq 0$, the maximum power coupling to port D is

$$P_{D,max} = P_0 \frac{\kappa^2}{\delta^2 + \kappa^2} \quad (4.5)$$

Therefore, most optical power will remain in the input waveguide if $\delta \gg \kappa$ without considering absorption.

The transmission characteristics of the two-waveguide switch in the ON and OFF states have been simulated using FIMMWAVE. The separation between the two waveguides and the coupling length are set to $0.6 \mu m$ and $365 \mu m$, respectively. The ON-state is characterized by $\Delta n_c = -0.005$ and $\Delta \alpha = 26/cm$ caused by the carrier density change $\Delta n_{e/h} = 1.8 \times 10^{18} cm^{-3}$ [35]. The simulation results are shown in Fig.4-8. Over 98% of the signal power is transmitted to port D in the OFF-state, and more than 91.5% to port C in the ON-state over the full C-Band, as shown in Fig.4-8(a) and (b). The ON-OFF ratio for output port D is around 20dB, which can be further increased by increasing the pump power.

We can estimate the required pump power consumption necessary to achieve such transmission characteristics. The pump power can be calculated by

$$P_p = \frac{hc}{\lambda_p \eta} V g_L \quad (4.6)$$

where h , c , λ_p , η and V are the Planck's constant, the speed of light, the pump wavelength, the transfer efficiency and the active region volume. The generation rate is calculated by using (5.32) to be $g_L = 2 \times 10^{17}/ns.cm^3$, assuming $\Delta n_{e/h} = 1.8 \times 10^{18}cm^{-3}$ and $\tau_{eff} = 10ns$. With $\lambda_p = 800nm$ and $\eta = 95\%$, a pump power of $10mW$ is obtained for the given volume V of the active region.

An estimate for the thermal effects can be given based on the pump power used. The 100 nm SiO_2 layer for the SRW dominates the thermal resistance $R_{therm} = d_{SiO_2}/(\kappa_{SiO_2} \cdot A_{eff}) + d_{Si.slub}/(\kappa_{Si} \cdot A_{eff.slub}) \simeq 250K/W$ (assuming the heat sinking is $1\mu m$ away from the waveguide). The temperature rises by $\Delta T = R_{therm} \cdot P_p = 2.5K$, with a corresponding index change $\Delta n_T = \Delta T \cdot 10^{-4}/K = 2.5 \times 10^{-4}$. Thus the index changes caused by carrier injection are much larger than those due to thermal effects, $\Delta n_c/\Delta n_T = 5 \times 10^{-3}/2.5 \times 10^{-4} \simeq 20$ times in this case. Therefore, heating effect can be neglected.

In summary, these calculations show that the two-waveguide switch has over 90% switching efficiency with a DC switching power as low as 10mW. Carrier effects dominate thermal effects by more than one order of magnitude, allowing for switching speeds as fast as the carrier lifetime (Sec.4.3.4). The low heating of 2 K in the ON state (Sec.4.3.1) allows for example the construction of large switch matrices with switching time approaching 10 ns.

4.3.3 Three-Waveguide All-Optical Switch

For many applications especially switching matrices often much slower switching times are acceptable. To cover the slow optical modulation bandwidth applications, the structure based on three waveguides, see Fig.4-7(b), is advantageous.

Similar to the two-waveguide switching structure, the coupling strength κ and

the detuning δ determine the power transfer characteristics. The working principle is shown in Fig.4-7(b). The three waveguides are designed to have the same propagation constant β_0 without carrier injection. By injecting carriers into the middle waveguide, the propagation constant in the middle waveguide can be detuned by δ . According to coupled mode theory [56], mode amplitudes in the output ports C and D (Fig.4-7(b)) are given by:

$$a_C(z) = e^{-j\beta_0 z} \times \left[\frac{1}{2} - \left(\frac{\delta}{8\beta'} - \frac{1}{4} \right) e^{-j\frac{\delta}{2}z - j\beta'z} + \left(\frac{\delta}{8\beta'} + \frac{1}{4} \right) e^{-j\frac{\delta}{2}z + j\beta'z} \right] \quad (4.7)$$

$$a_D(z) = e^{-j\beta_0 z} \times \left[-\frac{1}{2} - \left(\frac{\delta}{8\beta'} - \frac{1}{4} \right) e^{-j\frac{\delta}{2}z - j\beta'z} + \left(\frac{\delta}{8\beta'} + \frac{1}{4} \right) e^{-j\frac{\delta}{2}z + j\beta'z} \right] \quad (4.8)$$

where $\beta' = \frac{1}{2}\sqrt{\delta^2 + 8|\kappa|^2}$. For the OFF-State, substituting $a_A(z=0) = 1$, $|a_D(z=L)| = 1$ and $\delta = 0$ into (4.7) and (4.8), we obtain the half beat length $L = \pi/\sqrt{2}|\kappa|$. For maximum power transfer to port C in the ON-state, δ should be as large as possible to keep most of the power in the input waveguide.

Simulations have been done for the three-waveguide coupler to obtain power transfer characteristics. The separation and coupling length are set to $0.8 \mu m$ and $780 \mu m$, respectively. The active region index change is set to 0.01, a little larger than that for the two-waveguide coupler. Simulation results are shown in Fig.4-9. In the OFF-state over 97% of the signal power is transmitted to port D and about 90% to port C in the ON-state over the C-Band. The ON-OFF ratio of output port C is around 30dB, which is usually demanded for a switch, see (Fig.4-9 (c)).

Performing the same calculations as for the two-waveguide switch but using the carrier lifetime $\tau = 100 \text{ ns}$, the pump power is estimated to be 2mW and the temperature rise is only $\Delta T = 0.42K$. The induced index change is $\Delta n_T = \Delta T \times 10^{-4}/K = 4.2 \times 10^{-5}$. The index change caused by carrier injection is again significantly larger

than the that by the thermal effect, about $\Delta n_c/\Delta n_T = 10^{-2}/4.2 \times 10^{-5} \simeq 230$ times. Therefore, again heating effects can safely be neglected.

Comparing the modeling of two and three-waveguide switches, the two-waveguide switch may work well up to 100MHz, or even 1GHz. The three-waveguide switch can operate with slower switching speed because of its closed structure helping to hold the carriers inside the active region. The 30dB ON-OFF ratio of the three-waveguide switch with respect to port C is large enough for optical communications applications and its power consumption and thermal effect are at least 5 times smaller than that for the two-waveguide switch.

4.3.4 Switching Speed

As mentioned in the previous sections, the switching speed is determined by the effective carrier recombination time. If carriers are injected to bring the switch to the ON-state, it takes several carrier lifetimes, τ_{eff} , for the switch to recover to the unperturbed state. To design an all-optical switch with switching speeds of up to 100MHz, an effective recombination time of less than 1 ns is required. This value can be obtained by ion-bombarding the active zone [39]. Under this condition, the effective lifetime is shortened by the Shockley-Read-Hall (SRH) recombination [40]. To obtain a switch with speeds of around 1MHz an effective carrier lifetime of up to 100ns is needed. Usually for small thin waveguides, the surface recombination dominates the recombination process. In order to fabricate the designed switches, knowledge about the carrier lifetime for thin Si waveguides with different surface passivation is necessary.

4.4 Summary

A split-ridge waveguide was proposed and investigated. SRW is advantageous over the ridge and buried waveguides by providing better thermal sinking, broader bandwidth coupling and larger fabrication tolerance. SRW provides a heat sinking through the Si-slab while the buried waveguide is surrounded by low thermal conductance silicon

oxide and thus can not provide a good heat sinking. The coupling coefficient insensitive to waveguide width makes the coupler based on SRW broadband, covering the whole C-band with very low OFF-state insertion loss. The larger tolerance to fabrication error in SRW leads to 80 times optical power loss reduction at the Mach-Zehnder interferometer output compared with buried waveguides.

An electrically driven Mach-Zehnder waveguide modulator based on the high-index-contrast silicon split-ridge waveguide technology and electronic carrier injection was proposed. The excellent optical and carrier confinement in the high-index-contrast waveguide devices, together with good thermal heat sinking and forward biased operation, enables high speed modulation with a small signal modulation bandwidth of beyond 20GHz, a figure of merit of $V_{\pi} \cdot L = 0.5V \cdot cm$ and an insertion loss of about 4 dB. The modulator can be fabricated in a CMOS-compatible process.

Finally, all-optical switches based on optical carrier-injection in high index contrast Si/SiO_2 split-ridge-waveguide (SRW) couplers are analyzed. The waveguide devices are suitable for the construction of low-loss optical switch matrices as well as fast optical switching. These devices exhibit robustness against fabrication tolerances, improved heat sinking, good carrier confinement and high uniformity in transmission over the entire C-band of optical communications in contrast to comparable devices based on buried or ridge waveguides. Reasonably low electrical switching power of 1-10mW is predicted for switching frequencies of 1MHz-1GHz and faster switching speed can be achieved by reducing carrier lifetime. Carrier recombination measurements in thin Si-layers passivated with different oxide layers confirm the feasibility of the designed switches and modulators which will be shown in chapter 5.

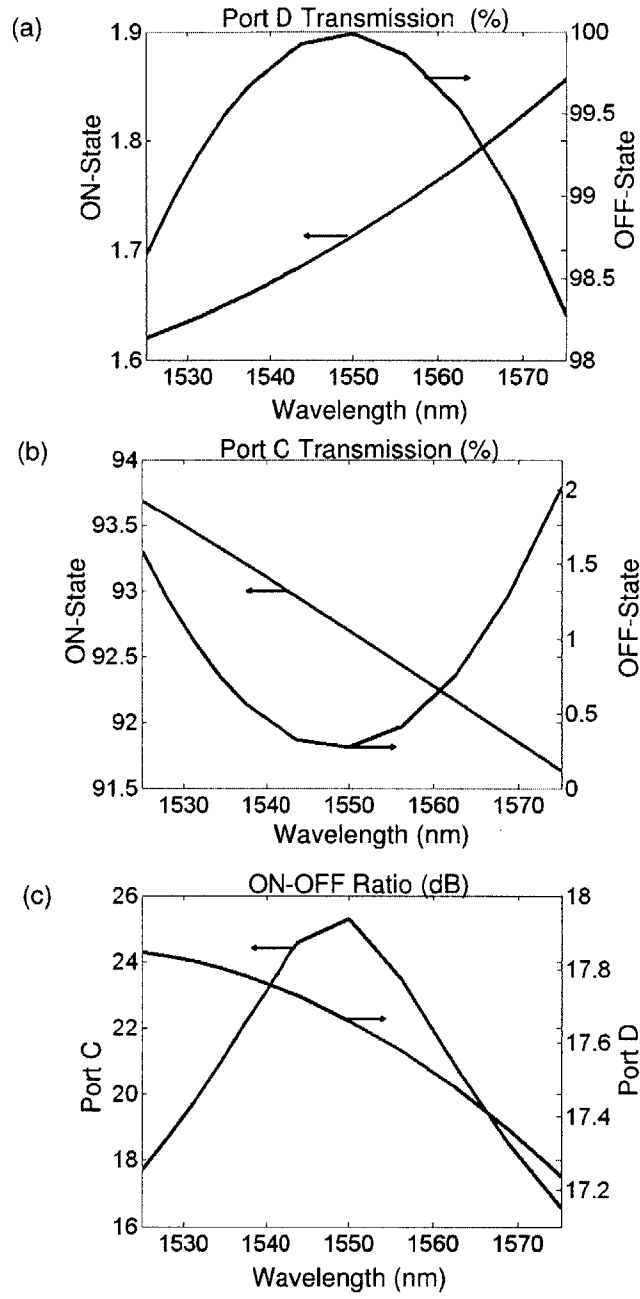


Figure 4-8: Two-waveguide switch performance (a) transmission to port D (%), (b) transmission to port C (%), (c) ON-OFF transmission ratio (dB)

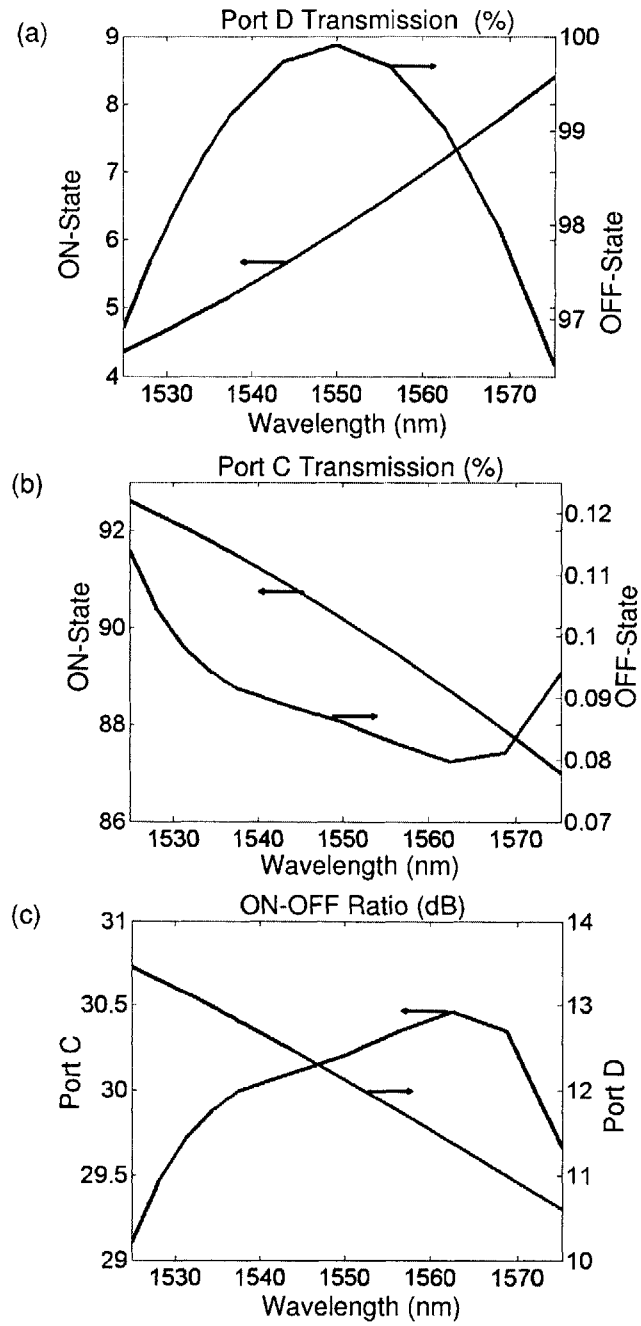


Figure 4-9: Three-waveguide switch performance (a) transmission of port D (%), (b) transmission of port C (%), (c) ON-OFF transmission ratio (dB)

Chapter 5

Carrier Lifetime in High-Index-Contrast Waveguides

The carrier lifetime in waveguides plays an important role in the performance of electro-optic modulators. In all-optical modulators as discussed in Sec.4.3 and forward-biased modulators without lifetime reduction as discussed in Sec.3.1.2, the highest modulation speed is determined by the carrier lifetime. In the modulators with lifetime reduction by ion implantation as discussed in Sec.3.1.1, the reduction of carrier lifetime effectively reduces the carrier density such that carriers can move at higher velocity and keep relatively low current, which enlarges modulation bandwidth. In contrast, smaller figure of merit can be achieved with longer carrier lifetime because it is inversely proportional to the carrier lifetime according to equation (3.12). Therefore, carrier lifetime in high-index-contrast waveguides needs special attention.

This chapter investigates carrier lifetime in bulk silicon and high-index-contrast waveguides. In particular, carrier recombination mechanisms in semiconductors and high-index-contrast waveguides are introduced in Sec.5.1. In Sec.5.2, the carrier lifetime measurement models are introduced. In Sec.5.3, the carrier lifetime and surface recombination velocity of bulk and SOI silicon wafers with different surface passivation is measured with the radio-frequency photo-conductance delay spectroscopy (RF-PCD) and the dual-color pump-probe method. In Sec.5.2, the carrier lifetime is calculated numerically in waveguides with PIN diodes.

5.1 Recombination Mechanisms

Four dominant recombination mechanisms: Auger recombination, radiative recombination, Shockley-Read-Hall(SRH) recombination and surface recombination, are discussed below.

5.1.1 Radiative Recombination

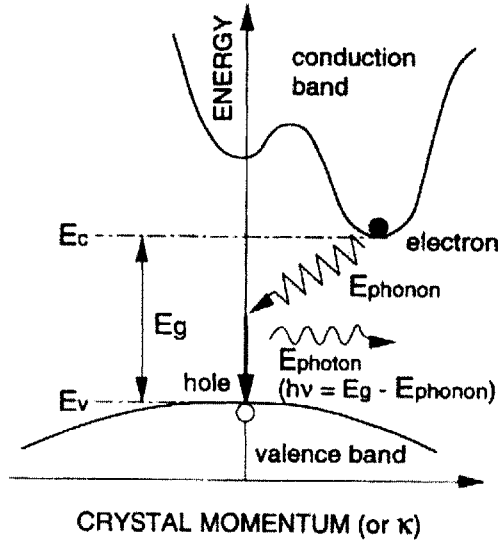


Figure 5-1: The Band-to-band radiative recombination processes in indirect band-gap semiconductors [20].

The band-to-band radiative recombination happens when conduction band electrons transit to the valence band, recombine with valence band holes, and release excess energy to photons. The transition of conduction band electrons to the valence band needs the assistance of a phonon in an indirect band-gap material, like Si and Ge, while it is not necessary in a direct band-gap material, like GaAs. As shown in Fig.5-1, the conduction band minimum locates differently from the valence band maximum in k -space for indirect band-gap materials. An electron in the conduction band minimum needs to change its momentum so as to transit to valence band maximum according to momentum conservation. The momentum change is usually

Table 5.1: B-values for silicon at 300°K

B-value (cm^3/s)	Publication year	References	Comments
6.0×10^{-14}	1967	cited in[58]	experimental data
1.2×10^{-15}	1969	cited in[58]	experimental data
1.1×10^{-14}	1972	[58]	experimental data

transferred to a phonon. Radiative recombination is a minor process for indirect band-gap materials.

The radiative band-to-band recombination rate can be written as [58]

$$U(t) = Bnp \quad (5.1)$$

where n and p are the density of free electrons and the density of free holes, respectively. Assuming that the electron density change is equal to the hole density change, we obtain

$$\begin{aligned}
\frac{dn}{dt} = \frac{dp}{dt} &= g(t) - U(t) \\
&= g(t) - B(n_0 + \Delta n)(p_0 + \Delta p) \\
&= g(t) - Bn_0p_0 - B(n_0 + p_0 + \Delta p)\Delta n \\
&= g'(t) - \frac{\Delta n}{\tau_{rad}}
\end{aligned} \quad (5.2)$$

where $g'(t) \equiv g(t) - Bn_0p_0$ is the effective generation rate. n_0 is the electron density and p_0 is the hole density in the equilibrium state. B is the radiative recombination coefficient. The radiative lifetime is defined as

$$\tau_{rad} = \frac{1}{B(p_0 + n_0 + \Delta p)} \quad (5.3)$$

The radiative recombination coefficient B is in units of cm^3/s . The radiative recombination coefficient B has been measured, as shown in Table 5.1. Even though the measured B-value differs in different measurements, a B-value of $B = 10^{-14}cm^3/s$ is reasonable for our calculation. The radiative lifetime is long at the low injection level

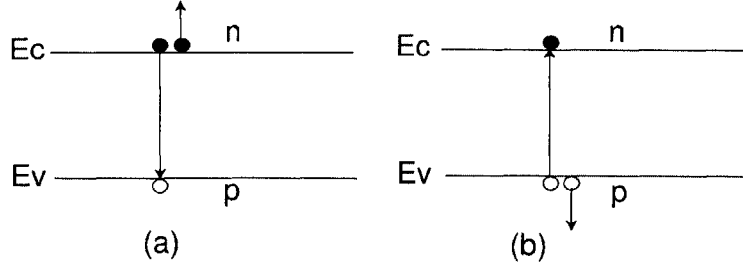


Figure 5-2: Band-to-band Auger recombination processes (a) eeh process and (b) ehk process [37].

of $\Delta n < 10^{18}/\text{cm}^3$, which is negligible. For example, $\tau_{rad} = 1\text{ms}$ when $\Delta n = 10^{17}/\text{cm}^3$ and $B = 10^{-14}\text{cm}^3/\text{s}$. The radiative recombination is important, however, for high injection $\Delta n > 10^{20}\text{cm}^3$, in which case $\tau_{rad} < 1\mu\text{s}$.

5.1.2 Auger Recombination

The band-to-band Auger recombination has two different processes: eeh-process in which one electron and one hole recombine and the energy is transferred to another electron as shown in Fig.5-2(a); ehk-process in which one electron and one hole recombine but the energy is transferred to another hole as shown in Fig.5-2(b). According to the Kerr's model in [59], the total Auger process recombination rate is given by the sum of the recombination rates of the two processes,

$$U_{Aur} = U_{eeh} + U_{ehk} = C_n n^2 p + C_p n p^2 \quad (5.4)$$

where n and p are the density of free electrons and the density of free hole, respectively. U_{eeh} is the eeh process recombination rate and U_{ehk} is the ehk process recombination rate. Assuming the electron density change is equal to the hole density change $\Delta p = \Delta n$ and substituting $n = n_0 + \Delta n$ and $p = p_0 + \Delta p$ into equation (5.4), the Auger lifetime for n-type semiconductors is

$$\tau_{aug} = \frac{\Delta n}{U_{Aur}} \simeq \frac{1}{C_n(n_0^2 + 2n_0\Delta p + \Delta n\Delta p) + C_p(n_0\Delta p + 2n_0p_0 + \Delta n\Delta p)} \quad (5.5)$$

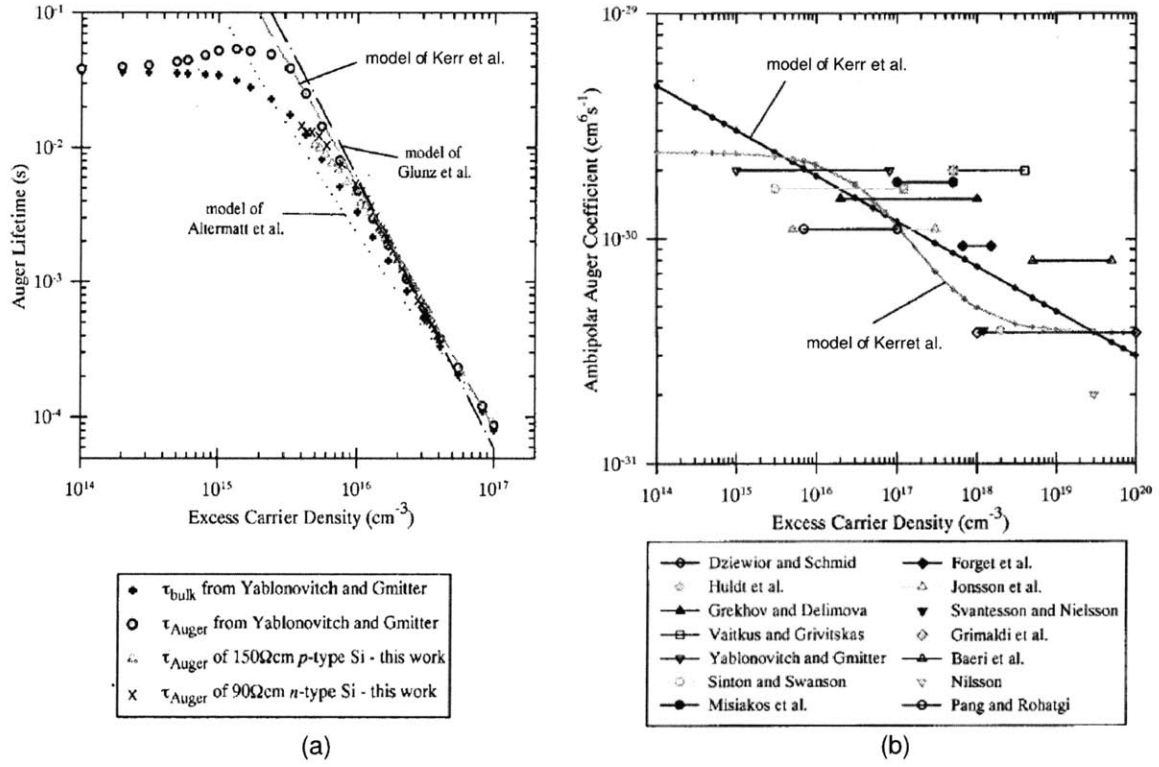


Figure 5-3: (a) High injection Auger lifetime measurements for high resistivity n-type silicon ($\sim 90V\text{cm}$) and p-type silicon ($\sim 150V\text{cm}$), (b) values for the ambipolar Auger coefficient as a function of excess carrier density [59].

where $\Delta p \gg p_0$ is used. The Auger lifetime for p-type semiconductors is

$$\tau_{\text{Aug}} = \frac{\Delta n}{U_{\text{Aug}}} \simeq \frac{1}{C_n(p_0\Delta n + 2n_0p_0 + \Delta n\Delta p) + C_p(p_0^2 + 2p_0\Delta p + \Delta n\Delta p)} \quad (5.6)$$

where $\Delta n \gg n_0$ is used. Therefore, the lifetimes for the low and high injection in the n-type semiconductor are,

$$\tau_{\text{Aug.low}} = \frac{1}{C_n N_D^2}, \quad \tau_{\text{Aug.high}} = \frac{1}{(C_n + C_p)\Delta p^2} \quad (5.7)$$

and for p-type semiconductor are

$$\tau_{\text{Aug.low}} = \frac{1}{C_n N_A^2}, \quad \tau_{\text{Aug.high}} = \frac{1}{(C_n + C_p)\Delta n^2} \quad (5.8)$$

where N_A and N_D are the densities of acceptors and donators, respectively.

The Auger recombination lifetime has been measured [59] on high resistivity n-type silicon ($\sim 90Vcm$) and p-type silicon ($\sim 150Vcm$). As shown in 5-3(a), the Auger lifetime is plotted as a function of the excess carrier density. A fitting of the measured Auger lifetime according to Kerr's model gives

$$\tau_{Aug} = \frac{1}{3 \times 10^{-27} \Delta p^{1.8}} \quad (5.9)$$

at an injection level of $\Delta p > 10^{15}/cm^3$. It can be seen that the Auger recombination only plays important roles in high injection $\Delta n > 10^{18}$ which gives $\tau_{Aug} < 1.3\mu s$. Using equations (5.8) and (5.9), the high injection Auger recombination coefficient can be evaluated as

$$C_a \equiv C_n + C_p = 3 \times 10^{-27} \Delta p^{-0.2} = 3 \times 10^{-27} \Delta n^{-0.2} \quad (5.10)$$

The measured Auger recombination coefficient agrees with the scaling in equation (5.9), as shown in Fig.5-3(b).

5.1.3 Shockley-Read-Hall (SRH) Recombination

The SRH recombination needs the assistance of residual recombination centers in the semiconductor. Four processes may happen with the existence of recombination centers, shown in Fig.5-4. The net recombination rates for electrons and holes are [37]

$$R_n - G_n = c_n N_t n (1 - f_t) - e_n N_t f_t \quad (5.11)$$

$$R_p - G_p = c_p N_t p f_t - e_n N_t (1 - f_t) \quad (5.12)$$

where n , p , N_t , and f_t are the density of free electrons, the density of free holes, the concentration of traps, the probability that traps are occupied, respectively. R_n and

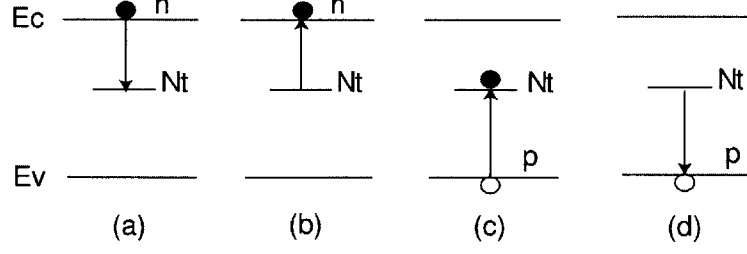


Figure 5-4: The energy band diagrams for the Shockley-Read-Hall generation-recombination processes: (a) electron capture $R_n = c_n n N_t (1 - f_t)$, (b) electron emission $G_n = e_n N_t f_t$, (c) hole capture $R_p = c_p p N_t f_t$, and (d) hole emission $G_p = e_p N_t (1 - f_t)$ [37].

R_p are the recombination rates of electrons and holes. G_n and G_p are the generation rates of electrons and holes. At steady state, if an excess electron-hole pair is created, another electron-hole pair will recombine through one of the four processes. Therefore, the net recombination rates for electrons and holes are equal,

$$R_n - G_n = R_p - G_p \quad (5.13)$$

Using equations (5.11), (5.12) and (5.13), the capture coefficient probability of the trap f_t is derived

$$f_t = \frac{c_n n + c_p p}{c_p p + c_n n + e_n + e_p} \quad (5.14)$$

Substituting (5.14) into equations (5.11) and (5.12), the net recombination rate becomes,

$$\begin{aligned} R_n - G_n = R_p - G_p &= \frac{c_n c_p n p - e_n e_p}{c_n n + e_n + c_p p + e_p} N_t \\ &= \frac{n p - \frac{e_n e_p}{c_n c_p}}{n \frac{1}{c_p N_t} + \frac{e_n}{c_n} \frac{1}{c_p N_t} + p \frac{1}{c_n N_t} + \frac{e_p}{c_p} \frac{1}{c_n N_t}} \\ &= \frac{n p - n_i p_i}{(n + n_1) \tau_p + (p + p_1) \tau_n} \end{aligned} \quad (5.15)$$

where substitutes have been made $\tau_n = \frac{1}{c_n N_t}$ (the electron lifetime), $\tau_p = \frac{1}{c_p N_t}$ (the hole lifetime), $n_1 = \frac{e_n}{c_n} = n_i e^{(E_t - E_i)/KT}$ and $p_1 = \frac{e_p}{c_p} = p_i e^{(-E_t + E_i)/KT}$ [60]. n_i , p_i , E_t

and E_i are the intrinsic electron and the hole densities, the trap energy level and the Fermi energy level, respectively. Under the quasi-neutral condition states $\Delta n \simeq \Delta p$, the SRH lifetime can be written as

$$\tau_{SRH} \equiv \frac{\Delta n}{R_n - G_n} = \Delta n \frac{(n + n_1)\tau_p + (p + p_1)\tau_n}{np - n_i p_i} \quad (5.16)$$

The SRH recombination is not important at low trap concentration because the lifetimes τ_n and τ_p are inverse to the trap concentration N_t , i.e., $\tau_n = \frac{1}{c_n N_t}$ and $\tau_p = \frac{1}{c_p N_t}$. The SRH recombination becomes significant at high trap concentration. Research has been done on control of SRH lifetime by increasing the trap concentration. Experimental exploration has found that the trap concentration can be increased by ion implantation [39]. At implantation doses above $3 \times 10^{14}/cm^2$ the measured lifetime reached a limit of $600 fs$.

5.1.4 Surface Recombination

The surface recombination becomes more and more important as the optical devices approach sub-micrometer size. Surface or inter surface recombination is induced by the surface states, i.e., dangling bonds at the inter surfaces [61]. A bulk silicon atom has four bonds and shares electrons through one bond with each of four surrounding silicon atoms. This is no longer true at the surface because the bonds are truncated and terminated with dangling bond. For purely clean surfaces, at least one bond for each atom is not occupied. For surfaces with impurities or defects, reconstruction happens at the surface and reduces the number of dangling bonds. Therefore, they have much less surface states, about two to three order of magnitude smaller than the intrinsic states in purely clean surface. This indicates that the surface recombination can be reduced by passivation with different materials and processes. The carrier lifetime due to passivation and carrier dynamic re-distribution are discussed in Sec.5.2 and the surface recombination velocity (SRV) is measured in Sec.5.3.

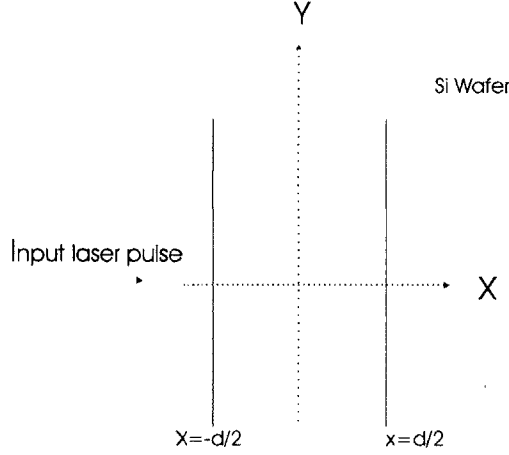


Figure 5-5: Sample configuration

5.2 Carrier Lifetime Measurement Model

5.2.1 Lifetime Model

In silicon, excess carriers can be generated by a visible pump laser. After generation, excess carriers start to re-distribute and recombine. The effective carrier lifetime is a measure of the time which carriers need to recombine until the carrier density becomes sufficiently low. During the recombination process, the dynamic carrier re-distribution causes the refractive index change which can be determined by measuring the reflectivity or transmission of a laser through the silicon film. In this section, the dynamics of carrier re-distribution and recombination with pump laser sources are investigated and the surface recombination lifetime as a function of silicon layer thickness, diffusion coefficient and surface recombination velocity (SRV) is derived.

A simplified 1-dimensional slab structure is used to model the process, as shown in Fig.5-5. A pump laser is injected into a slab Si wafer with a width of d . Here, we assume that the laser pulse is a unit Delta function,

$$G(t) = N_o \delta(t) \quad (5.17)$$

where N_o is the photon density in units of $/cm^2$, and $\delta(t) = 1$ at $t = 0$; $\delta(t) = 0$ at

$t \neq 0$. The excess minority carriers generated by the laser pulse can be written as [62]

$$n(x, 0) = g'_0(e^{-\alpha(x+d/2)} + Re^{-\alpha(-x+d/2)}) \quad (5.18)$$

where $g'_0 = N_0\alpha(1-R)/(1-(Re^{-\alpha d})^2)$ is the generation rate, α is the optical absorption coefficient and R is the reflection coefficient on the Si/air surface. The generated excess minority carriers have to satisfy the continuity equation

$$\frac{\partial n(x, t)}{\partial t} = D_n \frac{\partial^2 n(x, t)}{\partial x^2} - \frac{n(x, t)}{\tau_b} \quad (5.19)$$

with the boundary conditions at the two Si/air surfaces

$$D_n \frac{\partial n(x, t)}{\partial x} \Big|_{x=-d/2} = S \cdot n(x, t) \Big|_{x=-d/2} \quad (5.20)$$

$$D_n \frac{\partial n(x, t)}{\partial x} \Big|_{x=d/2} = -S \cdot n(x, t) \Big|_{x=d/2} \quad (5.21)$$

The general solution of equation (5.19) can be written as a linear superposition of many modes [62]

$$n_0(x, t) = e^{-t/\tau_b} \sum_n [A_n e^{-\alpha_n^2 D_n t} \cos(\alpha_n x) + B_n e^{-\beta_n^2 D_n t} \sin(\beta_n x)] \quad (5.22)$$

which is needed to satisfy the initial condition (5.18) and the boundary conditions (5.20) and (5.21). The coefficients A_n and B_n are determined by the initial condition (5.18)

$$A_n = \frac{2g'_0 \alpha_n}{\alpha_n d + \sin(\alpha_n d)} \int_{-\frac{d}{2}}^{\frac{d}{2}} (e^{-\alpha(x+\frac{d}{2})} + Re^{-\alpha d} e^{-\alpha(-x+\frac{d}{2})}) \cos(\alpha_n x) dx \quad (5.23)$$

$$B_n = \frac{2g'_0 \beta_n}{\beta_n d + \sin(\beta_n d)} \int_{-\frac{d}{2}}^{\frac{d}{2}} (e^{-\alpha(x+\frac{d}{2})} + Re^{-\alpha d} e^{-\alpha(-x+\frac{d}{2})}) \sin(\beta_n x) dx \quad (5.24)$$

Each term in solution (5.22) decays with its own characteristic decay constance

$$\frac{1}{\tau_{eff.n}} = \frac{1}{\tau_b} + \alpha_n^2 D_n \quad (5.25)$$

where $\tau_{eff.n}$ is the effective lifetime for the n th mode. Since the term $1/\tau_b$ is from the bulk recombination and the term $\alpha_n^2 D_n$ is induced by surface recombination, we can define the surface recombination lifetime to be $\tau_{sur} = 1/(\alpha_n^2 D_n)$. Because only one single mode is usually dominant in the measurement and the bulk lifetime τ_b is much longer than τ_{sur} such that τ_b can be neglected for thin silicon layer, the measured effective lifetime takes the simple form,

$$\frac{1}{\tau_{mea}} = \alpha_n^2 D_n \quad (5.26)$$

Substituting the n th-cosine and sine terms in (5.22) separately into (5.20) results in

$$\cot(\alpha_n d/2) = D_n \alpha_n / S \quad (5.27)$$

$$\tan(\beta_n d/2) = -D_n \beta_n / S \quad (5.28)$$

At low surface recombination velocities $S \rightarrow 0$, the right-hand side of (5.27) goes to infinity. Therefore, $\cot(\alpha_n d/2)$ can be approximated to $1/(\alpha_n d/2)$. Then (5.27) becomes $D_n \alpha_n^2 = 2S/d$. The measured lifetime becomes

$$\tau_{mea} = \frac{1}{D_n \alpha_n^2} = \frac{d}{2S} \quad (5.29)$$

When $S \rightarrow \infty$, equation (5.27) leads to $\cot(\alpha_n d/2) \rightarrow 0$, which means $\alpha_n d/2 \rightarrow$

$\pi/2$. Then the measured lifetime with fast surface recombination is

$$\tau_{mea} = \frac{d^2}{D_n \pi^2} \quad (5.30)$$

Therefore, the measured lifetime can be expressed as for large or small S

$$\tau_{mea} = \frac{d}{2S} + \frac{d^2}{D_n \pi^2} \quad (5.31)$$

The previous derivation uses unit Delta function source to analyze the decay of excess minority carriers. Using convolution technique, the analysis is readily to be generalized to various laser sources, such as step and Gaussian function. The detailed results can be found in [62].

5.2.2 Effective Lifetime

The relaxation of a carrier density change Δn_e in a silicon film is determined by four effects (Shockley-Read-Hall (SRH), radiative, Auger and surface recombination)[63]:

$$\frac{d\Delta n_e}{dt} = g_L(t) - \frac{\Delta n_e}{\tau_{eff}} \quad (5.32)$$

where

$$\begin{aligned} \frac{1}{\tau_{eff}} &= \frac{1}{\tau_{SRH}} + \frac{1}{\tau_{rad}} + \frac{1}{\tau_{Aug}} + \frac{1}{\tau_{sur}} \\ \frac{1}{\tau_b} &= \frac{1}{\tau_{SRH}} + \frac{1}{\tau_{rad}} + \frac{1}{\tau_{Aug}} \end{aligned}$$

with

$$\begin{aligned} \tau_{SRH} &= \Delta n \frac{(n + n_1)\tau_p + (p + p_1)\tau_n}{np - n_i p_i} \\ \tau_{rad} &= \frac{1}{B(n_0 + p_0 + \Delta p)} \\ \tau_{Aug} &= \frac{1}{3 \cdot 10^{-27} \Delta p^{1.8}} \\ \tau_{sur} &= \frac{d}{2S} + \frac{d^2}{D_a \pi^2} \end{aligned}$$

where g_L is the carrier generation rate, τ_{eff} and τ_b are the effective lifetime and the bulk lifetime, τ_{SRH} , τ_{rad} , τ_{Aug} and τ_{sur} are the recombination lifetimes of the SRH, the radiative, the Auger and the surface recombination, respectively, B is the radiative recombination coefficient, d is the sample thickness, S is the surface recombination velocity and D_a is the ambipolar diffusion coefficient.

Regular SRH recombination in pure crystalline silicon is on the order of $\tau_{SRH} \approx 10ms$ [64] and can be neglected. Radiative recombination in silicon is a minor process because of the indirect band gap [58], and Auger recombination plays only a significant role at very high carrier densities [59]. For thin semiconductor layers, surface recombination is the dominant effect. The lifetime due to surface recombination in a thin layer is given by [63]:

$$\tau_{sur} = \frac{1}{D_a} \left(\frac{d}{\pi} \right)^2 + \frac{d}{2S} \quad (5.33)$$

where d is the film thickness, S is the surface recombination velocity, and D_a is the ambipolar carrier diffusion coefficient. The derivation of equation (5.33) will be discussed in Sec.5.2. The first item in (5.33) dominates for strong surface recombination ($Sd/D_a \gg 1$) and the second term for weak surface recombination ($Sd/D_a \ll 1$). The surface recombination is determined by the surface recombination velocity S , which strongly depends on the surface passivation of the silicon film.

The effective lifetime τ_{eff} in equation (5.32) is determined by the bulk lifetime τ_b , the surface recombination velocity (SRV) S , the wafer thickness d and the diffusion coefficient [61], as plotted in Fig.5-6. When the SRV is very small, the bulk lifetime dominates the effective lifetime $\tau_{eff} = \tau_b$. When SRV is very large, the effective lifetime approaches to the diffusion lifetime $\tau_{eff} = d^2/D\pi$. For a medium level of SRV, the effective lifetime is surface-recombination dependent $\tau_{eff} = d/2S$. The surface recombination velocity can be extracted if the lifetime is located in the medium SRV domain, which will be applied to calculate SRV in the lifetime measurement in Sec.5.3. The effective lifetime with silicon film thicknesses of $650\mu m$ and $1.5\mu m$ is plotted in Fig.5-6(a) and (b), respectively.

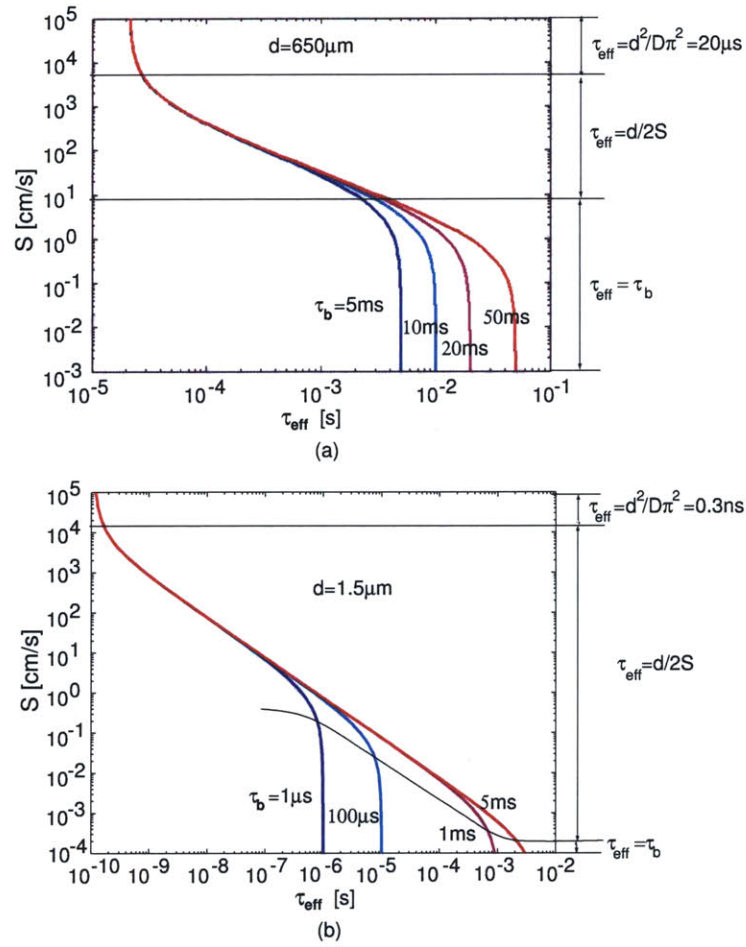


Figure 5-6: Effective lifetime with the wafer thickness (a) $d = 625 \mu\text{m}$ and (b) $d = 1.5 \mu\text{m}$.

5.3 Carrier Lifetime Measurement

To determine the surface recombination velocity, the carrier lifetimes in both $650 \mu\text{m}$ thick bulk crystalline silicon wafers, and SOI-wafers with a $1.5 \mu\text{m}$ low P-doped silicon layer (resistivity is around $1 - 30 \Omega\text{cm}$) on top of a $3 \mu\text{m}$ SiO_2 layer made by SIMOX [52], are studied. The carrier lifetimes for the bulk $650 \mu\text{m}$ thick silicon wafer and SOI-layers are expected to be on the order of $500 \mu\text{s}$ and less than $10 \mu\text{s}$, respectively, at low injection [63]. The surfaces of both kinds of samples are passivated with two common types of oxide: (i) a natural thermally grown silicon oxide and (ii) an oxide layer made by low temperature oxidation (LTO). All oxide layers were $1000 \text{ \AA}$ thick.

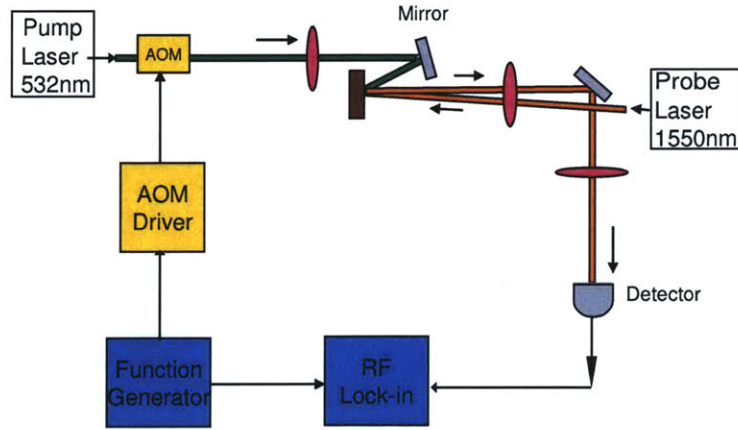


Figure 5-7: Implementation of the dual-color carrier lifetime measurement.

The natural oxide (i) was thermally grown at 1000°C under atmospheric pressure, the LTO layer (ii) was made at temperature 400°C and pressure 200mTorr.

First, we extracted the surface recombination velocity from the passivated $650\text{ }\mu\text{m}$ thick bulk silicon wafers by means of radio-frequency photo-conductance delay spectroscopy (RF-PCD) [65]. Due to the thick bulk wafer the carrier lifetime is in the range of $10\text{-}100\text{ }\mu\text{s}$, such that a slow measurement scheme like the RF-PCD is suitable for its characterization. The surface recombination velocity values for the two different oxides were found to range from a low 250 cm/s for the thermal oxide, that leads to a long carrier lifetime in the wafer of $137\text{ }\mu\text{s}$, to a high surface recombination velocity of 1600 cm/s for the LTO (Table 5.2) leading to an effective carrier lifetime of only $35\text{ }\mu\text{s}$. These values are in good agreement with previously published data [63]. Based on the values for the surface recombination velocity gained from the RF-PCD method, we expect overall carrier lifetimes τ_{tot} between 45 and 260 ns for the $1.5\text{ }\mu\text{m}$ thick silicon-on-insulator films, see Table 5.2. Thus, a much faster measurement scheme is necessary for their characterization.

A dual-color measurement technique [66] as schematically illustrated in Fig.5-7 was used to measure the carrier lifetime in the thin SOI layer. The method is capable of measuring lifetimes as short as 30 ns . The reflection from the surface of the wafer under test is measured with a low-power probe laser at 1550 nm , which is focused to a spot with $120\text{ }\mu\text{m}$ radius on the sample. A 532 nm laser with 80 mW of average

Table 5.2: Carrier lifetime measurement results

Passivation type (thickness)	Carrier Lifetime in 650 μm Si wafer (μs)*	Surface Recombination Velocity(cm/s)*	Carrier Lifetime in 1.5 μm Si layer of SOI (ns)	
			Expected*	Measured**
Thermal Oxide (1001Å)	137	250	260	150
LTO (1147Å)	35	1600	45	~ 15

* Radio-frequency photo-conductance decay spectroscopy(RF-PCD)

**Dual-Color Method

power is focussed to the same spot size on the sample and brought to overlap with the probe laser. The 532 nm light has an absorption length of 0.94 μm in the sample, such that about 80% of the incident light is absorbed in the 1.5 μm silicon film. The 532 nm beam is then modulated sinusoidally at frequencies ranging from 30 kHz to 10 MHz with an acousto-optic modulator (AOM), resulting in a time-dependent carrier injection due to the absorption in the sample. Due to the plasma effect (see Sec.4.3) the pump light causes a change in both absorption and refractive index of the thin silicon film. The reflection from the film is measured by lock-in detection. According to (5.32), for slow modulation of $g_L(t)$, the carrier density and the changes in refractive index and absorption can follow the modulated light. However, for fast modulation the carrier density cannot follow the modulation. Thus, the corner frequency of the transfer characteristic describing index and absorption changes due to pump light modulation is determined by the carrier lifetime in the sample. This low-pass effect is clearly observed in the characterization of the SOI-wafer passivated with a thermal oxide, see Fig.5-8. From the corner frequency of the normalized reflectivity change a carrier lifetime of 150ns is extracted. This value agrees within 50% with the value expected from the measured value for the surface recombination velocity, which constitutes a rather good agreement taking into account that the buried oxide of the SOI-wafer is a thermal oxide, which is grown during the manufacturing process of the SOI-wafer with the Smart Cut Technique [53]

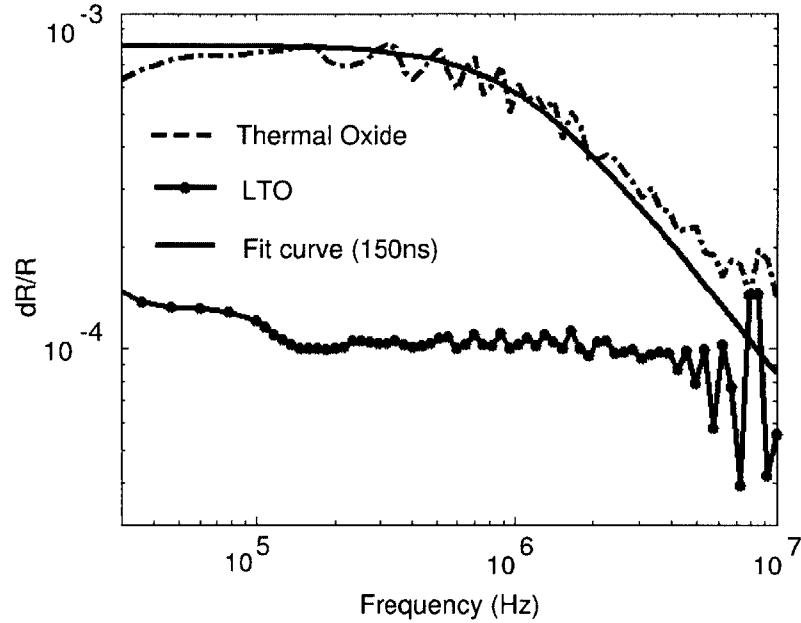


Figure 5-8: Modulated reflectivity versus frequency for $1.5\mu m$ thick SOI wafers passivated by thermal and LTO oxides, respectively.

In contrast, the carrier lifetime of the SOI-layers passivated with LTO could not be measured with the two-color pump-probe technique. As shown in Fig.5-8, in these samples the normalized change in reflectivity lies about one order of magnitude below the reflectivity change for the sample with the thermal oxide. This fact indicates a carrier lifetime about one order of magnitude lower, since the magnitude of the change in reflectivity is proportional to the carrier lifetime at low frequencies. From the data at low frequencies we can infer a carrier lifetime at least ten times shorter than in the SOI-wafer passivated with a thermal oxide, such that 15ns can be assumed as an upper boundary for the carrier lifetime in these samples.

In a silicon single-mode waveguide, the layer thickness will be even much thinner, on the order of 550 nm. For a 550 nm thin crystalline silicon layer with a thermally grown silicon oxide cover layer we estimate a carrier lifetime on the order of 50-100ns. This short carrier lifetime enables the fabrication of the two-waveguide switches with modulation frequencies 10 or 100MHz depending on which passivation

layer is used. These measurements verify the feasibility of the all-optical switches discussed in Sec.4.3.

5.4 Carrier Lifetime Extraction in a Waveguide with PIN Diode

Carrier lifetime in bulk silicon and SOI wafers has been investigated earlier in this chapter. In silicon waveguides with PIN diodes, the carrier lifetime may vary according to doping profile, dimensions and geometries. Except the dual-color method described in Sec.5.3, there are more methods to measure the lifetime in waveguides with PIN diodes, which are different from the measurement in silicon films because of the existence of metal contacts and PIN diodes.

Several methods have been used to measure carrier lifetime in PIN/PN diodes. Kuno proposed a method to measure the transient forward and reverse currents when the voltage applied to PIN/PN diodes is switched from forward-biased to reverse-biased from which carrier lifetime can be determined [67]. Secondly, the dependence of the reactive impedance of a PIN/PN diode on frequency is used in a RF technique to measure carrier lifetime. This method is also referred as Caverley's method. In addition, Szczesny [68] proposed a method based on the measurement of rapid changes of the voltage drop across a PIN diode. Oussalah [69] proposed another method to extract carrier lifetime by measuring the conductance at different forward-bias-voltages at a fixed frequency. A detailed discussion of the first two methods will be presented later in this section.

5.4.1 Kuno's Method (Reverse Recovery)

A reverse recovery method has been developed to measure the lifetime in a PIN diode. This method is widely used in high-power PIN diodes. The reverse current i_R as well as the forward current i_F are related to the charge Q stored in the base region by time constants τ_R and τ_F , respectively, as shown in Fig.5-9(b). The charge variation

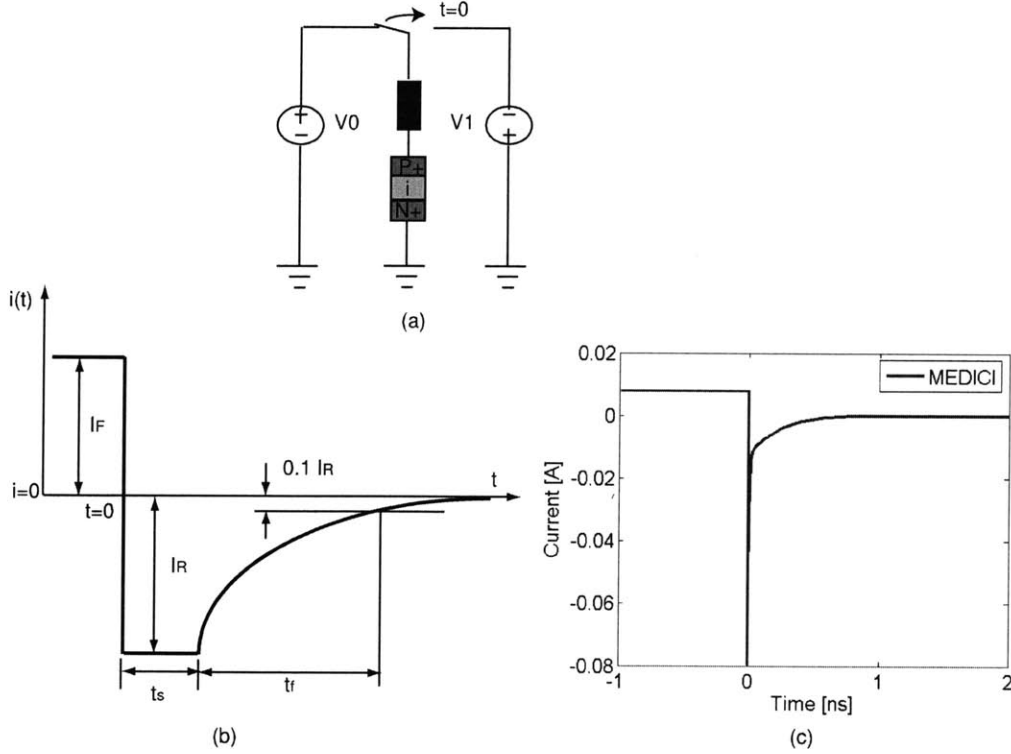


Figure 5-9: (a) Diode switching circuit, (b) transient reverse recovery of a PIN diode with thick base region[69] and (c) with a thin base region.

in the low doped base region in a PIN diode is

$$\frac{dQ}{dt} = -\frac{Q}{\tau} + I(t) \quad (5.34)$$

where τ and $I(t)$ are the carrier lifetime and current, respectively. In the steady-state, we have the initial conditions at $t = 0$

$$Q_F = I_F \tau$$

Switch the voltage from the forward bias to the reverse bias at time $t = 0$ as in Fig.5-9(a). It has been observed experimentally that the current keeps constant I_R for a duration t_s [67][69]. At time $t = t_s$, it's assumed that the stored charge is zero $Q_{t_s} = 0$ [69]. Substituting the conditions at $t = 0$ and $t = t_s$ into equation (5.34), we

can obtain the charge solution between $t = 0$ to $t = t_s$:

$$Q(t) = \frac{I_F \tau}{1 - e^{-t_s/\tau}} (e^{-t/\tau} - e^{-t_s/\tau}) \quad (5.35)$$

Putting current $I(t_s) = -I_R$, $t = t_s$ and equation (5.35) into equation (5.34), we get

$$\tau = \frac{t_s}{\ln(1 + \frac{I_F}{I_R})} \quad (5.36)$$

If a circuit inductance is used to prevent a rapid current change, another model can be derived [70].

Unfortunately, Kuno's method only works for a thick base thickness [71]. Fig.5-9(c) is a simulated result from 2D semiconductor simulator MEDICI for a PIN diode with a thin base thickness of $0.9\mu m$, as shown in Fig.3-1. t_s can not be clearly identified in the transient current plot and, therefore, Kuno's method is not appropriate for diodes with a thin base thickness.

5.4.2 Caverly's Method (RF Technique)

The capacitance and conductance of PIN diodes depends on frequency, the diode's geometry and electrical properties, such as carrier mobilities, carrier lifetime and etc. Microwave and RF measurements can be used to determine the carrier lifetime in silicon PIN RF switching diodes. Swept frequency measurements are used to determine the reactance minimum of the PIN diode. The frequency at which the PIN diode reactance reaches minimum is shown to be inversely proportional to the i-region carrier lifetime [72].

The PN diode theory is used to get insight into the extraction of the carrier lifetime from RF-measurement. The frequency-dependent conductance and capacitance of PN diodes are [33]

$$G_D = \frac{G_0}{\sqrt{2}} (\sqrt{1 + (\omega\tau)^2} + 1)^{\frac{1}{2}} \quad (5.37)$$

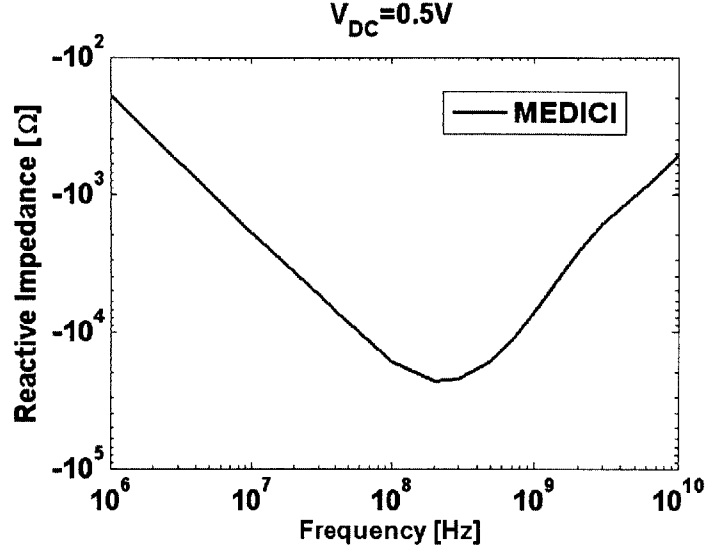


Figure 5-10: Reactive impedance as a function of frequency.

$$C_D = \frac{G_0}{\sqrt{2}\omega} (\sqrt{1 + (\omega\tau)^2} - 1)^{\frac{1}{2}} \quad (5.38)$$

where G_0 is the DC conductance, ω is the RF frequency and τ is the carrier lifetime. The impedance of the PN diode can be written as

$$Z = \frac{1}{G_D + j\omega C_D} = \frac{1}{\sqrt{2}G_0} \left[\frac{\sqrt{1 + (\omega\tau)^2} + 1}{\sqrt{1 + (\omega\tau)^2}} - j \frac{\sqrt{1 + (\omega\tau)^2} - 1}{\sqrt{1 + (\omega\tau)^2}} \right] \quad (5.39)$$

Differentiating the reactive impedance X , i.e., the imaginary part of the impedance Z , with respect to $\omega\tau$, we obtain

$$\frac{\partial X}{\partial(\omega\tau)} = 0 \quad (5.40)$$

The above equation has three solutions: $\omega\tau = 0$, 1 and ∞ . Therefore, the carrier lifetime can be extracted from RF measurement at the minimum $\tau = 1/2\pi f_{min}$.

A simulation has been done with MEDICI to calculate the reactive impedance of a PIN diode at different frequencies. The dimensions of the diode are chosen to be $w = 520nm$, $h = 240nm$, $w_r = 200nm$ and $h_s = 50nm$, as shown in Fig.3-1. In

the MEDICI simulation the lifetimes of electron and hole are set to $\tau_p = \tau_n = 1ns$, and the lifetime is SRH recombination dominant according to Sec.5.1. Therefore, the effective lifetime is

$$\begin{aligned}\tau_{eff} &= \frac{1}{\frac{1}{\tau_{SRH}} + \frac{1}{\tau_{rad}} + \frac{1}{\tau_{Aug}} + \frac{1}{\tau_{sur}}} \\ &= 1.9ns\end{aligned}\tag{5.41}$$

where we have assumed

$$\tau_{SRH} = \Delta n \frac{(n + n_1)\tau_p + (p + p_1)\tau_n}{np - n_i p_i} = 2ns$$

$$\tau_{rad} = \frac{1}{B(n_0 + p_0 + \Delta p)} = 0.3ms$$

$$\tau_{Aug} = \frac{1}{3 \cdot 10^{-27} \Delta p^{1.8}} = 83\mu s$$

$$\tau_{sur} = \frac{h}{2S} + \frac{h^2}{D_a \pi^2} = 0.125\mu s$$

and

$$\Delta n \simeq \Delta p \simeq n \simeq p \simeq 10^{17}/cm^3$$

$$n \gg n_1$$

$$p \gg p_1$$

$$B = 10^{-14} cm^3/s$$

$$S = 100cm/s$$

$$D_a = 20cm^2/s$$

The simulated reactive impedance is plotted as a function of frequencies in Fig.5-10. The minimum of the reactive impedance is at $f = 200MHz$ which gives the lifetime $\tau = 1/2\pi 200MHz = 1.6ns$. This is close to the lifetime set in the MEDICI simulation $\tau = 1.9ns$.

Chapter 6

Conclusions and Future Work

6.1 Conclusions

This thesis investigated electrically driven Mach-Zehnder modulators based on high-index contrast silicon waveguide technology and electronic carrier injection. The modulators based on four different structures, the forward-biased PiN diode with lifetime reduction, the forward-biased diode without lifetime reduction, the reverse-biased PIN/PN diode and metal-oxide-semiconductor (MOS) structure, were studied. For the first type modulator, simulations showed that with a graded doping profile, operating frequencies as large as 34GHz were reached while maintaining a low insertion loss of 3dB and a low figure of merit of $V_{\pi}L = 6Vmm$. The experiment has demonstrated a 3dB corner frequency of 1-2GHz which is still smaller than the designed bandwidth of 34GHz. This could result from the slow RC delay due to the high contact resistance and the external circuits. The second type of modulator, based on forward-biased PiN diodes without carrier lifetime reduction, has a low figure of merit, very low voltage and extremely low power in the low frequency regime. The measurement showed that the DC bias is around 1-2 V with 5-10 mA current. Around 75% modulation depth was obtained at low frequencies with RF voltage $V_{RF} = 0.1V$ and $P_{RF} = 1mW$. A very low DC figure of merit $V_{\pi} \cdot L = 0.02V \cdot cm$ illustrates the high modulation efficiency. At a frequency of 10GHz, a RF power consumption of 100mW for 25% modulation depth and a figure of merit of $V_{\pi} \cdot L = 0.28V \cdot cm$ were achieved.

A pre-compensation technique was used to increase the bandwidth. A high pass filter consisting of a parallel resistor and capacitor extends the modulator bandwidth from 100MHz to 5GHz experimentally. Thirdly, the reverse-biased PiN/PN modulator was designed to operate up to 80GHz while the measured f_{3dB} is only ~ 1 -2GHz. This discrepancy may be caused by the RC delay due to the experimental setup and imperfect fabrication. Finally, the performance of the MOS modulator was analyzed. The MOS structure is compatible with the current chip processes while the RF power is too large for most CMOS technologies.

A split-ridge waveguide (SRW) was proposed and investigated. The SRW provides good heat sinking through the Si-slab. The coupling coefficient insensitive to waveguide width makes the coupler based on SRW broadband, covering the whole C-band with very low OFF-state insertion loss. The larger tolerance to fabrication error in SRW leads to 80 times optical power loss reduction at the Mach-Zehnder interferometer output compared with buried waveguides. An electrically driven Mach-Zehnder waveguide modulator based on the high-index-contrast silicon split-ridge waveguide technology and electronic carrier injection has excellent optical and carrier confinement, together with good thermal heat sinking and forward biased operation. This electro-optic modulator exhibits a small signal modulation bandwidth of beyond 20GHz, a figure of merit of $V_\pi \cdot L = 0.5V \cdot cm$ and an insertion loss of about 4 dB.

Finally, all-optical switches based on optical carrier-injection in high index contrast Si/SiO_2 split-ridge-waveguide (SRW) couplers were analyzed. The waveguide devices are suitable for the construction of low-loss optical switch matrices as well as fast optical switching. These devices exhibit robustness against fabrication tolerances, improved heat sinking, good carrier confinement and high uniformity in transmission over the entire C-band of optical communications in contrast to comparable devices based on buried or ridge waveguides. Reasonably low electrical switching power of 1-10mW is predicted for switching frequencies of 1MHz-1GHz and faster switching speed can be achieved by reducing carrier lifetime.

6.2 Future Work

For the applications in optical interconnection, a modulator with low power consumption, large modulation efficiency and small size is required. A modulator based on resonators is found to have lower power dissipation, enhanced modulation efficiency, and relatively smaller size compared to the Mach-Zehnder interferometer structure. Resonators can be made of rings, photonic crystal or other structures [28]. Recently, a ring modulator attracts a lot of attention [73]. A ring modulator works for narrow bandwidth and needs very small electrical power, on the order of $10\mu W$, to modulate light. The power dissipation in a resonator can be reduced by increasing the finesse which is defined as the ratio of resonator free spectral range (FSR) over the resonator bandwidth [28].

For the applications in high-speed high-resolution optical sampling systems, a modulator needs to have low nonlinearity. Nonlinearity distorts signals by introducing additional higher harmonic terms besides the fundamentals. The nonlinearity in a Mach-zehnder interferometer is induced by the Mach-zehnder sinusoidal transmission characteristics, as shown in Chapter 2. The second order nonlinearity can be suppressed by setting the bias $\phi_b = \pi/2$. A dual-output modulator can be applied to suppress the light source noise and nonlinearity [8]. The nonlinearity can also be induced by the PIN diode itself because the forward-biased current is exponentially proportional to the biased voltage. The carrier distribution doesn't follow the external driving voltage linearly. A careful design on the doping profile and the geometry of PIN structures could reduce the nonlinearity.

The silicon modulator can achieve a bandwidth as large as 20GHz and a figure of merit of 4Vcm based on reverse-biased PN structure [14], and a figure of merit as low as 0.28Vcm at 10GHz based on forward-biased PIN structure [74]. It is difficult to obtain bandwidth as large as tens of gigahertz while keeping low figure of merit, such as less than 1Vcm. It is worth further investigating the pre-compensation scheme for the forward-biased PIN modulator to extend the bandwidth and novel designs for the reverse-biased PN modulator to reduce figure of merit.

Appendix A

DC/AC Characteristics of PIN Diode

Free Carrier Distribution with Unit Injection Efficiency

The free carrier distribution in the PIN intrinsic region is governed by the ambipolar diffusion equation

$$\frac{\partial n}{\partial t} = D_a \frac{\partial^2 n}{\partial x^2} - \frac{n}{\tau_a} \quad (\text{A.1})$$

with the boundary conditions

$$J_{+d} = 2qD_n \left. \frac{dn}{dx} \right|_{x=+d} \quad (\text{A.2})$$

$$J_{-d} = -2qD_p \left. \frac{dn}{dx} \right|_{x=-d} \quad (\text{A.3})$$

where the injection efficiency at both ends of the PIN diode is assumed to be unity and the electron current in p^+ -region and the hole current in n^+ -region are zero, as shown in Fig.2-8 of Chapter 2. Assuming the current density input is $J = J_0 + J_1 e^{j\omega t}$, the carrier density can be written as $n = n_0 + n_1 e^{j\omega t}$ by neglecting the higher order terms based on small signal assumption. Substituting n and J into the ambipolar

equation (A.1) and boundary conditions (A.2) (A.3), we obtain

$$\frac{\partial}{\partial t}(n_0 + n_1 e^{j\omega t}) = D_a \frac{\partial^2}{\partial x^2}(n_0 + n_1 e^{j\omega t}) - \frac{n_0 + n_1 e^{j\omega t}}{\tau_a} \quad (\text{A.4})$$

$$J_0 + J_1 e^{j\omega t} = 2qD_n \frac{d}{dx}(n_0 + n_1 e^{j\omega t})|_{x=+d} \quad (\text{A.5})$$

$$J_0 + J_1 e^{j\omega t} = -2qD_p \frac{d}{dx}(n_0 + n_1 e^{j\omega t})|_{x=-d} \quad (\text{A.6})$$

From equation (A.4) to (A.6), we have the equations for DC carrier distribution

$$\frac{\partial^2 n_0}{\partial x^2} - \frac{n_0}{L_a^2} = 0 \quad (\text{A.7})$$

$$J_0 = 2qD_n \frac{dn_0}{dx}|_{x=+d} \quad (\text{A.8})$$

$$J_0 = -2qD_p \frac{dn_0}{dx}|_{x=-d} \quad (\text{A.9})$$

and equations for AC carrier distribution

$$\frac{\partial^2 n_1}{\partial x^2} - \frac{n_1}{L_a^2}(1 + j\omega\tau_a) = 0 \quad (\text{A.10})$$

$$J_1 = 2qD_n \frac{dn_1}{dx}|_{x=+d} \quad (\text{A.11})$$

$$J_1 = -2qD_p \frac{dn_1}{dx}|_{x=-d} \quad (\text{A.12})$$

where $L_a = \sqrt{D_a\tau_a}$.

The DC solutions can therefore be found

$$n_0(x) = \frac{\tau_a J_0}{2qL_a} \left[\frac{\cosh(x/L_a)}{\sinh(d/L_a)} - B \frac{\sinh(x/L_a)}{\cosh(d/L_a)} \right] \quad (\text{A.13})$$

where $B = \frac{\mu_n - \mu_p}{\mu_n + \mu_p}$.

Putting $n_1 = e^{\frac{x}{L}}$ into equation (A.10), we get $L = \frac{L_a}{\sqrt{1+jw\tau_a}}$. Similarly, we can obtain AC solution by replacing L_a and J_0 by L and J_1 in equation (A.13):

$$n_1(x) = \frac{\tau_a J_1}{2qL} \left[\frac{e^{\frac{x}{L}} + e^{-\frac{x}{L}}}{e^{\frac{d}{L}} - e^{-\frac{d}{L}}} - B \frac{e^{\frac{x}{L}} - e^{-\frac{x}{L}}}{e^{\frac{d}{L}} + e^{-\frac{d}{L}}} \right] \quad (\text{A.14})$$

$$\bar{n} = \frac{\int_{x=-d}^{x=d} n_1(x) dx}{2d} = \frac{\tau_a J_1}{2dq} \quad (\text{A.15})$$

Free Carrier Distribution with Non-unit Injection Efficiency

In practice, the injection efficiency is not unity since the total current is not only due to the holes and electrons injected into the i-region, but also due to the minority holes into n^+ -region and the minority electrons into p^+ -region, as shown in Fig.A-1. The total forward-biased current density can be explicitly expressed as

$$J_F = J_{nP} + J_m + J_{pN} \quad (\text{A.16})$$

where J_{nP} and J_{pN} are the diffusion current densities due to the injection of electrons injected into the p^+ -region and holes injected into the n^+ -region [34]. J_m is due to the rate at which holes and electrons recombine within the i-region. So carrier distribution due to J_{nP} and J_{pN} can be regarded as constant along x . The carrier distribution due to J_m has hyperbolic distribution along x according to the ambipolar equation (A.1). Therefore, the carrier distribution in (A.13) and (A.14) is only due to J_m and doesn't include the contribution from J_{nP} and J_{pN} .

At the edge of i-region $x = -d$, the hole density is approximately equal to the high p-type doping, $p(-d) = N_A$, and the electron density depends on the applied

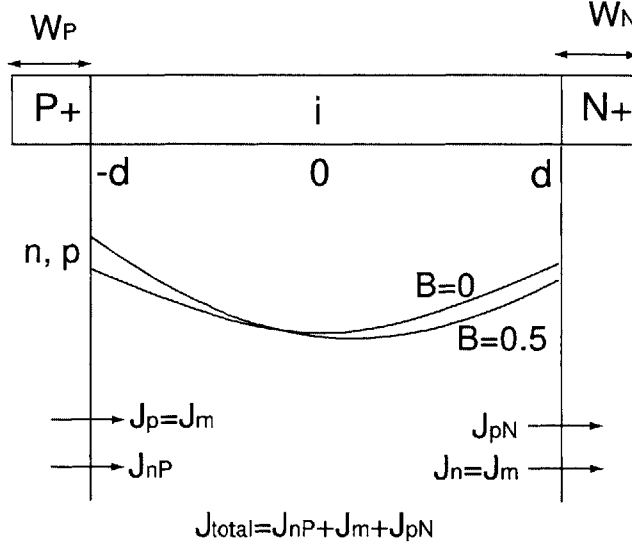


Figure A-1: Diagram of PIN diode structure, carrier distribution at intrinsic region and the current density at the edges $x = \pm d$ when assuming that the injection efficiency at both ends is not unity.

voltage $n(-d) = n_i^2/N_A e^{V/(kT/q)}$, so that

$$\frac{p(-d)n(-d)}{n_i^2} \simeq \frac{n(-d)^2}{n_i^2} = e^{V/(kT/q)} \quad (\text{A.17})$$

where $p(-d) \simeq n(-d)$ is assumed. We know that the diffusion current densities J_{nP} and J_{pN} are proportional to $e^{V/(kT/q)}$ [33], which is proportional to $n(-d)^2$ according to equation (A.17). $n(-d)$ is proportional to J_m according to equation (A.13). Thus we come to the relations $J_{nP} \propto J_m^2$ and $J_{pN} \propto J_m^2$. Equation (A.16) can be rewritten as

$$J_F = J_{nP} + J_m + J_{pN} = K_1 J_m^2 + J_m + K_2 J_m^2 = (K_1 + K_2) J_m^2 + J_m \quad (\text{A.18})$$

The values of the coefficients K_1 and K_2 have been studied in detail in [34]. Assuming $J_F = J_{F0} + J_{F1}e^{j\omega t}$ and $J_m = J_{m0} + J_{m1}e^{j\omega t}$, we have

$$\begin{aligned} J_{F0} + J_{F1}e^{j\omega t} &= (K_1 + K_2)(J_{m0}^2 + J_{m1}e^{j\omega t})^2 + J_{m0} + J_{m1}e^{j\omega t} \\ &\simeq (K_1 + K_2)(J_{m0} + 2J_{m0}J_{m1}e^{j\omega t}) + J_{m0} + J_{m1}e^{j\omega t} \end{aligned} \quad (\text{A.19})$$

where the DC and AC forward current densities are

$$J_{F0} = (K_1 + K_2)J_{m0}^2 + J_{m0} \quad (\text{A.20})$$

and

$$J_{F1} = (K_1 + K_2)2J_{m0}J_{m1} + J_{m1} \quad (\text{A.21})$$

Then the solution for (A.20) is

$$J_{m0} = \frac{\sqrt{1 + 4J_{F0}(K_1 + K_2)} - 1}{2(K_1 + K_2)} \simeq \frac{\sqrt{4J_{F0}(K_1 + K_2)}}{2(K_1 + K_2)} = \sqrt{\frac{J_{F0}}{K_1 + K_2}} \quad (\text{A.22})$$

and the solution for (A.21) is

$$J_{m1} = \frac{J_{F1}}{2J_{m0}(K_1 + K_2) + 1} = \frac{J_{F1}}{2\sqrt{J_{F0}(K_1 + K_2)} + 1} \quad (\text{A.23})$$

The average DC carrier distribution is

$$\bar{n}_0 = \frac{\tau_a J_{m0}}{2dq} = \frac{\tau_a}{2dq} \sqrt{\frac{J_{F0}}{K_1 + K_2}} \quad (\text{A.24})$$

and average AC carrier distribution is

$$\bar{n}_1 = \frac{\tau_a J_{m1}}{2dq} = \frac{\tau_a}{2dq} \frac{J_{F1}}{2\sqrt{J_{F0}(K_1 + K_2)} + 1} \quad (\text{A.25})$$

Appendix B

PIN Diode SPICE Model

SPICE Model Implementation [49]

The carrier distribution in the intrinsic region is governed by the ambipolar diffusion equation:

$$\frac{\partial^2 p}{\partial x^2} = \frac{p}{L_a^2} + \frac{1}{D_a} \frac{\partial p}{\partial t} \quad (\text{B.1})$$

where $D_a = 2D_n D_p / (D_n + D_p)$ is the ambipolar coefficient, $L_a = \sqrt{D_a \tau_a}$ is the ambipolar diffusion length and τ_a is the high-level carrier lifetime, respectively.

Applying the boundary conditions on the both sides of quasi-neutral region as in chapter 2, the DC carrier distribution in the quasi-neutral intrinsic region can be derived, as shown in Fig. B-1:

$$n(x) = p(x) = p_0 \sinh(w/L_a) \left[\frac{\cosh((x - w/2)/L_a)}{\sinh(w/2L_a)} - B \frac{\sinh((x - w/2)/L_a)}{\cosh(w/2L_a)} \right] \quad (\text{B.2})$$

where $B = (\mu_n - \mu_p) / (\mu_n + \mu_p)$, for silicon $b = \mu_n / \mu_p = 3$. The total charges can be integrated from equation (B.2):

$$q = 2A \int_0^w p(x) dx \quad (\text{B.3})$$

where A is the effective cross-section. The quasi-neutral intrinsic region has the same

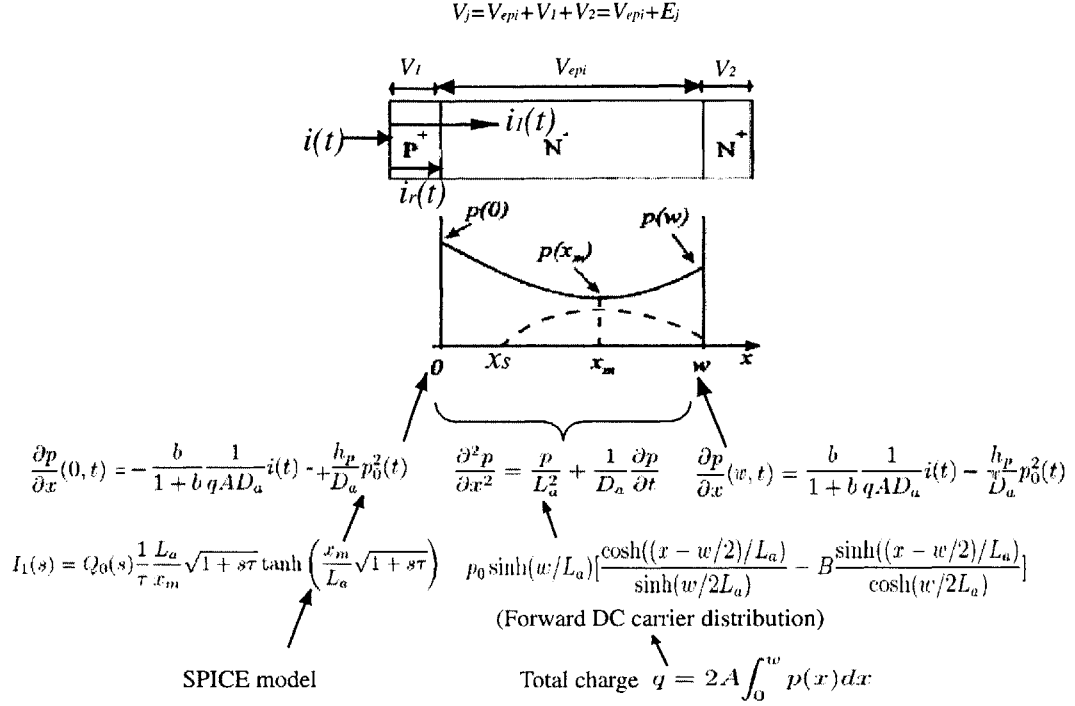


Figure B-1: Carrier distribution in the base region during forward conduction (solid) and reverse-recovery (dashed) [49].

number of holes and electrons and therefore gives a factor of 2.

To simplify the carrier distribution expression in (B.2), the boundary conditions are placed at $x = 0$ and $x = x_m$ [49]:

$$\frac{\partial p}{\partial x}(0, t) = -g(t) = -\left[\frac{b}{1+b} \frac{1}{qAD_a} i(t) - \frac{h_p}{D_a} p_0^2(t)\right] \quad (\text{B.4})$$

$$\frac{\partial p}{\partial x}(x_m, t) = 0 \quad (\text{B.5})$$

Apply Laplace transformation on equation (B.1):

$$\frac{\partial^2 P}{\partial x^2} = \frac{P}{L_a^2} + \frac{s}{D_a} \frac{\partial P}{\partial t} = \frac{1+s\tau}{L_a^2} P \quad (\text{B.6})$$

where and define $P(S)$ is the Laplace transform of $p(t)$. The general solution of

equation (B.6) can be expressed as

$$P = A \cosh\left(\frac{\sqrt{1+s\tau}}{L_a}x\right) + B \sinh\left(\frac{\sqrt{1+s\tau}}{L_a}x\right) \quad (\text{B.7})$$

where A and B are the constants determined by the boundary conditions. Using boundary condition (B.4)

$$\frac{\partial P}{\partial x}\bigg|_{x=0} = -G = B \frac{\sqrt{1+s\tau}}{L_a} \quad (\text{B.8})$$

where G is the Laplace transform of $g(t)$. We can determine

$$B = -G \frac{L_a}{\sqrt{1+s\tau}} \quad (\text{B.9})$$

Using boundary condition (B.5)

$$\frac{\partial P}{\partial x}\bigg|_{x=x_m} = 0 = A \frac{\sqrt{1+s\tau}}{L_a} \sinh\left(\frac{\sqrt{1+s\tau}}{L_a}x_m\right) + B \frac{\sqrt{1+s\tau}}{L_a} \cosh\left(\frac{\sqrt{1+s\tau}}{L_a}x_m\right) \quad (\text{B.10})$$

we have

$$A = -B \coth\left(\frac{\sqrt{1+s\tau}}{L_a}x_m\right) \quad (\text{B.11})$$

The solution for the ambipolar diffusion (B.1) is

$$\begin{aligned} P &= -B \coth\left(\frac{\sqrt{1+s\tau}}{L_a}x_m\right) \cosh\left(\frac{\sqrt{1+s\tau}}{L_a}x\right) + B \sinh\left(\frac{\sqrt{1+s\tau}}{L_a}x\right) \\ &= \frac{-B}{\sinh\left(\frac{\sqrt{1+s\tau}}{L_a}x_m\right)} \left[\cosh\left(\frac{\sqrt{1+s\tau}}{L_a}x_m\right) \cosh\left(\frac{\sqrt{1+s\tau}}{L_a}x\right) \right. \\ &\quad \left. - \sinh\left(\frac{\sqrt{1+s\tau}}{L_a}x_m\right) \sinh\left(\frac{\sqrt{1+s\tau}}{L_a}x\right) \right] \\ &= \frac{-B}{\sinh\left(\frac{\sqrt{1+s\tau}}{L_a}x_m\right)} \cosh \frac{\sqrt{1+s\tau}}{L_a}(x - x_m) \\ &= \frac{GL_a}{\sqrt{1+s\tau} \sinh\left(\frac{\sqrt{1+s\tau}}{L_a}x_m\right)} \cosh \frac{\sqrt{1+s\tau}}{L_a}(x - x_m) \end{aligned} \quad (\text{B.12})$$

At $x = 0$,

$$P_0 = \frac{G \cdot L_a}{\sqrt{1 + s\tau}} \coth\left(\frac{\sqrt{1 + s\tau}}{L_a} x_m\right) \quad (\text{B.13})$$

Substituting $q_0(t) = qAx_m^{\frac{1+b}{b}}p_0(t)$ and $i_1(t) = qA^{\frac{1+b}{b}}D_ag_1(t)$ into equation (B.13), equation (B.13) in S-domain is written as

$$\begin{aligned} I_1(s) &= Q_0(s) \frac{1}{\tau} \frac{L_a}{x_m} \sqrt{1 + s\tau} \tanh\left(\frac{\sqrt{1 + s\tau}}{L_a} x_m\right) \\ &= Q_0(s) \frac{1}{\tau} \frac{L_a^2}{x_m^2} \cdot X \tanh(X) \end{aligned} \quad (\text{B.14})$$

where $X = \frac{\sqrt{1+s\tau}}{L_a} x_m$. Making use of equation (B.14) and identity

$$X \tanh(X) = \frac{1}{X^{-2} + \frac{1}{3 + \frac{1}{5X^{-2} + \dots}}}, \quad (\text{B.15})$$

the transfer function can be written as

$$H(s) = \frac{I_1(s)}{Q_0(s)} = \frac{1}{Z_0 + \frac{1}{\frac{3}{T_0} + \frac{1}{5Z_0 + \dots}}} \quad (\text{B.16})$$

where $Z_0 = \tau/(1 + s\tau)$ and $T_0 = x_m^2/D_a$. Substituting $q_0(t) = qAx_m^{\frac{1+b}{b}}p_0(t)$ and $i_1(t) = qA^{\frac{1+b}{b}}D_ag_1(t)$ into boundary condition (B.4), the total current in the PIN diode is obtained

$$i(t) = i_1(t) + i_r(t) = i_1(t) + \left(\frac{q_0(t)}{\tau}\right)^2 \frac{1}{I_E} \quad (\text{B.17})$$

where $I_E = \frac{qAx_m^2}{h_p\tau^2} \frac{1+b}{b}$. The carriers $q_0(t)$ injected into base-region from P^+n and nN^+ junctions can be obtained from the diode DC characteristic as implemented in PSPICE [75]:

$$q_0(t) = \tau \frac{i_n(t)}{\sqrt{1 + i_n(t)/K_{KF}}} \quad (\text{B.18})$$

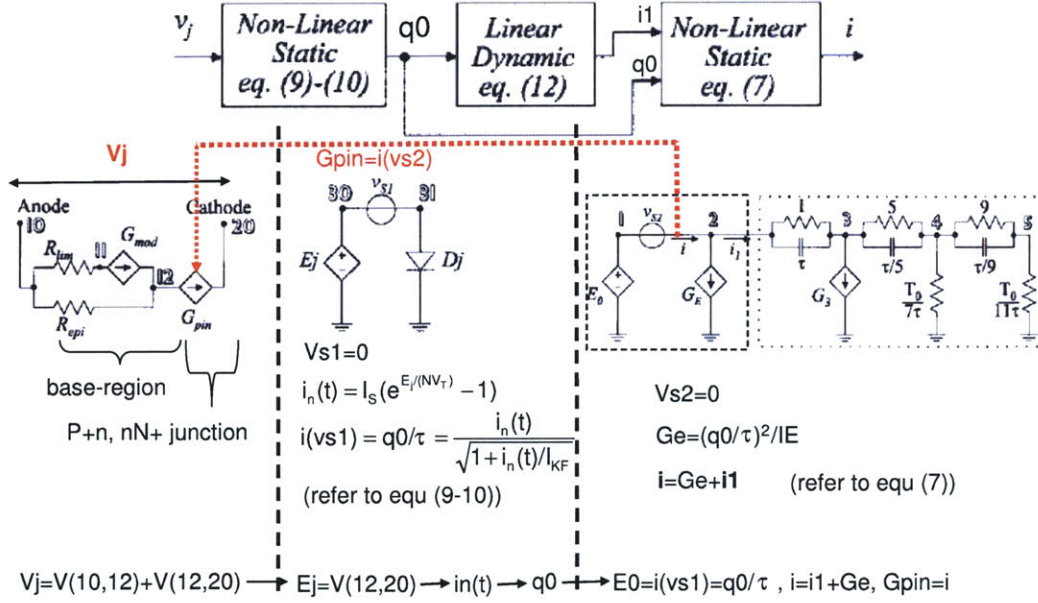


Figure B-2: The three-stage equivalent circuit and model equations interpretation (a linear distributed dynamic system and two nonlinear static blocks) for PIN Spice Model [49].

with $i_n(t) = I_S(e^{E_j/(NV_T)} - 1)$.

The characteristics of P^+n and nN^+ junctions in equation (B.18), the transfer function in (B.16) and the current model in equation (B.17) have been modeled with a three-stage equivalent circuit [49], shown in Fig.B-2. The first nonlinear static block uses applied voltage $V_j(t)$ as input and injected carriers $q_0(t)$ as output to model the characteristics of P^+n and nN^+ junctions. The linear dynamic block models the transfer function in (B.16) to describe the dynamic performance in the base-region. The second non-linear static block uses the outputs of the previous blocks as input and gives the output $i(t)$ which is the total current across the PIN diode.

SPICE Model Results

The large-signal analysis was done with sinusoidal and pulse voltage sources which may be used in analog and digital applications, respectively. Since the factor F_1 depends on voltage and frequency according to Sec.3.2, the factor F_1 was extracted by

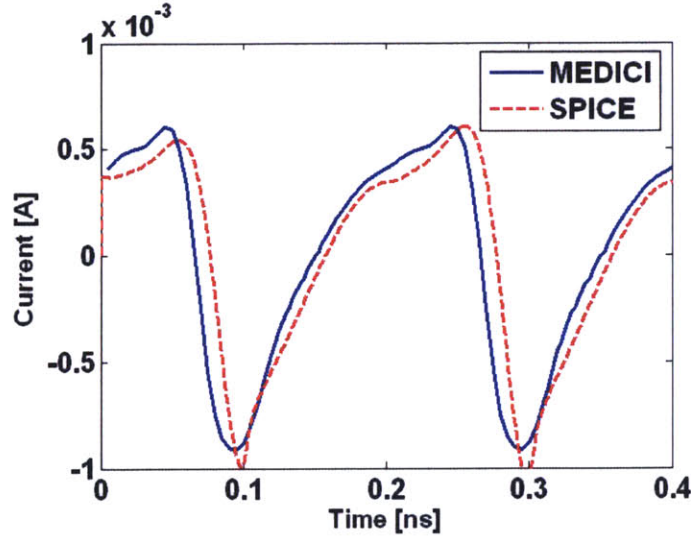


Figure B-3: Large-signal analysis with sinusoidal source $V(t) = V_0 + V_1 \sin(2\pi ft - \pi/2)$ ($V_0 = 0.4V$, $V_1 = 0.4V$).

comparing the SPICE and MEDICI simulation for different voltages and frequencies. F_2 is used to tune the reverse-recovery because HSPICE can't converge with nonlinear element G_3 in reference [49]. Appendix C gives the dependance of the factors F_1 and F_2 on the sinusoidal voltage source $V(t) = V_0 + V_1 \sin(2\pi ft - \pi/2)$. It can be seen that the factors approach to 1 as the frequency and voltage amplitude reduce. The results in low voltage amplitude further verify that it's reasonable to apply $F_1=1$ to the DC and small-signal analysis.

In digital applications, the pulse voltage source is used to simulate logic 1 and 0. The simulated current and total carriers have been done with MEDICI [45] and HSPICE [50] for voltage square pulse train between 1V and 0V and voltage square pulse train between 1V and -1V. In MEDICI modeling, the total carriers are obtained by integrating the carriers in the waveguide region, not including the thin slab regions in Fig. 3-1. However, in HSPICE simulations, part of injected carriers in every cycle are moved out of waveguide by current, part of those recombine and the other remain inside the waveguide. It's hard to know exactly how many carriers recombine and remain inside the waveguide region directly from HSPICE simulations. The total

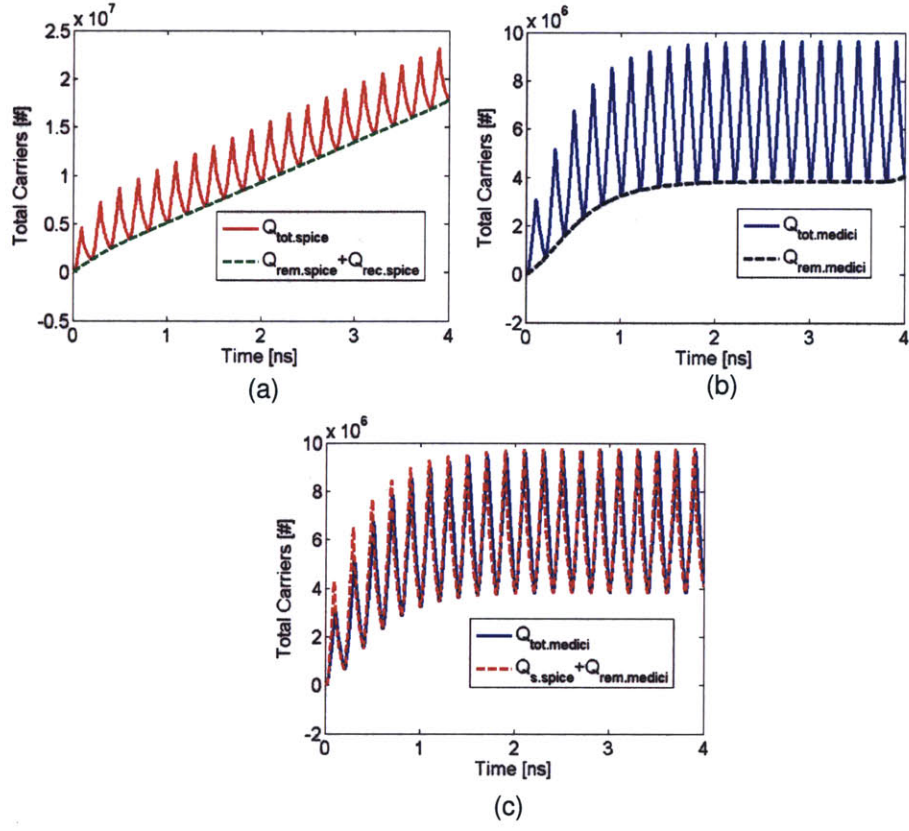


Figure B-4: (a) Carriers obtained from HSPICE simulations, (b) carriers obtained from MEDICI simulations, (c) comparison between carriers from HSPICE and MEDICI . The square pulse train starts with 1V and varies between +1V and 0V, the period is $T = 1/5GHz$ and PIN diode length is $L = 500\mu m$.

current across the diode is written as

$$I_{tot.spice}(t) = I_{s.spice}(t) + I_{rem.spice}(t) + I_{rec.spice}(t) \quad (B.19)$$

where $I_{s.spice}(t)$, $I_{rem.spice}(t)$ and $I_{rec.spice}(t)$ correspond to the carriers $Q_{s.spice}(t)$ injected and removed by current, $Q_{rem.spice}(t)$ remaining and $Q_{rec.spice}(t)$ recombining inside the diode respectively. The carriers can be obtained by integrating the current $Q(t) = \int_0^t I(t)dt/q$:

$$Q_{tot.spice}(t) = Q_{s.spice}(t) + Q_{rem.spice}(t) + Q_{rec.spice}(t) \quad (B.20)$$

It can be seen that the carriers $Q_{s.spice}(t)$ return to zero periodically. The carriers $Q_{rem.spice}(t) + Q_{rec.spice}(t)$ increase as time, and, in the first order, can be extracted by fitting all the minimum points in the plot of total carriers versus time in HSPICE simulations, as shown in Fig.B-4(a). Therefore the $Q_{s.spice}(t)$ can be obtained by, $Q_{s.spice}(t) = Q_{tot.spice}(t) - [Q_{rem.spice}(t) + Q_{rec.spice}(t)]$. It's hard to distinguish $Q_{rem.spice}(t)$ and $Q_{rec.spice}(t)$ in HSPICE simulations. Fortunately, MEDICI modeling can provide $Q_{rem.spice} = Q_{rem.medici}$ which is extracted by fitting all the minimum points in the plot of total carriers versus time in MEDICI modeling, as shown in Fig.B-4(b). Consequently, the integrated carriers in MEDICI $Q_{tot.medici}$ are equal to the stored carriers obtained from HSPICE $Q_{s.spice}(t) + Q_{rem.medici}(t)$, as shown in Fig.B-4(c).

Two kinds of pulse trains are used in SPICE modeling: voltage square pulse train between 1V and 0V and voltage square pulse train between 1V and -1V. These two pulse trains can be used to generate return-to-zero (RZ) and non-return-to-zero (NRZ) signals in digital applications. Figs. B-5, B-6, B-7 and B-8 give the current and total carriers variation versus time with the periods of $T = 1/5GHz$ and $T = 1/10GHz$. The HSPICE simulation results match the those from MEDICI very well when reaching steady-state.

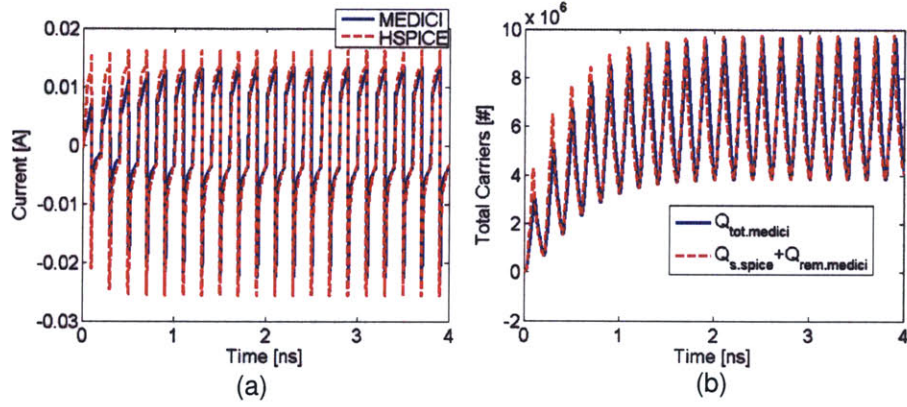


Figure B-5: (a) Modeled current versus time from MEDICI (solid) and HSPICE (dashed). (b) total carriers (half electrons and half holes) variation in the waveguide versus time of MEDICI (solid) and HSPICE (dashed). The square pulse train starts with 1V and varies between +1V and 0V, period is $T = 1/5GHz$ and PIN diode length is $L = 500\mu m$.

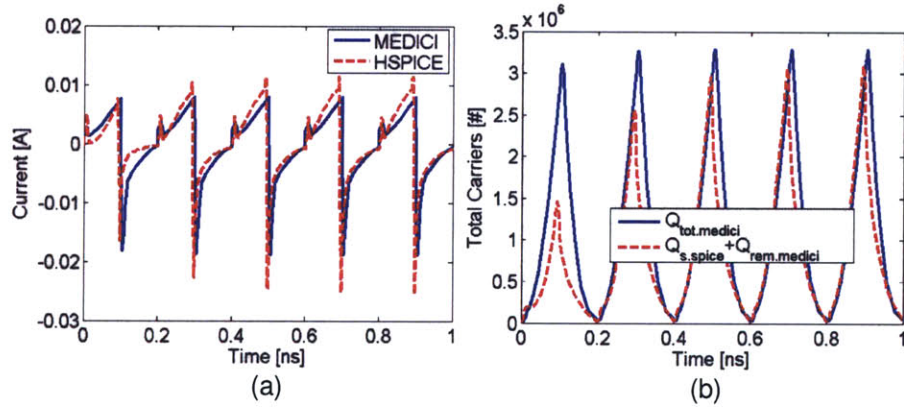


Figure B-6: (a) Modeled current versus time of MEDICI (solid) and HSPICE (dashed). (b) total carriers (half electrons and half holes) variation in the waveguide versus time of MEDICI (solid) and HSPICE (dashed). The square pulse train starts with 1V and varies between +1V and -1V, period is $T = 1/5GHz$ and PIN diode length is $L = 500\mu m$.

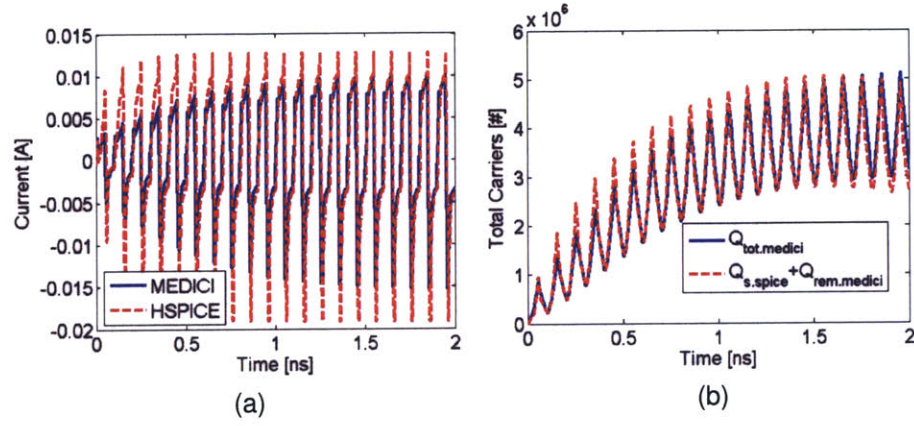


Figure B-7: (a) Modeled current versus time of MEDICI (solid) and HSPICE (dashed). (b) total carriers (half electrons and half holes) Variation in the Waveguide versus time of MEDICI (solid) and HSPICE (dashed). The square pulse train starts with 1V and varies between +1V and 0V, the period is $T = 1/10GHz$ and PIN diode length is $L = 500\mu m$.

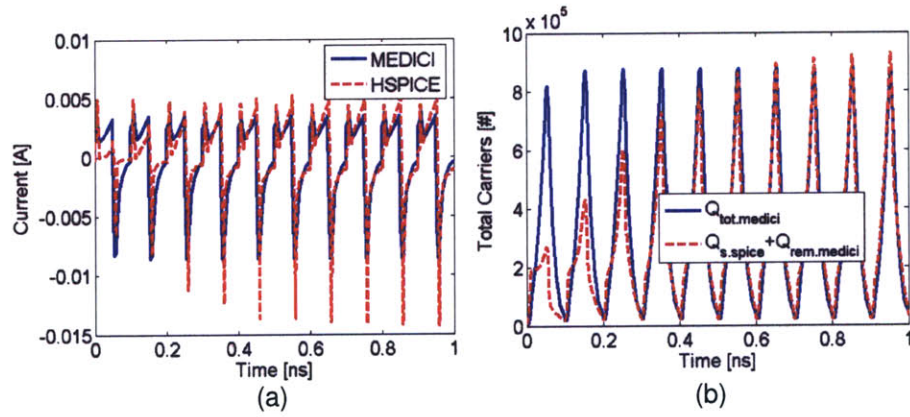


Figure B-8: (a) Modeled current versus time of MEDICI (solid) and HSPICE (dashed). (b) total carriers (half electrons and half holes) Variation in the waveguide versus time of MEDICI (solid) and HSPICE (dashed). The square pulse train starts with 1V and varies between +1V and -1V, period is $T = 1/10GHz$ and PIN diode length is $L = 500\mu m$.

Appendix C

PiN Diode SPICE Model Code

** Generated for: hspice

** Generated on: Jan. 5, 2007

**Fuwan Gan, fuwan@mit.edu

***** parameter table *****

*+++++

* Pulse square voltage source V0→V1→V0→V1 *

* freq=5GHz, V0/V1=1/0V, F1=15, F2=1.8, Lam=0.9 *

* freq=10GHz,V0/V1=1/0V, F1=5, F2=1.8, Lam=0.9 *

* freq=5GHz, V0/V1=+/-1V, F1=1, F2=1, Lam=0.9 *

* freq=10GHz,V0/V1=+/-1V, F1=0.9,F2=1, Lam=0.9 *

*+++++

* Sinusoidal voltagesource V=V0+V1.sin(wt) *

* freq=10GHz,V0/V1=0.5/0.5V, F1=30, F2=1, Lam=0.03 *

* freq=10GHz,V0/V1=0.5/0.25V,F1=29, F2=1, Lam=0.03 *

* freq=5GHz, V0/V1=0.5/0.5V, F1=23, F2=1, Lam=0.03 *

* freq=5GHz, V0/V1=0.5/0.25V,F1=23, F2=1, Lam=0.03 *

* freq=1GHz, V0/V1=0.5/0.5V, F1=3, F2=1, Lam=0.03 *

```

.PARAM
***** Modify parameters *****
+ Length=500e-6 F1=23 F2=1 Lam=0.03
*****

+ L0=500e-6 n=0.85 phi=.7 tau=0.001u w=0.9u VPT=10
+ is=5.78E-12 ik=3 ie=1e-3
+ rmin=1.8m rmax=80k cj=1.48E-14
+ ratio='Length/L0' to='w*w/.001935/4'
+ va='w*w/tau/0.075' af='to/tau'

* define PN junction diode
.MODEL PN D (IS='IS' IK='IK' N='N')

***** begin subckt definition *****

.subckt PiNModel anode cathode

* Main circuit
cjunc anode cathode 'cj*ratio'
gpinF anode cathode value='i(v2)*F1*ratio'

*PiN diode
eF net10 net20 value='v(anode,cathode)'
rmax net10 net12 'rmax'
rmin net10 net11 'rmin'
*grmod net11 net12 value='v(net11,net12)*((1-Lam)*v(net2,net3)+Lam*V(net2))/va'
grmod net11 net12 VCCS POLY(3) net11 net12 net2 net3 net2 0 0 0 0 0 0 '(1-
Lam)/va' 'Lam/va'
gpin net12 net20 value='i(v2)'
vgnd net20 0 0

```

```

* the next resistance helps convergence
rpin net10 0 1e12

*Pn, nN junctions
*ej net30 0 value='v(net12,net20)'
ej net30 0 VCVS net12 net20 1

v1 net30 net31 0
* two different junction diodes
dpi net31 net32 PN
din net32 0 PN

* 10th order H(s) approximation
*e0 net1 0 value='i(v1)'
hireg net1 0 C CVS v1 1

v2 net1 net2 0
*GE net2 0 value='pos(V(net2)*V(net2))/ie'
GE net_dummy2 0 VCCS POLY(1) net_dummy2 0 0 0 '1/ie'

* memorize current
v_dummy1 net_dummy2 net_dummy1 0
v_dummy2 net2 net_dummy2 0

rp1 net_dummy1 net3 1
cp1 net_dummy1 net3 'tau'

```

```

rs1 net3 0 'af/3*F2'
* Grs1 net3 0 value='3*V(net3)/af/hh(V(net20,net12),-i(v2))'
* hspice can't converge with nonlinear element Grs1,
* therefore Factor F2 is used to tune the reverse-recovery.

rp2 net3 net4 5
cp2 net3 net4 'tau/5'
rs3 net4 0 'af/7'
rp4 net4 net5 9
cp4 net4 net5 'tau/9'
rs5 net5 0 'af/11'
rp6 net5 net6 13
cp6 net5 net6 'tau/13'
rs6 net6 0 'af/15'
rp7 net6 net7 17
cp7 net6 net7 'tau/17'
rs7 net7 0 'af/19'
.ends PiNModel
***** End subckt definition *****

```


Bibliography

- [1] Global Roadmap for ceramics 2006 report of the 1st international congress on ceramics, 25-29 June 2006, Toronto, Canada.
- [2] International technology roadmap for semiconductors 2005 edition.
- [3] P. Muthana, M. Swaminathan, R. Tummala, V. Sundaram, L. Wan, S. K Bhattacharya, P. M. Raj, Packaging of multi-Core microprocessors: tradeoffs and potential solutions, in *proceedings of the 55th electronic components and technology conference*, pp.1895-1903, Lake Buena Vista, FL, May 31- June 3 2005.
- [4] C. S. Chang, K. A. Monnig, M. Melliar-Smith, Interconnection challenges and the national technology roadmap for semiconductors, in *Proceedings of the IEEE International Interconnect Technology Conference*, pp.3-6, 1998.
- [5] E. Guizzo and H. Goldstein, Expressway To Your Skull, *IEEE Spectrum*, August 2006.
- [6] Jin Au Kong, *Electromagnetic Wave Theory*, EWM Publishing, 2000.
- [7] G. Served and D. Deschacht, On-chip crosstalk evaluation between adjacent interconnections, in *Proceeding of the 7th IEEE International Conference on Electronics, Circuits and Systems (ICECS)*, vol.2, pp.827-834, December 2000.
- [8] P. W. Juodawlkis, J. C. Twichell, G. E. Betts, J. J. Hargreaves, R. D. Younger, J. L. Wasserman, F. J. O'Donnell, K. G. Ray and R. C. Williamson, Opti-

cally Sampled Analog-to-Digital Converters, *IEEE Transactions on Microwave Theory and Techniques*, Vol.49, No.10, pp.1840, October 2001.

- [9] Private communication with Vladimir M. Stojanovic.
- [10] F. X. Kärtner, S. Akiyama, G. Barbastathis, T. Barwicz, H. Byun, D. T. Danielson, F. Gan, F. Grawert, C. W. Holzwarth, J. L. Hoyt, E. P. Ippen, M. Kim, L. C. Kimerling, J. Liu, J. Michel, O. O. Olubuyide, J. S. Orcutt, M. Park, M. Perrott, M. A. Popovic, P. T. Rackich, R. J. Ram, H. I. Smith and M. R. Watts, Electronic Photonic Integrated Circuits for High Speed, High Resolution, Analog to Digital Conversion, in *SPIE Photonics West* (San Jose, California, USA, January 21-26, 2006)
- [11] D. M. Tennanta, R. Fullowan, H. Takemura, M. Isobe and Y. Nakagawa, Evaluation of a 100 kV thermal field emission electron-beam nanolithography system, *J. Vac. Sci. Technol. B.*, vol.18, no.6, pp.3089-3094, November/December 2000.
- [12] A. Liu, R. Jones, L. Liao, D. Samara-Rubio, D. Rubin, O. Cohen, R. Nicolaescu, M. Paniccia, High-speed silicon optical modulator based on a metal-oxide-semiconductor capacitor, *Nature*, Vol.427, February 2004.
- [13] C. Gunn, Silicon photonics poised to invade local network *Photonics Spectra*, pp.62-67, March 2006.
- [14] A. S. Liu, L. Liao, D. Rubin, H. Nguyen, B. Ciftcioglu, Y. Chetrit, N. Izhaky and M. Paniccia, High-speed optical modulation based on carrier depletion in a silicon waveguide, *Optic Express*, Vol.15, Iss.2, pp.660-668, January 2007.
- [15] J. F. Liu, GeSi Photodetectors and Electro-absorption Modulators for Si Electronic-photonic Integrated Circuits, PhD thesis, Massachusetts Institute of Technology, February 2007.

- [16] M. Paniccia, V. Krutul, R. Jones, O. Cohen, J. Bowers, A. Fang, H. Park, A Hybrid Silicon Laser - Silicon photonics technology for future tera-scale computing, *Intel White Paper*, 2006.
- [17] Vilson R. Almeida, Carlos A. Barrios, Roberto R. Panepucci and Michal Lipson, All-optical control of light on a silicon chip, *Nature*, V. 431, October, 2004.
- [18] K. Kidoo Lee, Transmission and routing of optical signals in on-chip waveguides for silicon microphotonics, PhD thesis, Massachusetts Institute of Technology, 2001.
- [19] Y. Li, J. Yu, S. Chen, Y. Li, Y. Chen, Submicrosecond rearrangeable nonblocking silicon-on-insulator thermo-optic 4x4 switch matrix, *Opt. Lett.*, Vol.32(6), pp.603-4, March 2007.
- [20] M. Fukuda, *Optical Semiconductor Devices*, John Wiley & Sons, Inc., 1999.
- [21] R. A. Soref and J. P. Lorenzo, All-silicon active and passive guided-wave components for $\lambda = 1.3\mu m$ and $1.6\mu m$, *IEEE Journal of Quantum Electronics*, vol.22 no.6 pp.873 -879, June 1986.
- [22] J. P. Lorenzo and R. A. Soref, $1.3 \mu m$ Eletrooptic Silicon Switch, *Applied physics letters*, vol. 51 no.1, pp.6-8, July 1987.
- [23] L. Friedman, R. A. Soref and J. P. Lorenzo, Silicon double-injection electro-optic modulator with junction gate control, *Journal of applied physics*, vol.63, iss.6, pp.1831 -1839, March 1988.
- [24] B. R. Hemenway, O. Solgaard and D. M. Bloom, All-silicon integrated optical modulator for $1.3 \mu m$ fiber-optic interconnects, *Appl. Phys. Lett.*, vol. 55, no. 4, pp.349, 1989.
- [25] P. Dainesi, A. Küng, M. Chabloz, A. Lagos, Ph. Flückiger, A. Ionescu, P. Fazan, M. Declercq, Ph. Renaud and P. Robert, CMOS compatible fully inte-

- grated Mach-Zehnder interferometer in SOI technology, *IEEE Photon. Technol. Lett.*, Vol.12, 660-662, 2000.
- [26] G. V. Treyz, P. G. May and Jean-Marc Halbout, Silicon Mach-Zehnder waveguide interferometers based on the plasma dispersion effect, *Appl. Phys. Lett.*, 59 (7), pp.771, August 1991.
 - [27] C. Angulo Barrios, V. R. Almeida, R. R. Panepucci, B. S. Schmidt and M. Lipson, Compact Silicon Tunable Fabry-Prot Resonator With Low Power Consumption, *IEEE Phot. Tech. Lett.*, VOL.16, NO.2, pp.506-508, February 2004.
 - [28] A. A. Melloni, F. Morichetti and M. Martinelli, Linear and nonlinear pulse propagation in coupled resonator slow-wave optical structures, *Optical and Quantum Electronics*, Vol.35, pp.365-379, 2003.
 - [29] S. Schmitt-Rink, D. S. Chemla, W. H. Knox and D. A. B. Miller, How fast is excitonic electroabsorption *Opt. Lett.*, Vol.15, 60-62, 1990.
 - [30] R. S. Jacobsen, K. N. Andersen, P. I. Borel, J. Fage-Pedersen, L. H. Frandsen, O. Hansen, M. Kristensen, A. V. Lavrinenko, G. Moulin, H. Y. Ou, C. Peucheret, B. Zsigri and A. Bjarklev, Strained silicon as a new electro-optic material, *Nature*, Vol.441, pp.199-202, May 2006.
 - [31] L. L. Gu, W. Jiang, Y. Q. Jiang, X. N. Chen and R. T. Chen, Photonic-crystal-waveguide-based Silicon Mach-Zehnder Modulators, in *Integrated Photonics Research and Applications/Nanophotonics 2006 Technical Digest* (Optical Society of America), Uncasville, Connecticut, USA, April 24-26, 2006.
 - [32] F. Y. Gardes, G. T. Reed, N.G. Emerson, C.E. Png, A sub-micron depletion-type photonic modulator in Silicon On Insulator, *Optics Express*, Vol.13, No.22, pp.8845-8854, October 2005.
 - [33] R. F. Pierret, *Semiconductor Device Fundamentals*, Addison Welsley Publishing Company, Inc., 1996.

- [34] S. K. Ghandhi, *Semiconductor Power Devices*, A Wiley-Interscience Publication, 1977.
- [35] R. A. Soref and B. R. Bennett, Electrooptical Effects in Silicon, *IEEE J. Quantum Electron.*, vol.QE-23(1), 1987.
- [36] R. A. Soref, Silicon-based optoelectronics, in *Proceedings of the IEEE*, Vol.18, No.12, December 1993.
- [37] S. L. Chuang, *Physics of Optoelectronic Devices*, John Wiley & Sons, Inc., 1995.
- [38] Wen Luh Yang, Tan Fu Lei and Chung Len Lee, Contact Resistivities of Al and Ti on Si Measured by a Self-aligned Vertical Kelvin Test Resistor Structure, *Solid-State Electronics*, Vol.32, No.11, pp.997-1001, 1989.
- [39] F. E. Doany, D. Grischkowsky, C. -C. Chi, Carrier lifetime versus ion-implantation dose in silicon on sapphire, *Appli. Phys. Lett.*, Vol.50(8), pp.460-482, 1987.
- [40] F. Gan, F. X. Kärtner, High-Speed Silicon Electrooptic Modulator Design, *IEEE Photonics Technology Letter*, VOL. 17, NO. 5, May 2005.
- [41] M. W. Geis, S. J. Spector, M. E. Grein, R. T. Schuelein, J. U. Yoon, D. M. Lennon, S. Deneault, F. Gan, F. X. Kaertner and T. M. Lyszczarz, CMOS-Compatible All-Si High-Speed Waveguide Photodiodes With High Responsivity in Near-Infrared Communication Band, *IEEE Photonics Technology Letters*, Vol.19, No.3, pp.152-154, February, 2007.
- [42] D. V. Lang, M. Jaros, Complex nature of gold-related deep levels in silicon, *Physical Review B*, Vol.22, No.7, pp.3918-3934, October 1980.
- [43] S. Yegnanarayanan and M. Naydenkov, Method to realize fast silicon-on-insulator (SOI) optical device, Patent application No. 11/411,059, April 2006.

- [44] B. Van Zeghbroeck, *Principles of Semiconductor Devices*, <http://ece-www.colorado.edu/~bart/book/>.
- [45] *MEDICI User Guide*, Synopsis Inc., Mountain View, CA, USA, 2005.
- [46] S. J. Spector, M. E. Grein, R. T. Schulein, M. W. Geis, J. U. Yoon, D. M. Lennon, F. Gan, F. X. Kätner and T. M. Lyszczarz, Compact Carrier Injection Based Mach-Zehnder Modulator in Silicon, in *Integrated Photonics and Nanophotonics Research and Applications (IPNRA) 2007 Technical Digest*, (Optical Society of America, Uncasville, Salt Lake City, Utah, USA, July 8-11, 2007.)
- [47] L. Liao, D. Samara-Rubio, M. Morse, A. Liu, D. Hodge, D. Rubin, U. D. Keil and T. Franck, High speed silicon Mach-Zehnder modulator, *Optics express*, vol.13 iss.8 pp.3129-3135, April 2005.
- [48] A. Liu, D. Samara-Rubio, L. Liao and M. Paniccia, Scaling the modulation bandwidth and phase efficiency of a silicon optical modulator, *IEEE Journal of Selected Topics in Quantum Electronics*, Vol.11, No.2, pp.367-372, March/April 2005.
- [49] A. G. M. Strollo, A New SPICE Model of Power P-I-N Diode Based on Asymptotic Waveform Evaluation, *IEEE Transactions on Power Electronics*, Vol. 12, No.1, pp.12-20, January 1997.
- [50] *HSPICE Applications Manual, June 2006*, Synopsis Inc., Mountain View, CA, USA, June 2006.
- [51] Photon Design, FIMMWAVE, Version 3.41.9, 2002.
- [52] R. A. Soref, E. Cortesi, F. Namavar and L. Friedman, Vertically Integrated Silicon-on-Insulator Waveguides, *IEEE Photonics Technology Letters*, Vol.3, No.1, January 1991.

- [53] M. Bruel, Silicon on insulator material technology, *Electronics Letters*, Vol. 31, No.14, July 1995.
- [54] R. A. Soref, J. Schmidtchen and K. Pertermann, Larger Single-Mode Rib Waveguides in GeSi-Si and Si-onSiO₂, *IEEE Journal of Quantum Electronics*, Vol.27, No.8, August 1991.
- [55] J. Schmidtchen, A. Splett, B. Schuppert, K. Pertermann, Low Loss Single-mode Optical Waveguides with Large Cross-section in Silicon-On-Insulator, *Electronics Letters*, Vol.27, No.16, August 1991.
- [56] H. A. Haus, *Waves and Fields in optoelectronics* , Prentice-Hall, Inc., 1984.
- [57] A. Hache and M. Bourgeois, Ultrafast all-optical switching in a silicon-based photonic crystal, *Appl. Phys. Lett.*, Vol.77, pp.4089-4091, 2000.
- [58] W. Gerlach, H. Schlangenotto and H. Maeder, On the Radiative Recombination Rate in Silicon, *Phys. Stat. Sol. A*, vol.13:277, 1972.
- [59] M. J. Kerr and A. Cuevas, General parameterization of Auger recombination in crystalline silicon, *J. Appl. Phys.*, Vol.91(4), 2002.
- [60] W. Shockley and W. T. Read, Jr., Statistics of the recombinations of holes and electrons, *Physical Review*, Vol.87(5), 1952.
- [61] H. M'saad, The role of surface and bulk perfection in the processing and performance of crystalline silicon, PhD thesis, Massachusetts Institute of Technology, 1994.
- [62] L. L. Luke, Analysis of the interaction of a laser pulse with a silicon wafer: Determination of bulk lifetime and surface recombination velocity, *J. Appl. Phys.* , Vol.61(6), pp.2282-2293, March 1987.
- [63] R. Hull, *Properties of Crystalline Silicon*, INSPEC, The Institution of Electrical Engineers, London, United Kingdom, 1999, <http://www.knovel.com/knovel2/Toc.jsp?BookID=1148>.

- [64] E. Yablonovitch and T. Gmitter, Auger recombination in silicon at low carrier densities, *Appl. Phys. Lett.*, vol.49(10), 1986.
- [65] G. L. Miller, D. A. H. Robinson and J. D. Wiley, *Rev. Sci. Instrum.*, 47, 799(1976); also see US Patent no. 4,286,215 to G. L. Miller, Method and apparatus for the contactless monitoring carrier lifetime in semiconductor materials, August, 1981.
- [66] P. -C. Yang and S. S. Li, Measurements of film carrier lifetimes in silicon-on-insulator wafers by a contactless dual-beam optical modulation technique, *Appl. Phys. Lett.*, vol.61(12), 1992.
- [67] H. J. Kuno, Analysis and Characterization of P-N Junction Diode Switching, *IEEE Trans. IEEE Transactions on Electron Devices*, Vol.11, Iss.1, p.8-14, 1964.
- [68] J. Szczensny and A. Jelenski, a New Method for Carrier Lifetime Measurements in PIN diodes, *Solid-State Electronics*, Vol.32, No.2, pp.125-130, 1989.
- [69] S. Oussalah and R. Jerisian, Carrier Lifetime Recombination Measurement by Conductivity modulation for power P-i-N Diode, in *Proceeding of Electron Devices Meeting*, Hong Kong, 1999.
- [70] Y. C. Kao and J. R. Davis, Correlations Between Reverse Recovery Time and Lifetime of p-n Junction Driven by a Current Ramp, *IEEE Transactions on Electron Devices*, Vol.17, Iss.9, pp.652-657, 1970.
- [71] S. C. Jain and R. J. Van Overstraeten, Discussion of the validity of Kunos relation to determine the base lifetime from a reverse recovery experiment, *J. Appl. Phys.*, Vol.55(2), 1984.
- [72] R. H. Caverly, RF technique for determining ambipolar carrier lifetime in pin RF switching diodes, *Elec. Lett.*, pp.2277-2278, Vol.34, No.23, November 1998.

- [73] Q. F. Xu, B. Schmidt, S. Pradhan and M. Lipson, Micrometre-scale silicon electro-optic modulator, *Nature*, Vol.435, May 2005.
- [74] F. Gan, S. J. Spector, M. W. Geis, M. E. Grein, R. T. Schuelein, J. U. Yoon, T. M. Lyszczarz and F. X. Kätner, Compact, Low-Power, High-Speed Silicon Electro-Optic Modulator, in *the Conference on Lasers and Electro-Optics 2007 Technical Digest*, (Optical Society of America, Baltimore, Maryland, USA, May 6-11, 2007.)
- [75] *The Design Center User's Guide*, MicroSim Corporation, 20 Fairbanks, Irvine, USA, 1993.
- [76] R. W. Boyd, Nonlinear Optics, second edition, Academic Press, 2003.
- [77] S. M. Sze, Physics of Semiconductor Devices, 2nd edition, New York: Wiley, 1981.
- [78] Y.-H. Kuo, Y. K. Lee, Y. Ge, S. Ren, J. E. Roth, T. I. Kamins, D. A. B. Miller and J. S. Harris, Strong quantum-confined Stark effect in germanium quantum-well structures on silicon, *Nature*, Vol.437, No.27, pp.1334-1336, October 2005.
- [79] Rajeev Ram, Physics for Solid State Applications, Lecture notes, Massachusetts Institute of Technology, 2005.
- [80] F. J. Blatt, Physics of Electronic Conclusion in Solids, Mc Graw-Hill Book Company, 1968.
- [81] D. K. Schroder, Semiconductor Material and Device Characterization, A Wiley-Interscience publication, 1990.
- [82] S. G. Kim, G. Barbastathis and H. L. Tuller, MEMS for Optical Functionality, *Journal of Electroceramics*, Vol.12, pp.133-144, 2004.
- [83] S. Ahderom, M. Raisi, K. Lo, K.E. Alameh and R. Mavaddat, Applications of liquid crystal spatial light modulators in optical communications, *High Speed*

Networks and Multimedia Communications 5th IEEE International Conference on 3-5 July 2002, pp.239-242.

- [84] J. Kondo, A. Kondo, K. Aoki, M. Imaeda, T. Mori, Y. Mizuno, S. Takatsuji, Y. Kozuka, O. Mitomi and M. Minakata, 40-Gb/s X-cut LiNbO₃ optical modulator with two-step backslot structure. *Journal of Lightwave Technology*, Vol.20, No.12, pp.2110-2114, 2002.
- [85] L. R. Friedman, R. A. Soref, J. B. Khurgin, Linear and quadratic electrooptic effects in symmetrical and asymmetric quantum-well structures, *IEEE journal of quantum electronics*, vol.31, iss.2, pp.219 -227, February 1995.
- [86] B. R. Bennett and R. A. Soref, Electrorefraction and Electroabsorption in InP, GaAs, GaSb, InAs and InSb, *IEEE Journal of Quantum Electronics*, Vol.QE-23, No.12, pp.2159-2166, December 1987.
- [87] D. A. B. Miller, D. S. Chemla, T. C. Damen, A. C. Gossard, W. Wiegmann, T. H. Wood and C. A. Burrus, Band-edge electroabsorption in quantum well structures: the quantum-confined Stark effect, *Phys. Rev. Lett.*, Vol.53, pp.2173-2176, 1984.
- [88] Y. Shibata, N. Kuze, K. Kaya, M. Kanai and T. Kawai, Piezoelectric LiNbO₃ and LiTaO₃ Films for SAW Device Applications, *1996 IEEE Ultrasonics Symposium*, pp.247-254.
- [89] M. Salib, L. Liao, R. Jones, M. Morse, A. S. Liu, D. Samara-Rubio, D. Alduino, M. Paniccia, Silicon Photonics, *Intel Technology Journal*, Volume 08 Issue 02, May 10, 2004. <http://developer.intel.com/technology/itj/index.htm>
- [90] R. Nagarajan, C. H. Joyner, R. P. Schneider, Jr., Jeffrey S. Bostak, T. Butrie, A. G. Dentai, V. G. Dominic, P. W. Evans, M. Kato, M. Kauffman, D. J. H. Lambert, S. K. Mathis, A. Mathur, R. H. Miles, M. L. Mitchell, M. J. Missey, S. Murthy, A. C. Nilsson, F. H. Peters, S. C. Pennypacker, J. L. Pleumeekers, R. A. Salvatore, R. K. S., R. B. Taylor, H. S. Tsai, M. F. Van Leeuwen,

- J. Webjorn, M. Ziari, D. Perkins, J. Singh, S. G. Grubb, M. S. Reffle, D. G. Mehuys, F. A. Kish and D. F. Welch, Large-Scale Photonic Integrated Circuits, *IEEE Journal of Selected Topics in Quantum Electronics*, VOL.11, NO.1, pp.50-64, January/February 2005.
- [91] Y. L. Liu, E. Liu, G. Z. Li, S. L. Zhang, J. S. Luo, F. Zhou, M. Q. Cheng, B. C. Li and H. Ge, Novel silicon waveguide switch based on total internal reflection, *Appl. Phys. Lett.*, Vol.64 (16), pp.2079-2080, April 1994.
- [92] C.K. Tang and G. T. Reed, Highly efficient optical phase modulator in SOI waveguides, *Elec. Lett.*, Vol.31, No.6, pp.451-452, March 1995.
- [93] A. Cutolo, M. Iodice, P. Spirito and L. Zeni, Silicon Electro-Optic Modulator Based on a three Terminal Device Integrated in a Low-Loss Single-Mode Optical Waveguide, *Journal of Lightwave Technology*, Vol.15, No.3, pp.505-518, March 1997.
- [94] A. Irace, G. Coppola, M. Iodice and A. Cutolo, A high-efficiency silicon optoelectronic modulator based on a Bragg Mirror and integrated in a low-loss Silicon On Insulator waveguide, *SPIE proceeding*, Vol. 3847, pp.94-101, September 1999.
- [95] M. Lee, H. E. Katz, C. Erben, D. M. Gill, P. Gopalan, J. D. Heber, D. J. McGee, Broadband Modulation of Light by Using an Electro-Optic Polymer, *Science*, Vol.298, pp.1401-1403, November 2002.
- [96] D. M. Gill and A. Chowdhury, Electro-Optic Polymer-Based Modulator Design and Performance for 40 Gb/s System Applications, *Journal of Lightwave Technology*, VOL.20, NO.12, pp.2145-2153, December 2002.
- [97] D. Samara-Rubio, L. Liao, A. Liu, R. Jones, M. Paniccia, D. Rubin and O. Cohen, A gigahertz silicon-on-insulator Mach-Zehnder modulator, in *Proceeding of Optical Fiber Communication Conference*, (Optical Society of America, Washington, D.C., 2004), Vol.2, pp.3-5.

- [98] Q. F. Xu, S. Manipatruni, B. Schmidt, J. Shakya and M. Lipson, 12.5 Gbit/s carrier-injection-based silicon micro-ring silicon modulators, *Optics Express*, Vol.15, Iss.2, pp.430-436, January 2007.
- [99] D. Guckenberger, D. Kucharski and A. Narasimha, A Fully Integrated 20-Gb/s Optoelectronic Transceiver Implemented in a Standard 0.13-um CMOS SOI Technology, *IEEE Journal of Solid-state Circuits*, Vol.41, No.12, pp.2945-2955, December 2006.